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Questions

Q1.

The scientist Carl Woese proposed the three-domain system of classification.

The photograph shows *Haemanthus albiflos*, a flowering plant native to South Africa.



(Source: © COULANGES/Shutterstock)

(a) (i) Which of the following is the domain for this plant?

(1)

- ☒ **A** Archaea
- ☒ **B** Bacteria
- ☒ **C** Eukarya
- ☒ **D** Prokaryota

(ii) Describe the evidence that Carl Woese used to support his system of classification.

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(b) A pollen grain can germinate on the stigma of a flower.

After germination, a pollen tube grows through the style.
Explain the function of a pollen tube in the fertilisation of flowering plants.

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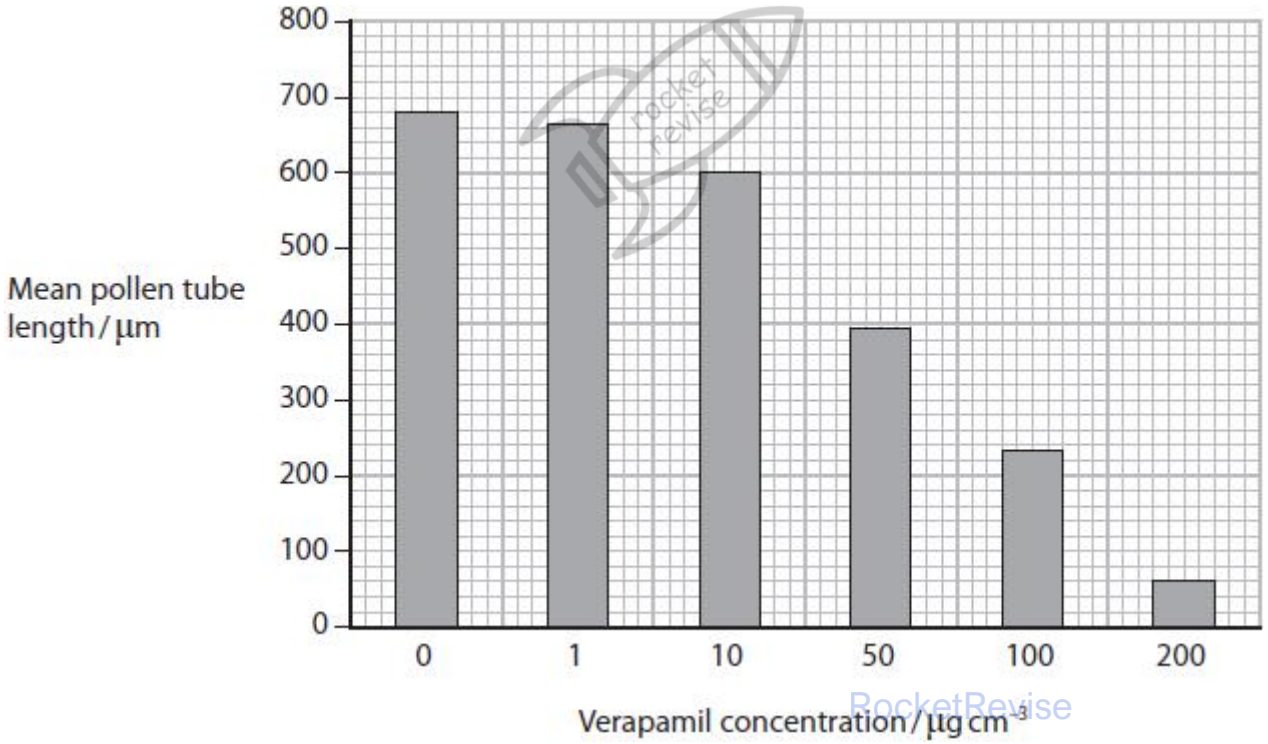
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(c) Verapamil is a substance that inhibits the uptake of calcium ions by the pollen tube.

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An investigation was carried out to study the effect of Verapamil on the growth of pollen tubes.
Each pollen sample was exposed to a different concentration of Verapamil solution.
The lengths of the pollen tubes in each sample were measured after one hour.
The graph shows the mean length of the pollen tubes after one hour in each concentration of Verapamil.



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(i) Determine the effect of Verapamil concentration on the growth of pollen tubes.

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(ii) Explain why Verapamil can reduce the strength of cell walls.

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(Total for question = 11 marks)

Q2.

Farmers sometimes spray herbicides on their fields. Herbicides kill plants that the farmer does not want to grow.

The effect of the concentration of herbicide on the number of species of insects in three fields was investigated.

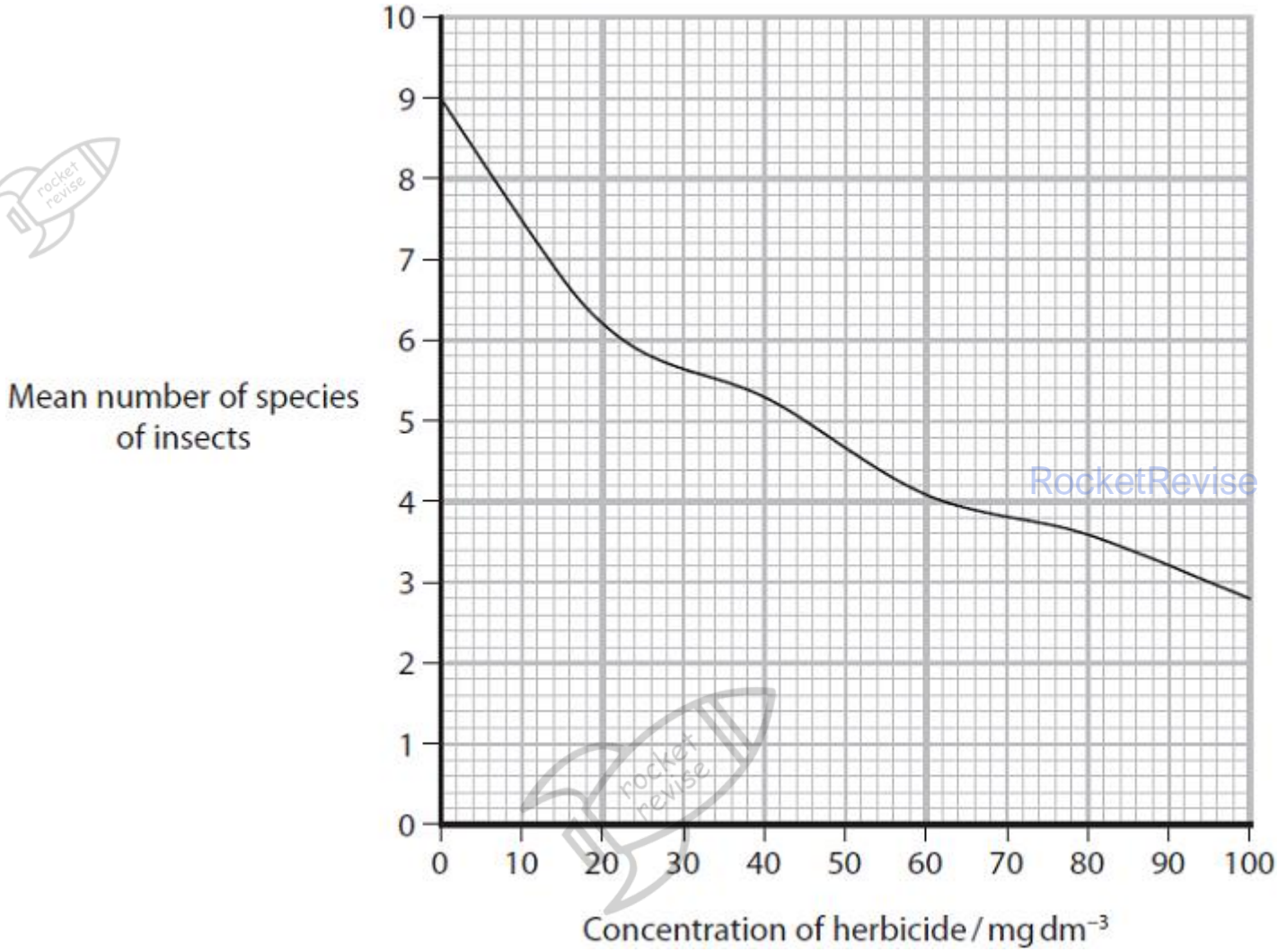
The mean number of species for the three fields was calculated.

The graph shows the results of this investigation.

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(a) State what is meant by the term species.

(1)

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(b) Explain the relationship between the concentration of herbicide and the mean number of species of insects in the three fields.
Use the information in the graph to support your answer.

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(Total for question = 5 marks)

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Q3.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Organisms can be classified using the three-domain system.

Molecular evidence is used to place an organism into one of the three domains.

(a) (i) Give **one** example of the molecular evidence used to support the three-domain system.

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(ii) Describe the role of the scientific community in evaluating the evidence for this system of classification.

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(b) The table shows some features of living organisms.

Which box in each row shows whether the feature is found in these domains?

(4)

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Feature	Archaea only	Archaea and Bacteria only	Archaea and Eukarya only	Archaea and Bacteria and Eukarya
absence of a nuclear envelope	☐	☐	☐	☐
presence of circular DNA	☐	☐	☐	☐
presence of a cell membrane	☐	☐	☐	☐
presence of ribosomes	☐	☐	☐	☐

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(Total for question = 7 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The Ulin tree (*Eusideroxylon zwageri*), is native to Brunei, Indonesia and the Philippines.

The photograph shows an Ulin tree.

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(Source: © Pacific Imagica / Alamy Stock Photo)

(a) This species of tree is endangered and the export of wood from these trees is forbidden from some countries.

Suggest **three** reasons why this species of tree is endangered.

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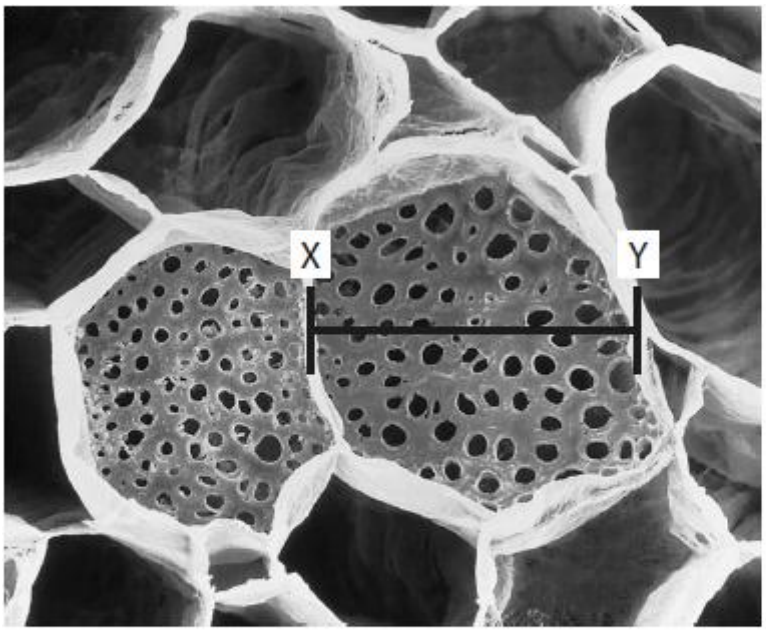
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(b) Trees, and other plants, contain phloem vessels and sclerenchyma fibres.

The photograph shows sieve plates in the phloem of a plant.

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(Source: © Science Photo Library / Alamy Stock Photo)

magnification × 200

(i) Calculate the width of the phloem vessel along the line XY in micrometres.
Give your answer in standard form.

(3)

Answer ..... μm

(ii) Describe the role of phloem.

(2)

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(iii) How many of the following statements about mature sclerenchyma fibres are correct?

- they contain cellulose microfibrils
- they contain pits
- they contain cytoplasm

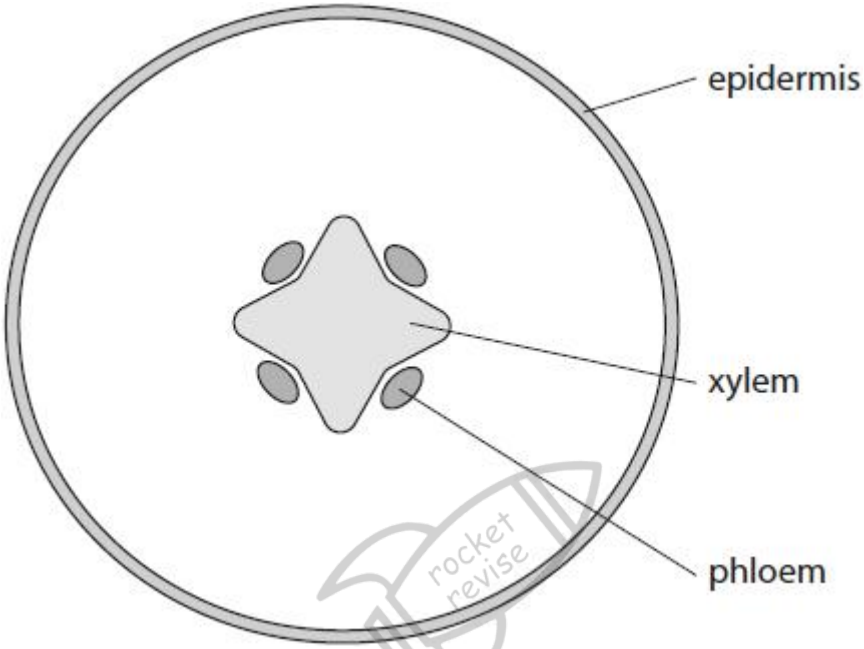
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(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(c) The roots of plants contain phloem and xylem vessels, as shown in the diagram.



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Give **three** differences between the distribution of phloem and xylem in the root compared with their distribution in the stem.

(3)

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(Total for question = 12 marks)

Q5.

Beavers were hunted to extinction in Scotland in the 16th century.

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In mainland Europe, populations of beavers have become isolated from each other.

In 2009, 11 beavers were reintroduced to Scotland from Norway.

Beavers are adapted to live on land and in water. They can cut down trees. They gnaw branches from trees, which they use to build dams.

The photograph shows two beavers and a pond that has been created due to a beaver dam.



(a) Explain how the reintroduction of beavers resulted in a change in the biodiversity in Scotland.

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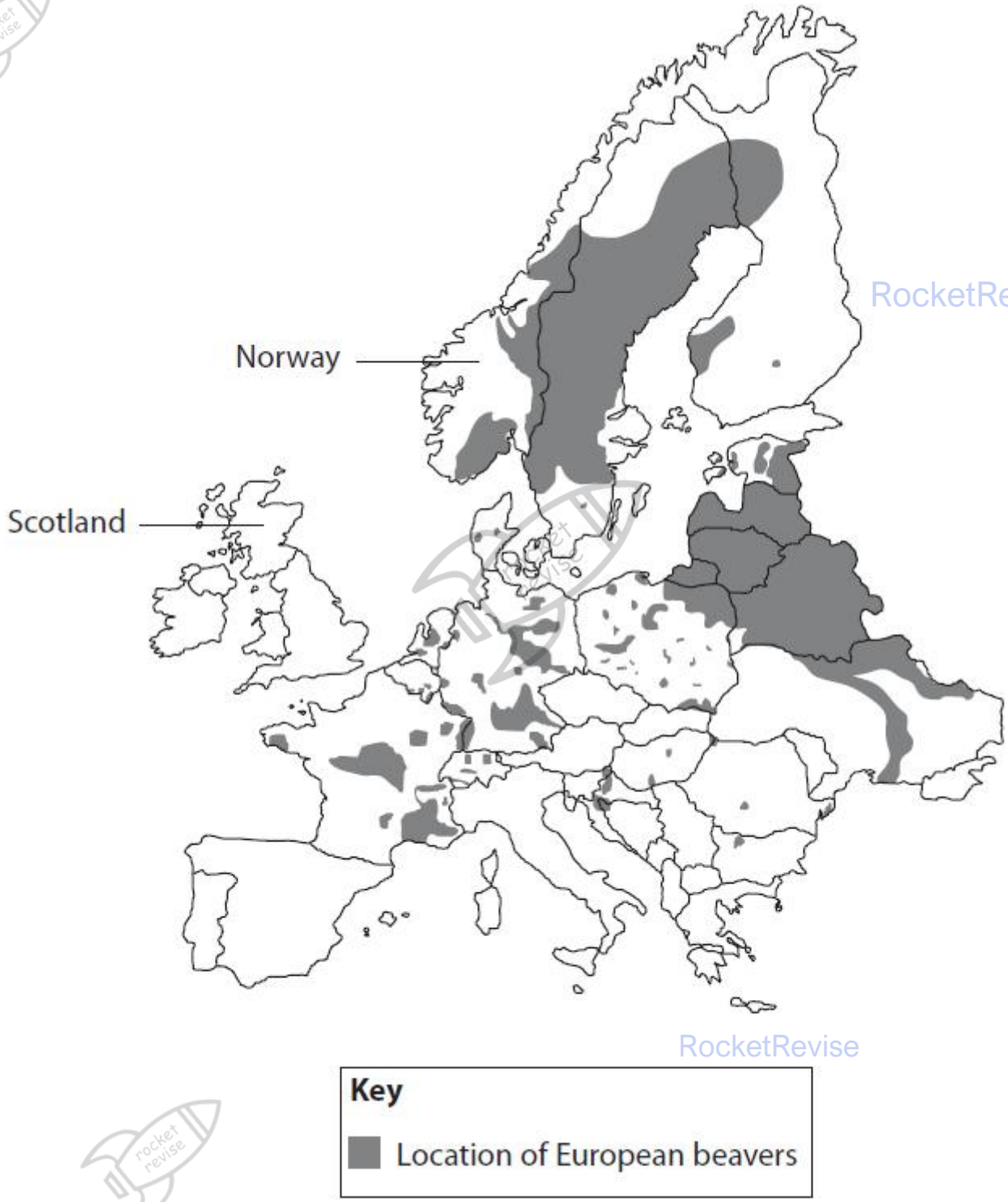
*(b) Some scientists were concerned that the low numbers of beavers reintroduced into Scotland from Norway could affect the genetic diversity of the population in Scotland. The scientists proposed two solutions to this concern:

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- to introduce more beavers from Norway
- to introduce beavers from several different European locations.

The map shows locations of European beavers before they were reintroduced to Scotland.



Discuss the solutions, proposed by these scientists, to overcome the concern of introducing only 11 beavers into Scotland from Norway.
Use the information in the map to support your answer.

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(c) In a beaver population, the frequency of a recessive homozygous genotype is 0.09.

Calculate the percentage of beavers in this population that are homozygous for the dominant allele, using the equation

$$p^2 + 2pq + q^2 = 1$$

(3)

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Answer ..... %

(Total for question = 13 marks)

Q6.

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The photograph shows a strawberry poison dart frog.



(Source: © Dirk Ercken. Shutterstock (PAL))

The strawberry poison dart frog is found in parts of Nicaragua, Costa Rica and Panama only. This frog has several features that are adaptations to its environment.

Three of these features are:

- bright red skin with black spots
- production of a powerful poison that can cause paralysis
- adhesive discs at the end of each toe.

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(a) Which of the following is a term used to describe the fact that this frog is found only in these countries?

(1)

- ☒ **A** diversity
- ☒ **B** endemic
- ☒ **C** polygenic
- ☒ **D** specific

(b) Complete the table to show the type of adaptations shown by this frog.

(3)

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Feature	Type of adaptation
bright red skin with black spots	
production of poison	
adhesive discs on toes	

(c) Suggest the advantages for the frog of these features.

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(d) Suggest how natural selection could have given rise to frogs that can produce poison.

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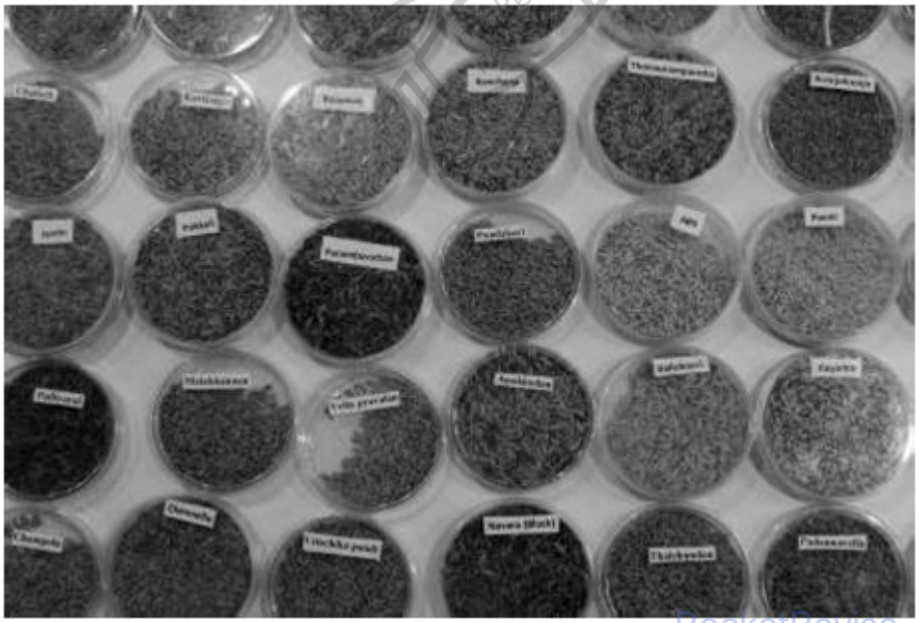
Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

One of the aims of seed banks is to maintain genetic diversity.

Seeds are stored in the Navdanya Seed Vault, one of over a hundred seed banks in India.
This seed bank stores over 4000 varieties of rice.

The photograph shows some of these varieties of rice.



(a) (i) Which of the following can be used to calculate genetic diversity?

(1)

- ☒ **A** Hardy–Weinberg equation
- ☒ **B** heterozygosity index
- ☒ **C** index of diversity
- ☒ **D** mitotic index

(ii) Give **two** differences between genetic diversity and species richness.

(2)

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(b) Suggest **two** reasons why seed banks store seeds instead of growing whole plants.

(2)

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(c) Bananas are a popular food and seed banks are involved in the development of disease-resistant varieties.

The table shows the mass of bananas imported by three countries in 2007 and in 2011.

Country	Mass imported in 2007 / tonnes $\times 10^5$	Mass imported in 2011 / tonnes $\times 10^5$
China	3.3	8.2
Iran	6.3	6.2
Japan	9.8	1.1

(i) Calculate the percentage increase in the mass of bananas imported by China.

(2)

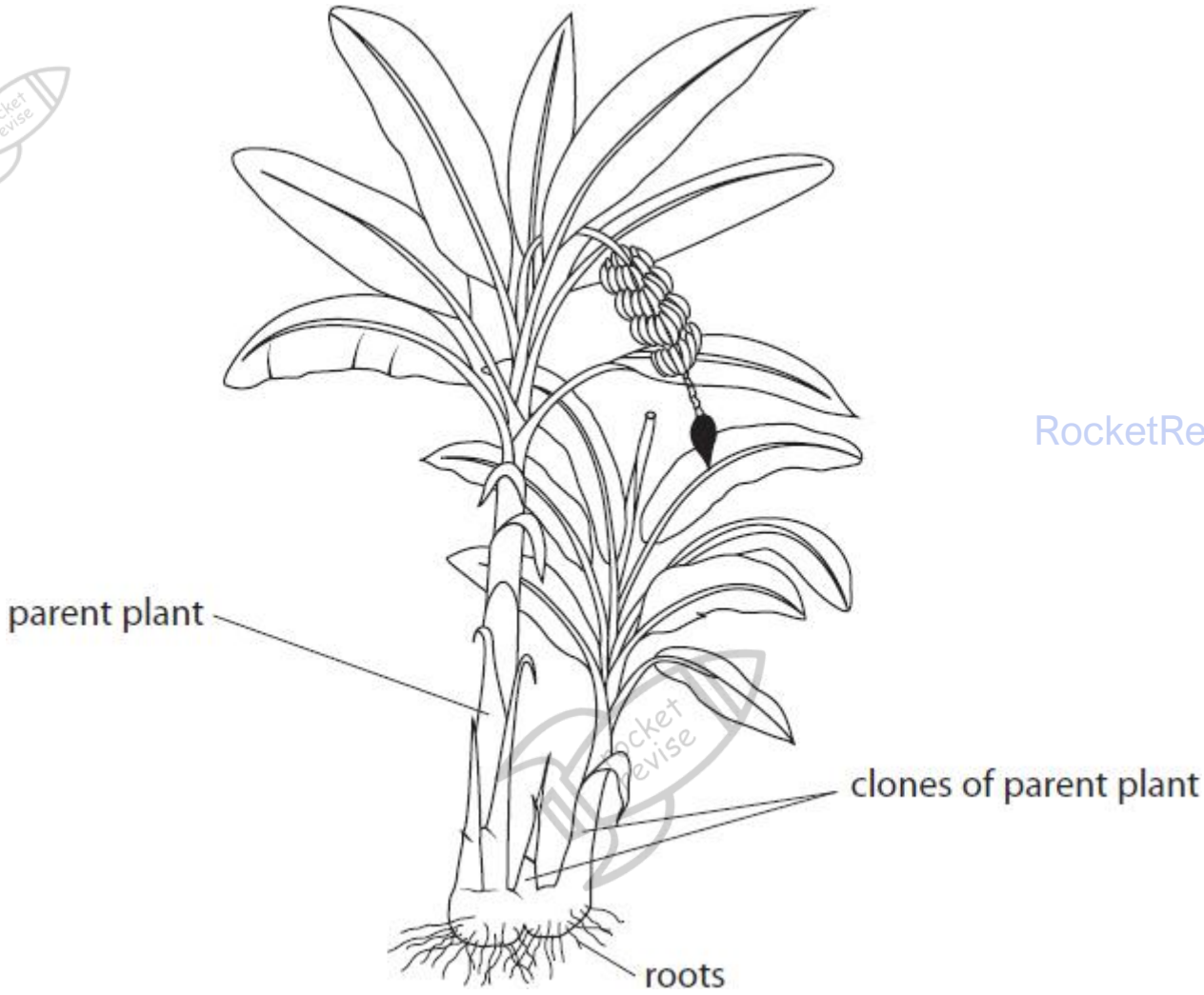
Answer ..... %

(ii) Banana plants can reproduce asexually.
Most of the bananas grown by farmers around the world are clones from a

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single parent banana plant called 'The Cavendish'.
The diagram shows a parent banana plant and two clones of this parent plant.



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These clones can be removed from the parent plant and grown into new banana plants.
Panama disease is caused by a fungus. It has destroyed Cavendish banana plants in the Philippines. Cavendish banana plants are not resistant to this fungus.
Seed banks around the world have seeds from different varieties of banana.
Scientists are trying to improve the Cavendish banana by breeding it with different varieties of banana.
Explain why scientists are breeding the Cavendish banana with different varieties of banana.

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(Total for question = 11 marks)

Q8.

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Tendrils are produced by climbing plants, such as pea plants.

The photograph shows the tendrils on a growing pea shoot.



(Source: © Jerry Lin/Shutterstock)

A tendril is a modified leaf that can coil around anything it touches.

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Cells in the tip of the tendril divide by mitosis.

(a) A student investigated the mitotic index of cells in the tips of tendrils.

Acetic orcein stain was added to the tissues from the tips of the tendrils.

(i) Give **one** reason why the cells were stained in this investigation.

(1)

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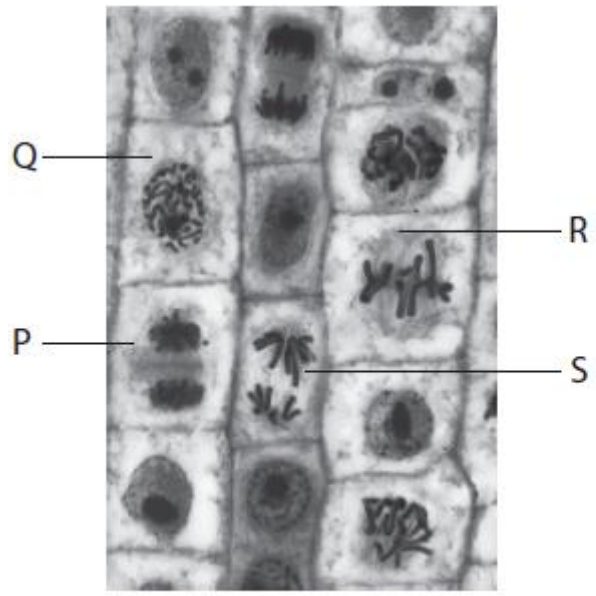
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(ii) The photograph shows these cells, as seen using a light microscope.

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(Source: © Steve Gschmeissner/Science Photo Library)

Which labelled cell is in the metaphase stage of mitosis?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R
- ☒ **D** S

(iii) The table shows the results of this investigation.

Stage	Number of cells
Interphase	117
Prophase	15
Metaphase	3
Anaphase	10
Telophase	5

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Calculate the mitotic index.

(2)

Answer .....

(b) Suggest the advantages of tendrils for the pea plant.

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(c) Pea plants produce flowers.

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Some cells in the flowers undergo meiosis.
Explain how the separation of chromosomes in meiosis gives rise to genetic variation in the offspring of pea plants.

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(Total for question = 10 marks)

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Q9.

The photograph shows a Tristan albatross.

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(Source: © blickwinkel/Alamy Stock Photo)

The Tristan albatross is endemic to a few small islands in the South Atlantic Ocean.

The adult birds nest on these islands and fly long distances in search of food. They feed on fish, octopus and squid.

The Tristan albatross is critically endangered, with 4 500 of these birds left in the wild.

(a) (i) Describe **two** anatomical adaptations of the Tristan albatross that enable it to occupy its niche.

Use the information in the photograph to support your answer.

(2)

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(ii) It has been predicted that the population will continue to decline by 5.3% per year.
Calculate the predicted albatross population after one year.

(2)

Answer .....

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(b) The Tristan albatross (*Diomedea dabbenena*) was once classified as the same species as the wandering albatross
(*Diomedea exulans*).

The photograph shows a wandering albatross.



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(Source: © Rebecca Jackrel/Alamy Stock Photo)

Suggest why the Tristan albatross and wandering albatross were once classified as the same species.

(1)

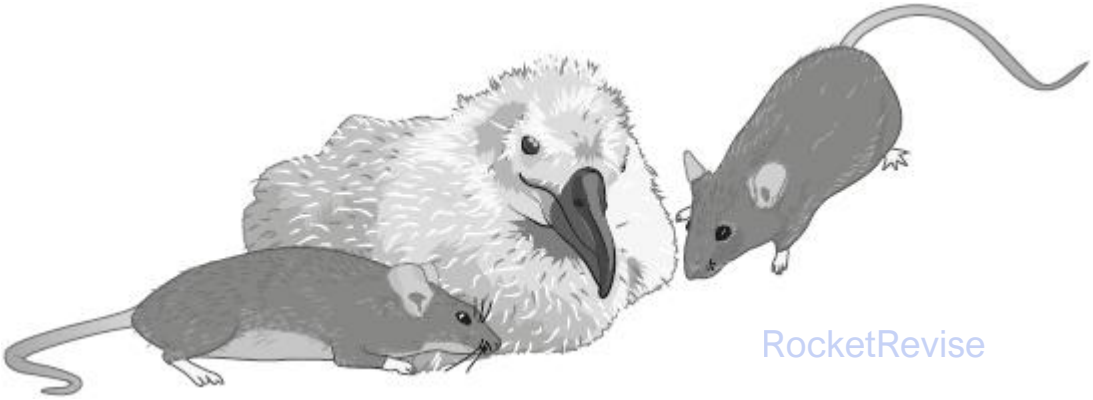
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(c) Tristan albatross chicks in nests on one island have been found to be at risk of predation from mice.

The diagram shows an albatross chick and two mice on this island.



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Over a long period of time, mice on this island have become 50% larger than normal mice.
It has been suggested that the mice from this island are a new species.
Suggest how the mice on this island have evolved to become a new species.

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*(d) Scientists have suggested that the Tristan albatross could be conserved.

Some suggestions are conservation strategies on the island where the mice are found.
Some suggestions involve zoos in other countries.
Explain how the Tristan albatross could be conserved.
Use the information in the question to support your answer.

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(Total for question = 15 marks)

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Q10.

The photograph shows a Madagascan pygmy kingfisher (*Ceyx madagascariensis*).



(Source: © Nature Picture Library/Alamy Stock Photo)

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This kingfisher is endemic to forests in Madagascar.

The diet of this kingfisher is mainly frogs and insects.

(a) (i) State what is meant by the term **endemic** with reference to this kingfisher.

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(ii) State the domain in which this kingfisher is classified.

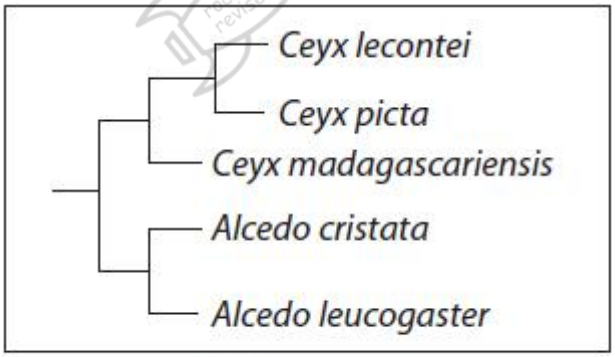
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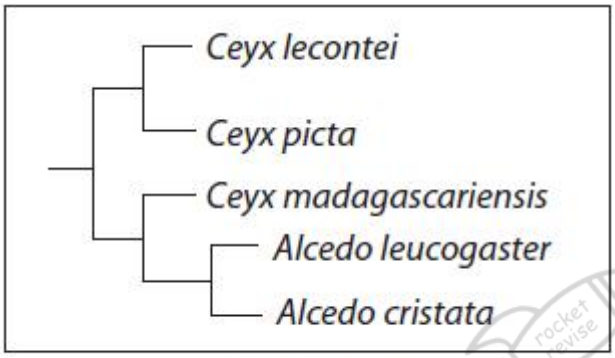
(b) A scientist recently collected new data about the classification of kingfishers, using molecular evidence.

The data were used to propose the reclassification of some kingfishers, including *Ceyx madagascariensis*.

The diagrams show an old classification and the proposed reclassification of five species of kingfishers.



Old classification



Proposed reclassification

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(i) Using the proposed reclassification, explain which species of kingfisher are most closely related to *Ceyx madagascariensis*.

(2)

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(ii) Explain how molecular evidence led to the reclassification of species and how the scientific community would have reviewed the evidence before it was accepted.
Use the information in the photograph and the diagrams to support your answer.

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(Total for question = 10 marks)

Q11.

The photograph shows a river and mangrove trees in Bhitarkanika National Park in India.



(Source: © Mariano Gaspar/Shutterstock)

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(a) (i) The biodiversity of this area of Bhitarkanika National Park could be measured by calculating species richness.

Which of the following describes species richness in Bhitarkanika National Park?

(1)

☒ **A** the number of animal species

☒ **B** the number of plant species

☒ **C** the number of organisms in each of the species

☒ **D** the number of species

(ii) Describe **one** other way in which the biodiversity of this area of Bhitarkanika National Park could be measured.

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(2)

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(b) Saltwater crocodiles are found in the rivers in this area.

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The numbers of this endangered species have decreased over the past 40 years.
Explain the possible causes of this decrease.

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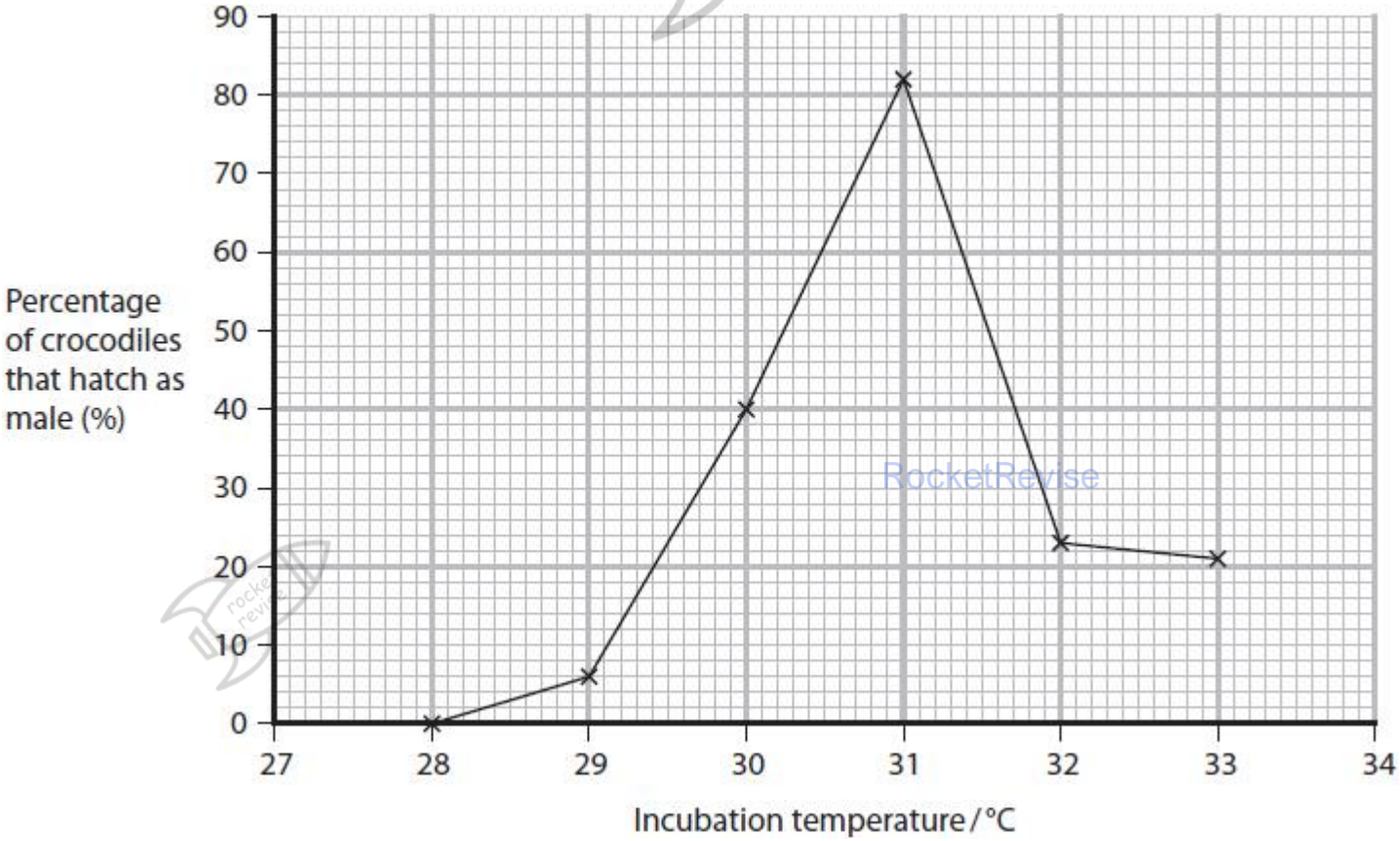
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(c) Scientists are involved in conservation projects to increase the numbers of endangered saltwater crocodiles, without causing a reduction in the genetic diversity of future populations.

One project involves collecting the eggs from nests in different parts of Bhitarkanika National Park.
The eggs are incubated at a suitable temperature.
The incubation temperature influences the sex of the developing crocodiles.
The crocodiles are released into the park when they have hatched.
The graph shows the effect of temperature on the percentage of crocodiles that hatch as male.



(i) Calculate the number of female crocodiles that would develop from 350 eggs incubated at 32 °C.

(2)

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Answer .....

(ii) Explain how this project could increase the number of crocodiles rapidly, without reducing the genetic diversity of future populations in Bhitarkanika National Park.
Use the information in the graph to support your answer.

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(Total for question = 14 marks)

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Q12.

The Galapagos Islands are located off the west coast of South America.

The biodiversity on these islands has changed over the past 40 years due to human activity.

One of these islands is called Santa Cruz.

(a) The population of Santa Cruz increases by approximately 6.4% per year.

The majority of the population, 85%, live in urban areas.
The population of Santa Cruz living in urban areas in 2020 was 17 000.
Predict the total population of Santa Cruz in 2025.
Assume that the rate of increase stays constant.

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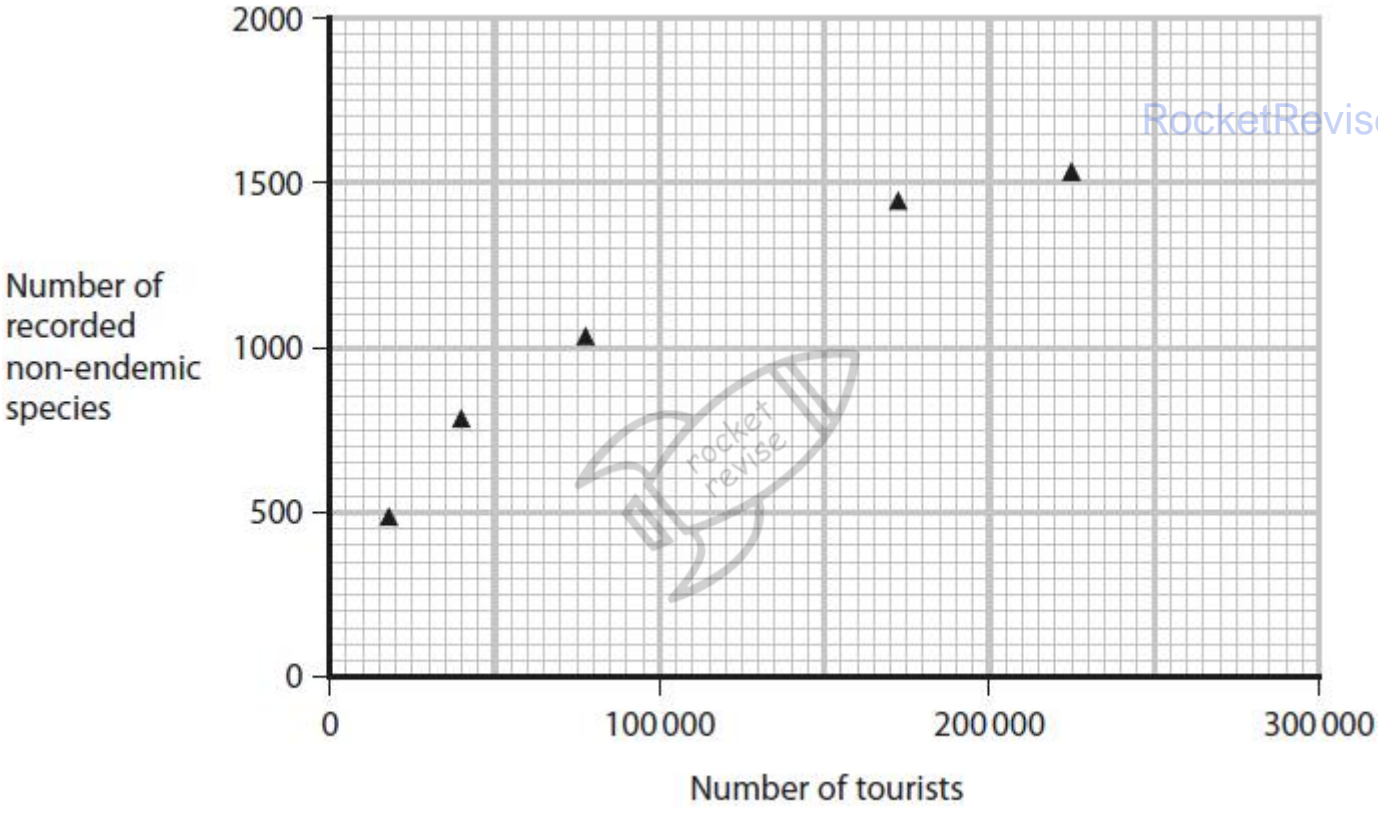
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(3)

Answer .....

(b) Non-endemic species have been introduced to the Galapagos Islands.

The graph shows the number of tourists to all the islands and the number of recorded non-endemic species.



https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0184379

State the relationship shown in the graph.

(1)

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(c) The wild blackberry was introduced to Santa Cruz in 1970 and has since spread over much of the island.

This has resulted in the reduction of the native endemic forest.
In 1960, the number of species in an area of the native forest was measured.
The table shows these results.

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Species	Number of individuals (n)	(n – 1)	n(n – 1)
A	21	20	
B	2	1	2
C	4	3	12
D	13	12	
E	54	53	
F	15	14	210
G	6	5	30
H	32	31	
	Total (N) =		Σn(n – 1) =

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(i) Calculate the index of diversity (D) for this area of the forest using the formula

$$D = \frac{N(N-1)}{\Sigma n(n-1)}$$

Use the table to help you.
Give your answer to **one** decimal place.

(3)

Answer .....

(ii) Explain why the spread of the wild blackberry affects the biodiversity of the island.

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(Total for question = 12 marks)

Q13.

Many female organisms produce egg cells containing cortical granules.

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(a) Explain how cortical granules ensure that the egg cell is diploid after fertilisation.

(4)

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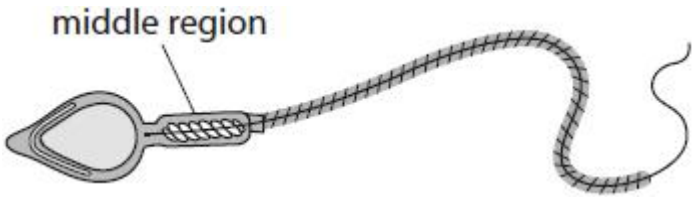
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*(b) Organisms have evolved different mating strategies such as:

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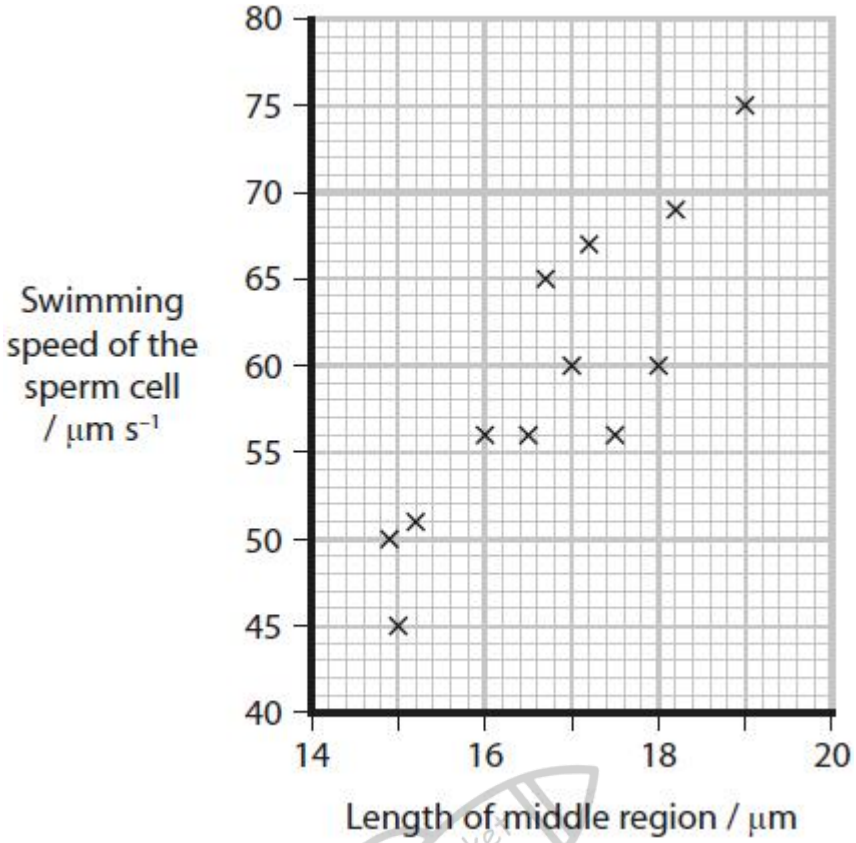
- Some male animals have sperm cells with longer middle regions, as shown in the diagram.



The females of these species produce multiple egg cells and mate with many males in a short period of time.
The graph shows the effect of the length of the middle region on the swimming speed of the sperm cell.

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- Some male animals, such as the zebra longwing butterfly, produce spermatophores that remain inside the reproductive system of the female after mating.

The spermatophore contains sperm cells and nutrients for the female. The spermatophore also releases chemicals that reduce the attractiveness of this female butterfly to other males.

- Some female animals, such as the eastern box turtle, mate with numerous males and then store the sperm for a number of years. This allows fertilised egg cells to be laid even when the female has not mated in that year.

Comment on the effect that these three mating strategies would have on the reproductive success of the males and females, and the genetic diversity of the offspring.

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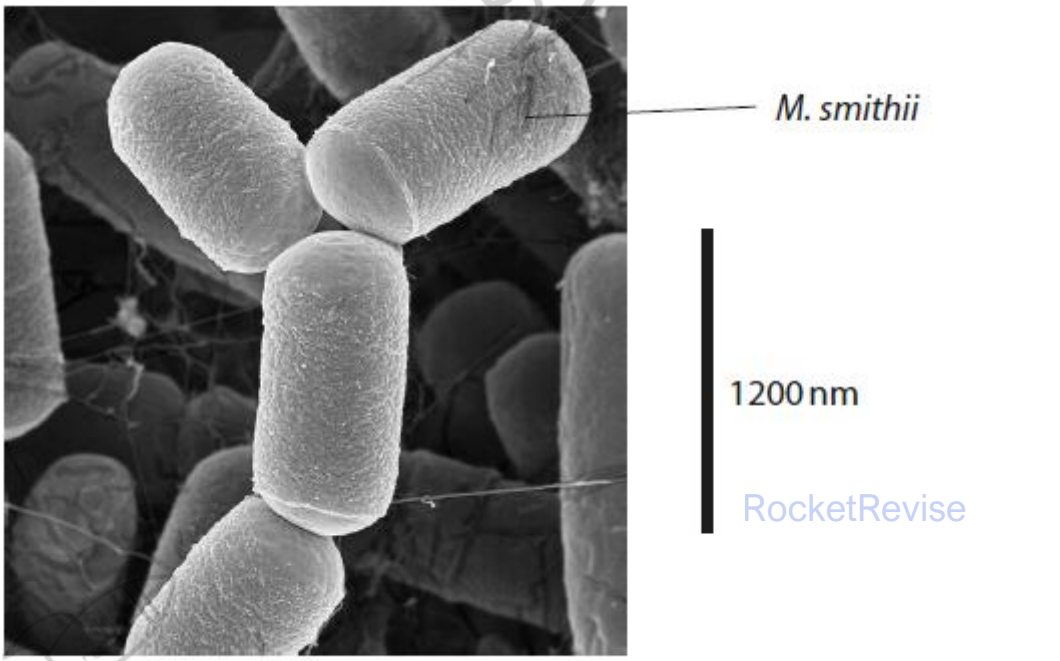
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(Total for question = 10 marks)

Q14.

Methanobrevibacter smithii (*M. smithii*) is a single-celled microorganism found in the human intestine.

The photograph shows *M. smithii*.



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(© DENNIS KUNKEL MICROSCOPY / SCIENCE PHOTO LIBRARY)

(a) (i) Calculate the magnification of this photograph.

Give your answer in standard form to two significant figures.

(2)

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Answer .....

(ii) Explain which type of microscope was used to take this photograph.

(2)

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(b) Scientists classified this microorganism as a species in the domain Archaea.

(i) Describe the information the scientists would have used to classify *M. smithii* into the Archaea domain.

(2)

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(ii) Scientists have identified a similar microorganism in the human mouth.
This microorganism is called *Methanobrevibacter oralis* (*M. oralis*).
Explain how the scientists could confirm that *M. smithii* and *M. oralis* are different species of Archaea.

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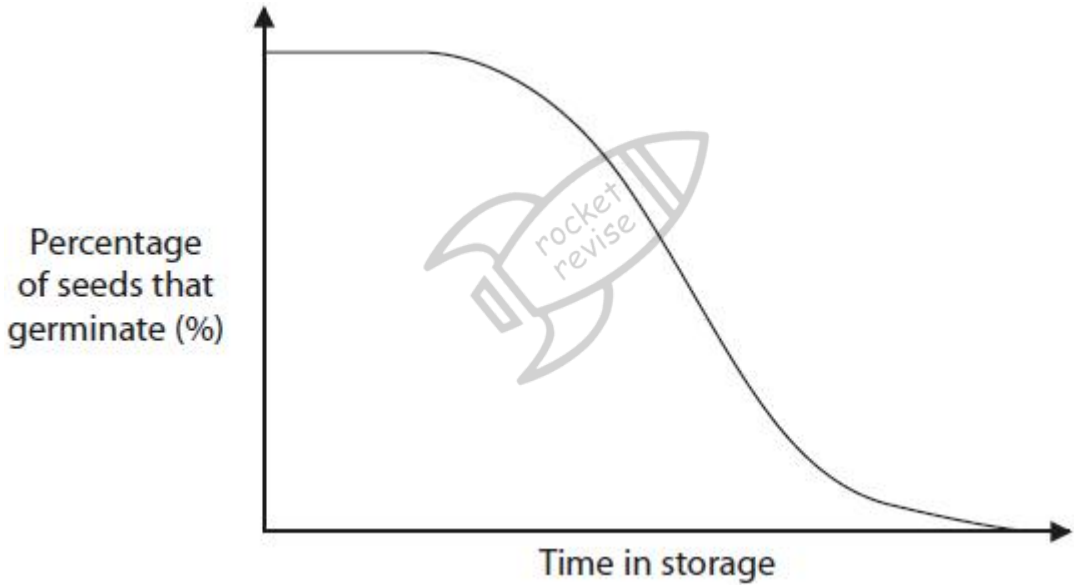
(Total for question = 9 marks)

Q15.

Seed banks are used to store seeds from a wide variety of plants. Some plants are a sustainable source of bioplastics.

(a) The Germplasm Bank of Wild Species is the largest seed bank for wild species of plants in Asia.

Seeds from 8 500 species have been collected and stored.
Seeds were dried before being stored at temperatures below 0 °C.
The effect of length of time in storage on seed germination is shown in the graph.



(i) Describe the effect of length of time in storage on the germination of these seeds.

(2)

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(ii) Two of the methods used in seed banks are:

- store seeds from many different plants of the same species, instead of many seeds from just one plant
- regularly germinate samples of the stored seeds, allowing them to grow into adult plants.

Explain the advantages of these methods.

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(b) The photograph shows bioplastic straws and cutlery, produced from the seeds of avocados.



© Facinadora / Alamy Stock Photo

(i) In 2018, a company in Mexico produced 130 000 kg of bioplastic cutlery and straws per month.
40% of the products were straws.
Calculate the mass of cutlery produced per year by this company.
Give your answer to **two** significant figures.

(2)

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Answer .....

(ii) The use of these plant-based products is more sustainable than the use of cutlery and straws made from oil-based plastic.
Explain what is meant by the term sustainable, with reference to the cutlery produced from the seeds of avocados.

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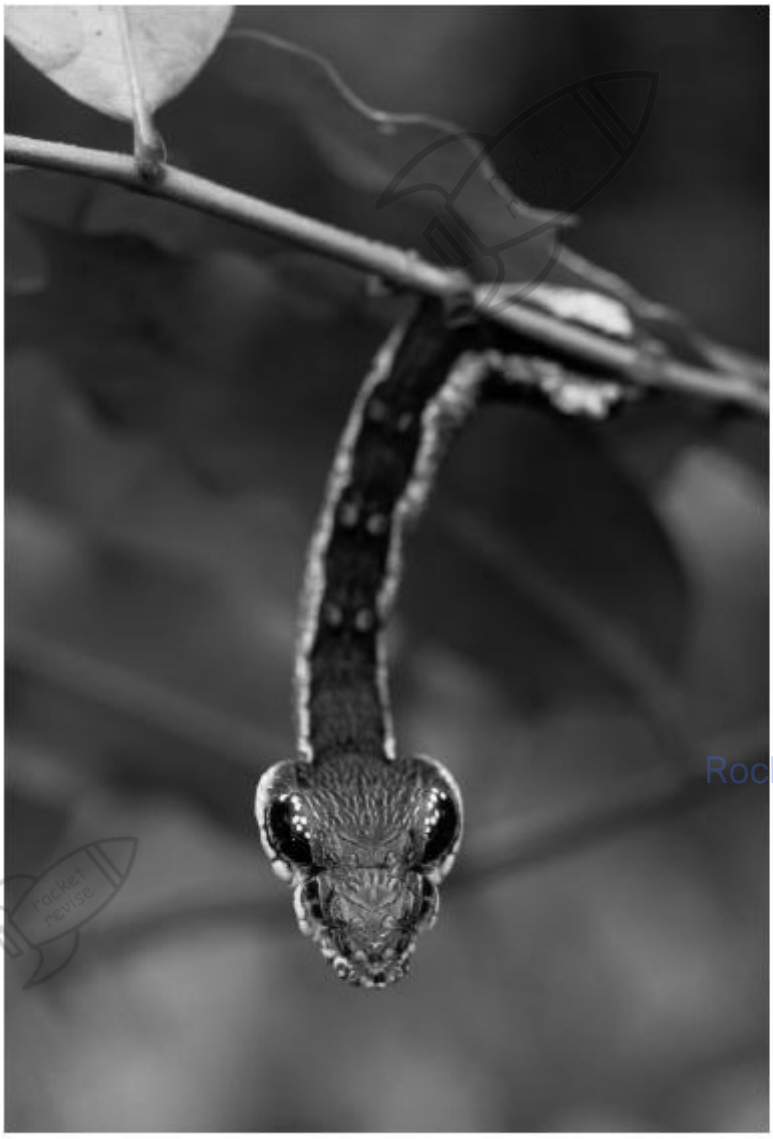
(Total for question = 10 marks)

Q16.

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The photograph shows the caterpillar of a moth from Costa Rica.

The caterpillar looks like a pit viper snake. It will try to strike potential predators.



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(Source: © Avalon/Photoshot License / Alamy Stock Photo)

(a) This caterpillar is adapted to its niche.

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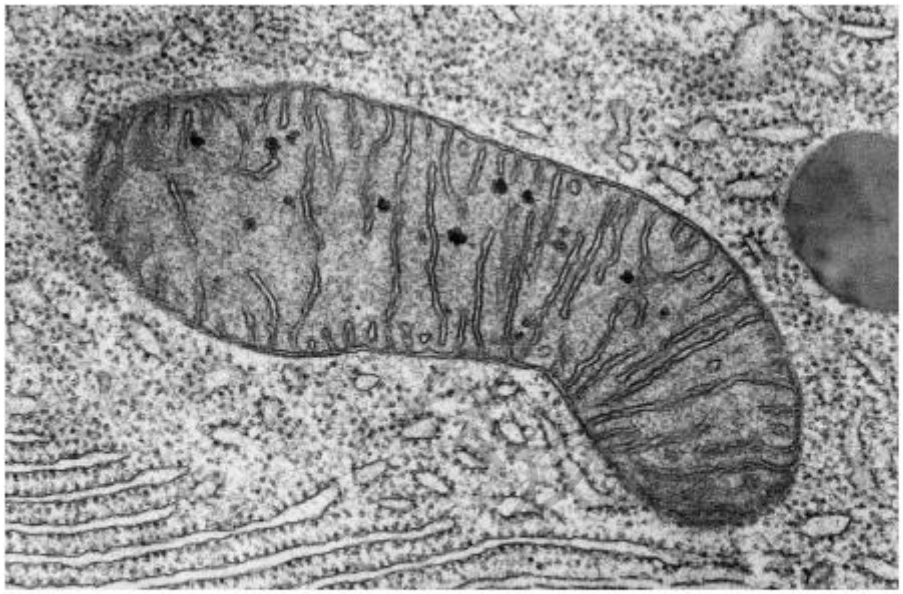
www.RocketRevise.com

Complete the table to show the type of adaptations shown by this caterpillar.

(2)

Feature	Type of adaptation
looks like a pit viper snake	.....
tries to strike potential predators	.....

(b) The photograph shows an organelle that can be found in caterpillar cells, as seen using a microscope.



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(Source: © Science History Images / Alamy Stock Photo)

Magnification × 72 500

(i) State the type of microscope used to view this organelle.

(1)

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(ii) Describe the function of this organelle.

(3)

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(c) Rough endoplasmic reticulum is another organelle found in caterpillar cells.

Draw a labelled diagram of rough endoplasmic reticulum.

(3)

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(Total for question = 9 marks)

Q17.

The phenotype of an organism is the result of interactions between the genotype and the environment.

Phenotypes can show continuous or discontinuous variation in a population.

(a) Sketch a graph to show the distribution for a phenotype showing continuous variation in a population.

(2)

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(b) The presence or absence of teeth in chickens is one example of a discontinuous phenotype.

Chicken genomes contain the genes for tooth production, but chickens do not normally develop teeth. These genes are switched off as the embryo develops.
Explain how a mutation could cause the development of teeth in a chicken embryo.

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(c) The photograph shows two Himalayan goats.

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(Source: © Rainer Pauschert / Alamy Stock Photo)

The fur colour of a goat depends on its genotype.
Gene B controls fur colour in these goats.
The allele for brown fur is dominant to the allele for white fur in goats.
The allele **B** codes for brown fur and the allele **b** codes for white fur.
In one population, there were 470 brown goats and 140 white goats.
Calculate the frequency of the allele **B** in this population, using the equation

$$p^2 + 2pq + q^2 = 1$$

Give your answer to **two** decimal places.

(3)

Answer .....

(Total for question = 9 marks)

Q18.

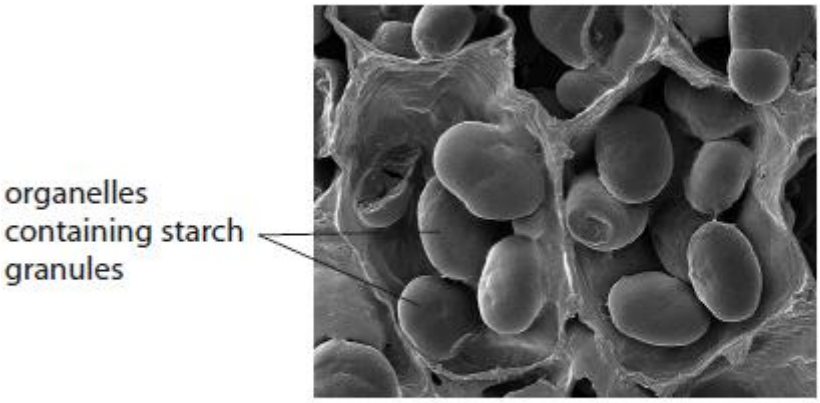
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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The cassava root is an important food crop grown in the tropics.

The photograph shows organelles containing starch granules inside cassava root cells, as seen using an electron microscope.



(Source: © Science Photo Library / Alamy Stock Photo)

(a) (i) Name these organelles found in cassava root cells.

(1)

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(ii) Explain how the structure of starch relates to its function.

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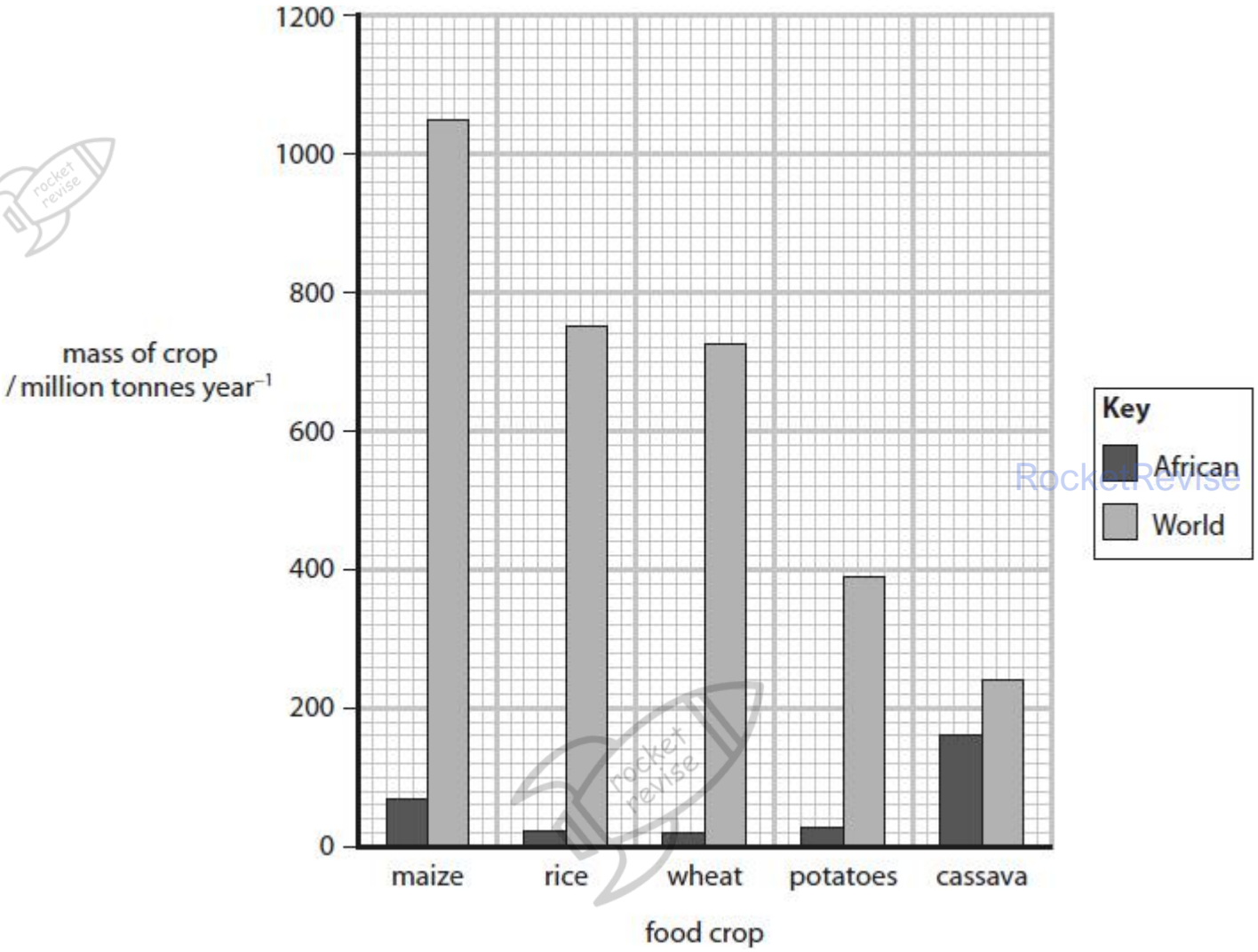
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(b) The graph shows the mass of five food crops grown in Africa. It also shows the world production of these crops.

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(i) Which is the percentage of the world cassava crop that is grown in Africa?

(1)

- ☒ **A** 33 %
- ☒ **B** 34 %
- ☒ **C** 66 %
- ☒ **D** 67 %

(ii) It is predicted that the world maize production will increase by 2.3 % in the following year.
Calculate the world production of maize in the following year.
Give your answer to the nearest whole number.

(1)

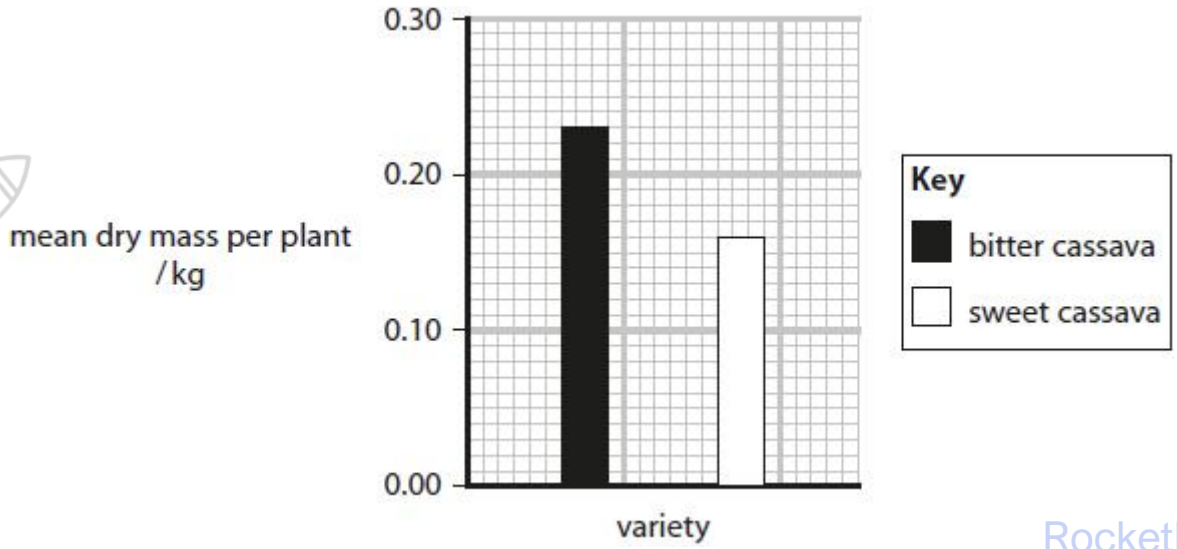
Answer ..... million tonnes

* (c) Some plants contain chemicals that protect them from being eaten by animals.

These chemicals taste bitter and can be converted by enzymes into hydrogen cyanide.
Hydrogen cyanide inhibits respiration and can cause death.
Cassava can be grouped into two main varieties: bitter cassava and sweet cassava. These varieties differ in the concentration of the bitter-tasting chemicals.
Processing of the cassava removes most of these chemicals.
The graph shows the mean dry mass of these cassava varieties when grown in the same conditions.

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The table gives information about three crop plants.

Crop plant	Water requirements	Minimum level of nutrients required in soil	Typical mass of bitter tasting chemicals / mg kg^{-1}	Typical mass of carbohydrate / 100 g of dry plant matter	Typical mass of protein / 100 g of dry plant matter
Bitter cassava	low to medium	low	100 to 500	94.70	1.80
Sweet cassava	low to medium	low	10 to 50	94.62	1.84
Maize	medium to high	medium to high	1 to 2	77.46	8.75

Discuss the advantages and disadvantages of growing these three crop plants for human food.
Use the information in the question to support your answer.

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(Total for question = 12 marks)

Q19.

The common wombat is endemic to Australia and Tasmania.

The photograph shows a common wombat.



(Source: © Albert Ziraj / Alamy Stock Photo)

magnification × 0.1

(a) Common wombats mark areas of their habitat to signal to other wombats of the same species that they are living there. These marked areas are called territories.

Common wombats produce cube-shaped faeces. It is thought that these cubes can be stacked to mark territory without rolling away.

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(i) Calculate the surface area : volume ratio of one cube with a length of 2.5 cm.

(2)

Answer .....

(ii) State what is meant by the term **habitat**.

(1)

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(iii) State what is meant by the term **species**.

(1)

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(b) The height of a wombat is a result of polygenic inheritance.

State what is meant by the term **polygenic inheritance** with reference to wombat height.

(1)

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(c) Common wombats dig burrows to live in. They have adaptations to help them to do this successfully.

Describe **two** anatomical adaptations of a common wombat that help it to dig burrows to live in.
Use the information in the photograph to support your answer.

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(Total for question = 7 marks)

Q20.

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Sehuencas water frogs are endemic to Bolivia.

It was once believed that this frog was extinct in the wild.

This was due to human activity and a disease caused by a fungus.

(a) Suggest how human activity could cause the extinction of the Sehuencas water frog in the wild.

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(b) In 2008, there was only one Sehuencas water frog in captivity.

In 2018, five Sehuencas water frogs were discovered in a Bolivian mountain forest. Three of these frogs were male.
Scientists suggested that these five frogs were resistant to the fungal disease.
Explain how resistance to this fungus could develop in a population of Sehuencas water frogs.

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(c) One way of studying the genetic diversity within a species is to calculate the heterozygosity index.

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Describe how the scientists could calculate the heterozygosity index for a population of Sehuencas water frogs.

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*(d) Describe the role of zoos in the conservation of endangered Sehuencas water frogs.

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(Total for question = 15 marks)

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Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The Hood Island giant tortoise (*Chelonoidis hoodensis*) is only found on one of the Galapagos Islands.

The photograph shows two Hood Island giant tortoises fighting over territory.



(Source: © Minden Pictures / Alamy Stock Photo)

(a) Which of the following is a term that would be used to describe the fact that this species of giant tortoise is found only on the Galapagos Islands?

(1)

- ☒ **A** diversity
- ☒ **B** endemic
- ☒ **C** isolation
- ☒ **D** linkage

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(b) The Hood Island giant tortoise has adaptations to its environment.

Two features are:

- long neck
- shell that arches above the neck.

When there is a dispute over territory, the tortoises use these features to make themselves look as large as possible.

(i) Complete the table to show the type of adaptations shown by this tortoise.

(3)

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Feature	Type of adaptation
long neck	
shell that arches above neck	
making themselves look as large as possible	

(ii) Suggest one selection pressure that results in the development of one of these features.

(1)

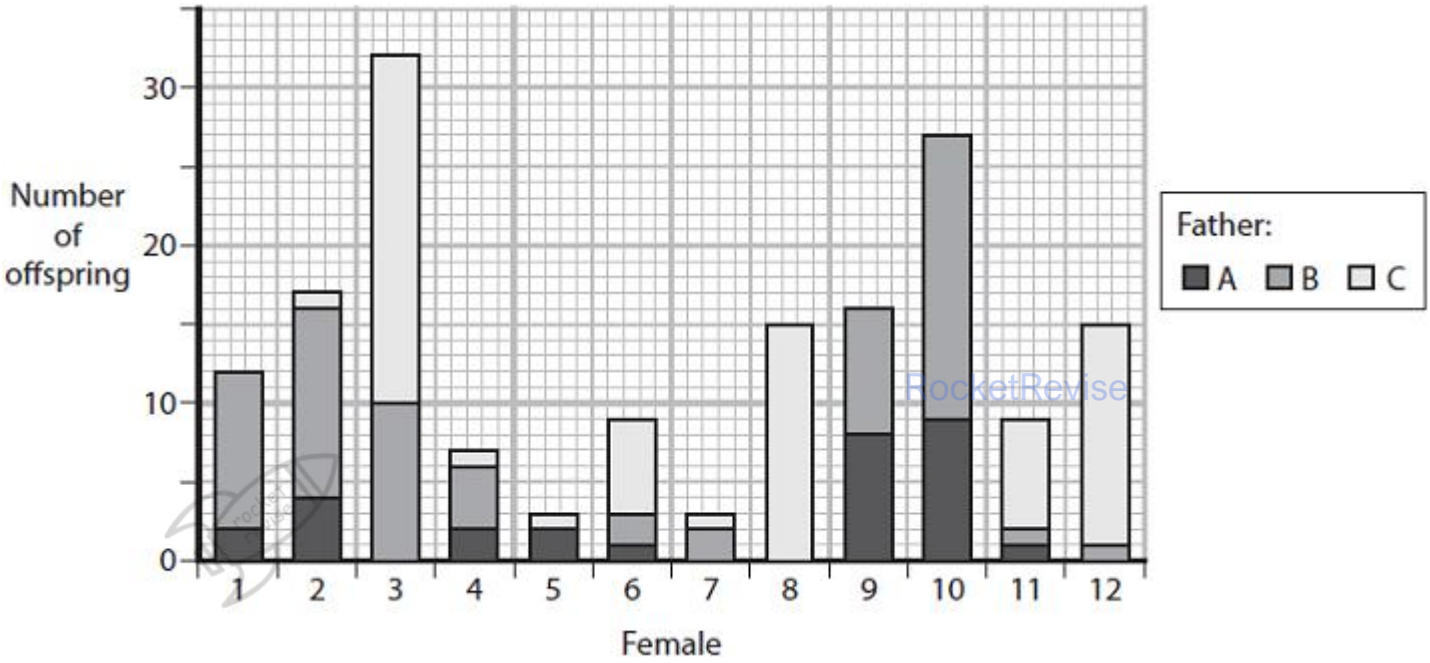
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(c) In 1963, there were only three males and 12 females of *C. hoodensis* left in the wild.

These 15 tortoises were used in a breeding programme.
The diagram shows the number of offspring from each female tortoise and the fathers of these offspring.



(i) Calculate the percentage of female 10's offspring that were fathered by male B.

(1)

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Answer ..... %

(ii) The wild population of C. hoodensis is now 1800 individuals.
Calculate the percentage increase in the wild population of *C. hoodensis tortoises*.
Give your answer to **one** decimal place.

(2)

Answer ..... %

(iii) A zoo tested its captive-bred giant tortoise and determined that it was a *C. hoodensis tortoise*.
Explain how the zoo determined that its giant tortoise was a *C. hoodensis tortoise*.

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(d) Scientists thought that a recessive allele was changing in frequency in the tortoise population.
Explain how the change in frequency of this allele could be determined.

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(Total for question = 13 marks)

Q22.

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Elk are one of the largest species of deer. The photograph shows three elk grazing in Yellowstone National Park.

They prefer to feed on young tree shoots, such as aspen and cottonwoods, rather than grass.



(Source: ©Westend61 GmbH/Alamy Stock Photo)

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(a) Wolves were introduced into Yellowstone National Park in 1995.

Wolves are predators of elk.

Before the introduction of wolves, elk grazed undisturbed, eating all the young tree shoots in the forest and riverside habitats.

The table shows how the percentage of young shoots eaten by the elk changed after the introduction of wolves.

Year	Percentage of young tree shoots eaten by elk (%)	
	Forest habitat	Riverside habitat
1996	100	100
1998	100	100
2000	100	92
2002	90	81
2004	86	45
2006	63	11
2010	19	2

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Explain why the percentage of young tree shoots eaten by the elk changed after the introduction of

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wolves.

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(b) The photograph shows a beaver.

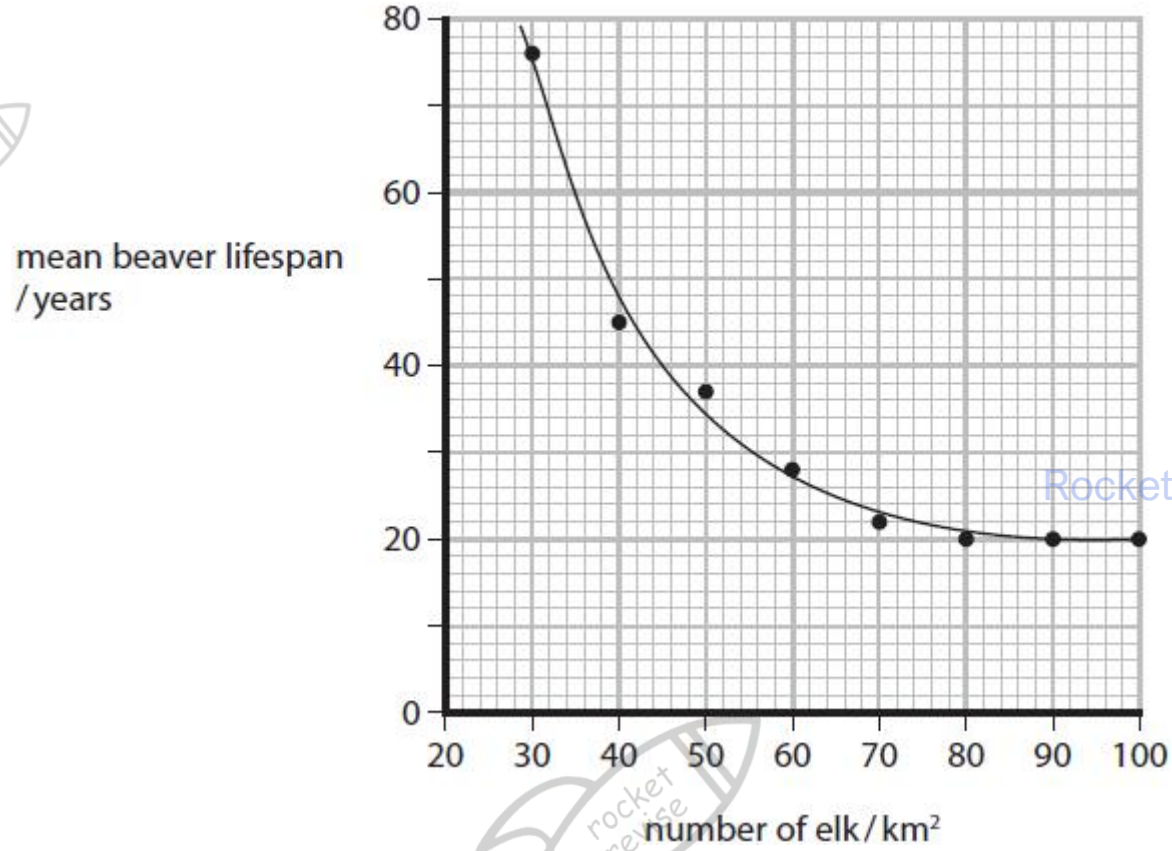


(Source: © Ian MacNicol/Getty Images)

The graph shows the mean beaver lifespan in areas with different numbers of elk per km^2.

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Describe the relationship shown in the graph.

(3)

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(c) Scientists investigated the effect of the introduction of wolves on the biodiversity of Yellowstone National Park.

(i) Suggest what information the scientists would need to collect in order to calculate species richness in a habitat in Yellowstone National Park.

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(ii) Write a formula that could be used to determine the biodiversity of a habitat in Yellowstone National Park.

(1)

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* (d) Beavers and elk eat the shoots of young trees.

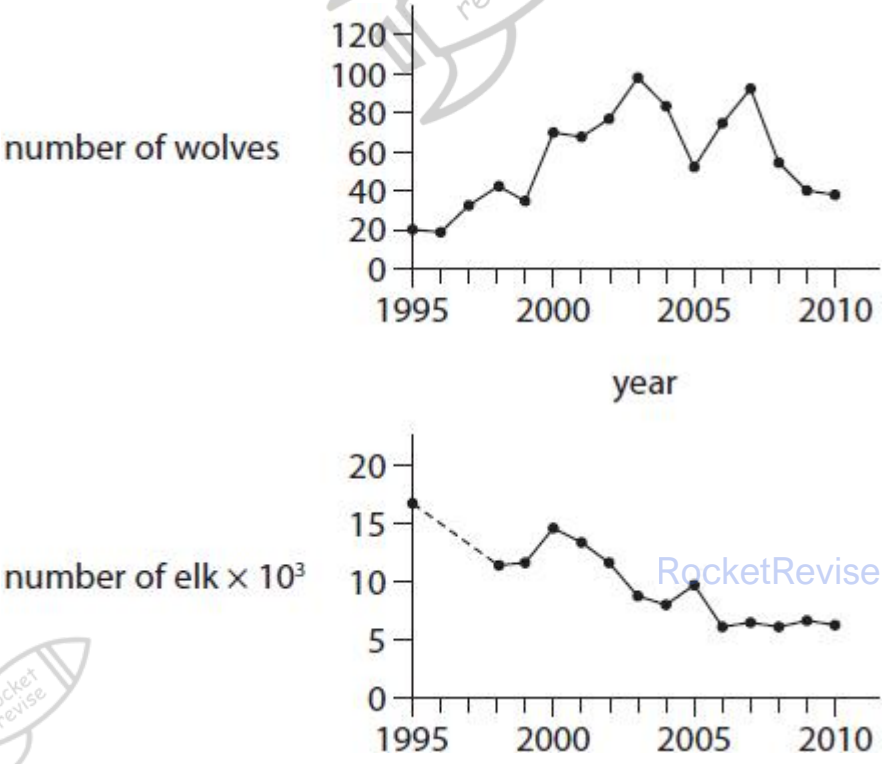
Beavers use older trees to build dams in the rivers, which create new habitats, such as ponds.

Beavers live in colonies of up to 8 individuals.

After the introduction of the wolves, scientists recorded the number of wolves, elk and beaver colonies for 15 years.

They also recorded the number of young cottonwood shoots that were taller than 5 cm.

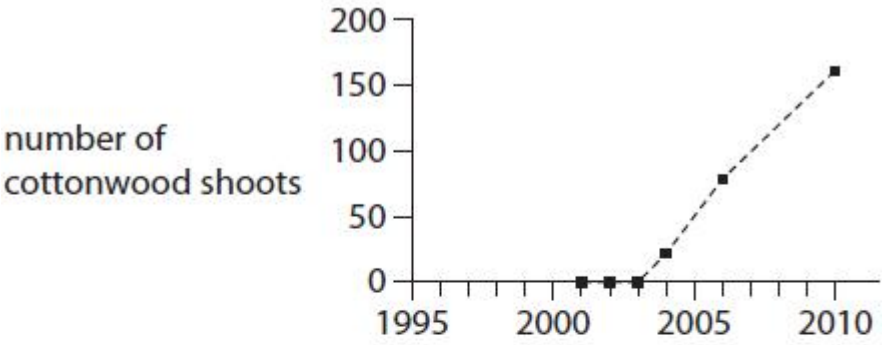
The graphs show the data collected by these scientists.

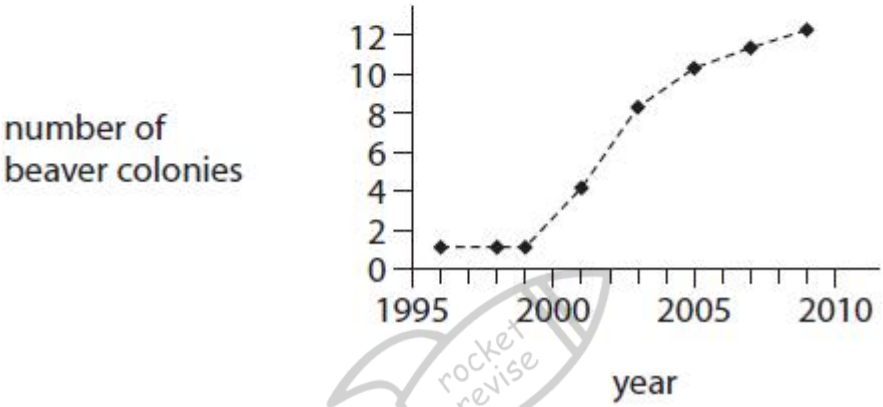


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Discuss the effect of the introduction of wolves on the biodiversity of Yellowstone National Park.

Use all the information in the question to support your answer.

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(Total for question = 15 marks)

Q23.

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There are 11 species of muntjac in Asia.

One of these is the Indian muntjac.

The photograph shows an Indian muntjac.



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(Source: © blickwinkel / Alamy Stock Photo)

(a) The female Indian muntjac body cells have three pairs of chromosomes.

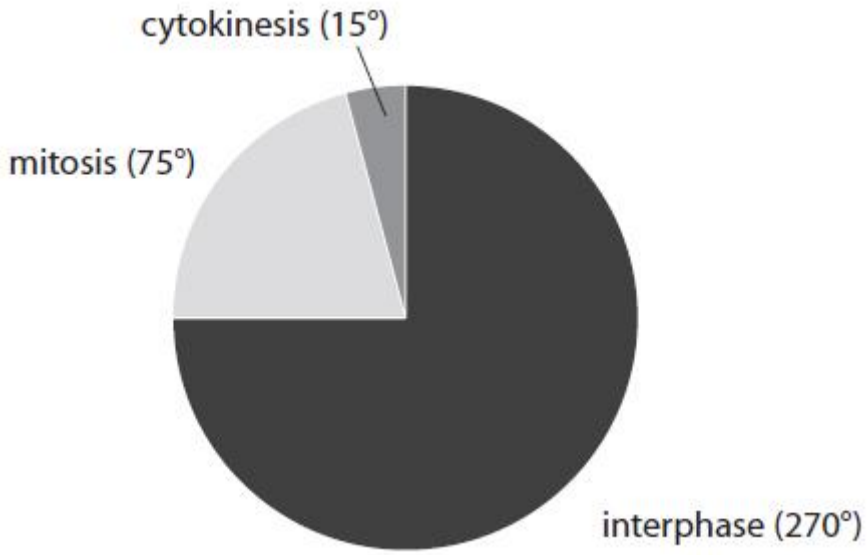
(i) Dividing cells can be taken from the body of a female Indian muntjac.
Draw one of these body cells showing the arrangement of chromosomes in the anaphase stage of mitosis.

(2)

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(ii) The diagram shows the relative proportions of time that a cell spends in each part of the cell cycle.



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The cell was in interphase for nine hours.
The cell was in anaphase for 20 minutes.
Calculate the angle that would represent anaphase plotted on this pie chart.

(2)

Answer ..... °

(b) The diagram shows one of the chromosomes in an Indian muntjac cell.



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(i) State what is meant by the term **locus**.

(1)

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(ii) Explain how the chromosomes of an Indian muntjac egg cell could differ from those in a body cell.

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(c) Chinese muntjac cells contain 23 pairs of chromosomes.

The photograph shows a Chinese muntjac.



(Source: © Design Pics Inc / Alamy Stock Photo)

The Chinese muntjac looks similar to the Indian muntjac. However, if they breed together, they produce offspring that are infertile.
Suggest why the offspring would be infertile.

(2)

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(Total for question = 10 marks)

Q24.

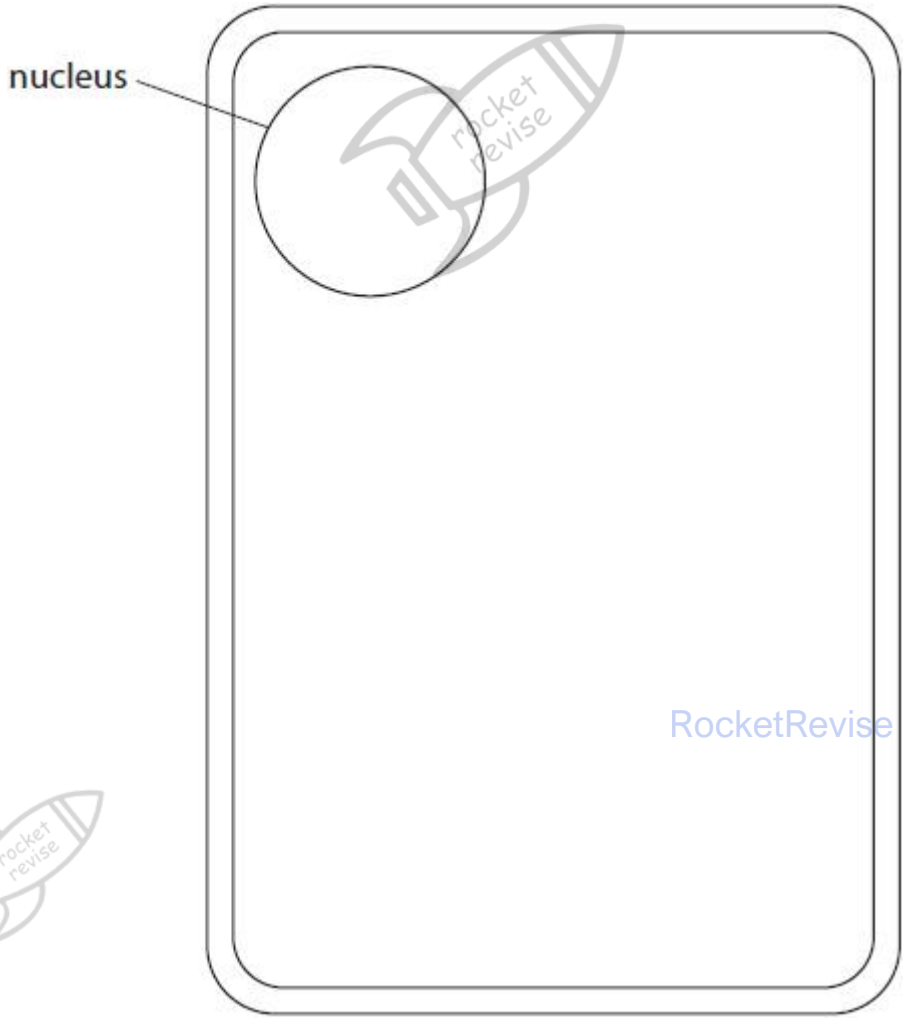
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Woese classified organisms into a three-domain system.

Plant cells were classified into the Eukarya domain.

The diagram shows an incomplete cell from a plant.



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(a) Complete the diagram by drawing an amyloplast, a chloroplast and the tonoplast to show their location and relative size in this cell.

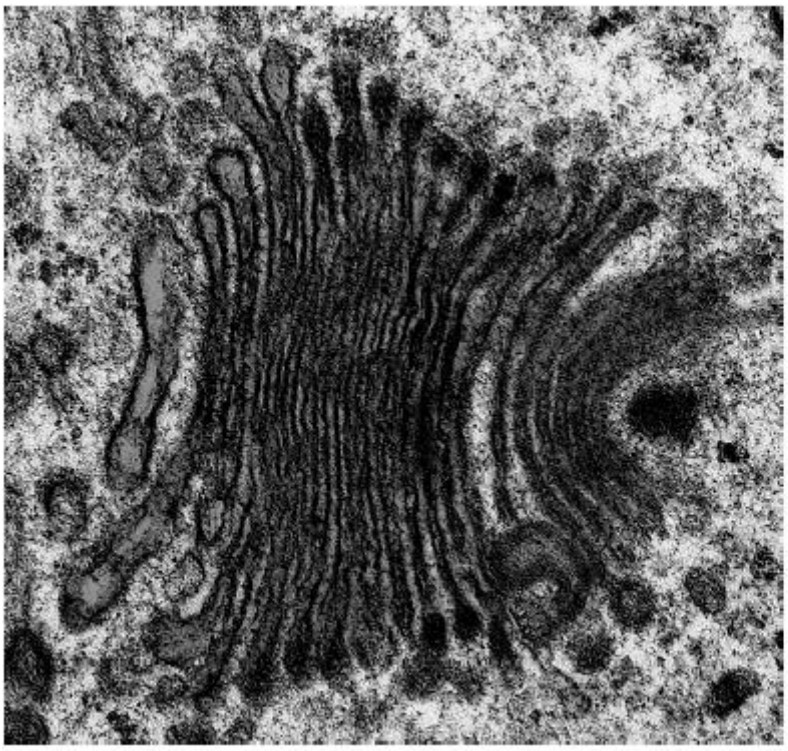
Label these three structures.

(3)

(b) The photograph shows an organelle found in a plant cell, as seen using an electron microscope.

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(Source: © SCIENCE PHOTO LIBRARY)

State the function of this organelle.

(1)

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(c) The other domains in the three-domain system are Archaea and Bacteria. Complete the table to show in which of the domains the following structures would be found.

(4)

Structure	Archaea only	Bacteria only	Eukarya only	More than one domain
cell membrane	☐	☐	☐	☐
nucleolus	☐	☐	☐	☐
cell wall	☐	☐	☐	☐
70S (small) ribosome	☐	☐	☐	☐

(Total for question = 8 marks)

Q25.

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Answer the questions with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Woese classified organisms into a system with three domains.

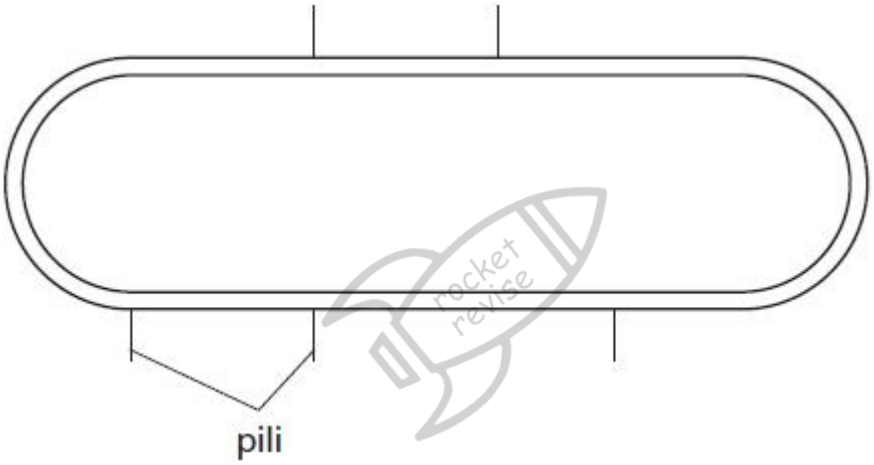
(a) How many of the domains contain prokaryotic organisms?

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(b) The diagram shows part of a prokaryotic cell, as drawn by a student.

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(i) Complete the diagram by drawing and labelling a flagellum and a plasmid.

(2)

(ii) State the function of a plasmid.

(1)

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(c) (i) The ribosomes in prokaryotic cells are different from the ribosomes in eukaryotic cells.
Give **one** difference between prokaryotic ribosomes and eukaryotic ribosomes.

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(1)

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(ii) How many of the following statements about eukaryotic ribosomes are correct?

- ribosomes are located on the surface of both types of endoplasmic reticulum
- ribosomes are located inside chloroplasts and mitochondria
- ribosomes are involved in the process of transcription

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(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(Total for question = 6 marks)

Q26.

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The Hawaiian islands have 18 species of honeycreeper birds.

The Hawaii amakihi honeycreeper and the Hawaii palila honeycreeper are two species found on these islands.

(a) The table gives information about these two species of honeycreeper.

Information	Hawaii amakihi honeycreeper	Hawaii palila honeycreeper
Photograph	 (Source: © NPS Photo/Alamy Stock Photo)	 (Source: © Minden Pictures/Alamy Stock Photo)
Habitat	lowland and mountainous forest and shrubland	mountainous forest
Number of Hawaiian islands with a population	3	1
Estimated population size	<10000	<1000
Food source	nectar, tree sap, spiders and insects	seeds and berries

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(i) The Hawaii palila honeycreeper is found only in the mountainous forest on one island.
Which of the following is the term used to describe this distribution?

(1)

- ☒ **A** diversity
- ☒ **B** endemic
- ☒ **C** polygenic
- ☒ **D** specific

(ii) In 2003, it was estimated that there were 7100 Hawaii palila honeycreeper birds on this island. In 2020, it was estimated that there were 950 of these birds.

Calculate the percentage change in the estimated populations of these birds.

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Give your answer to **two** significant figures.

(2)

Answer .....%

* (b) It is suggested that both of these Hawaiian honeycreeper species evolved from the same common ancestor.

Explain how these two species of honeycreeper could have evolved from the same common ancestor.

Use the information in the table, and your own knowledge, to support your answer.

(6)

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(c) Explain how scientists could determine which of the 17 other species of honeycreeper on these islands is the most closely related to the Hawaii palila honeycreeper species.

(2)

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(Total for question = 11 marks)

Q27.

Great lakes and crater lakes can be found in Nicaragua, a country in Central America. These two types of lake differ in their species richness.

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(a) State what is meant by the term **species richness**.

(1)

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* (b) Lake Nicaragua and Lake Managua are great lakes.

Crater lakes were formed from the craters of volcanoes.
The photographs show a great lake and a crater lake in Nicaragua.



Great lake

(Source: © mauritius images GmbH/Alamy Stock Photo)



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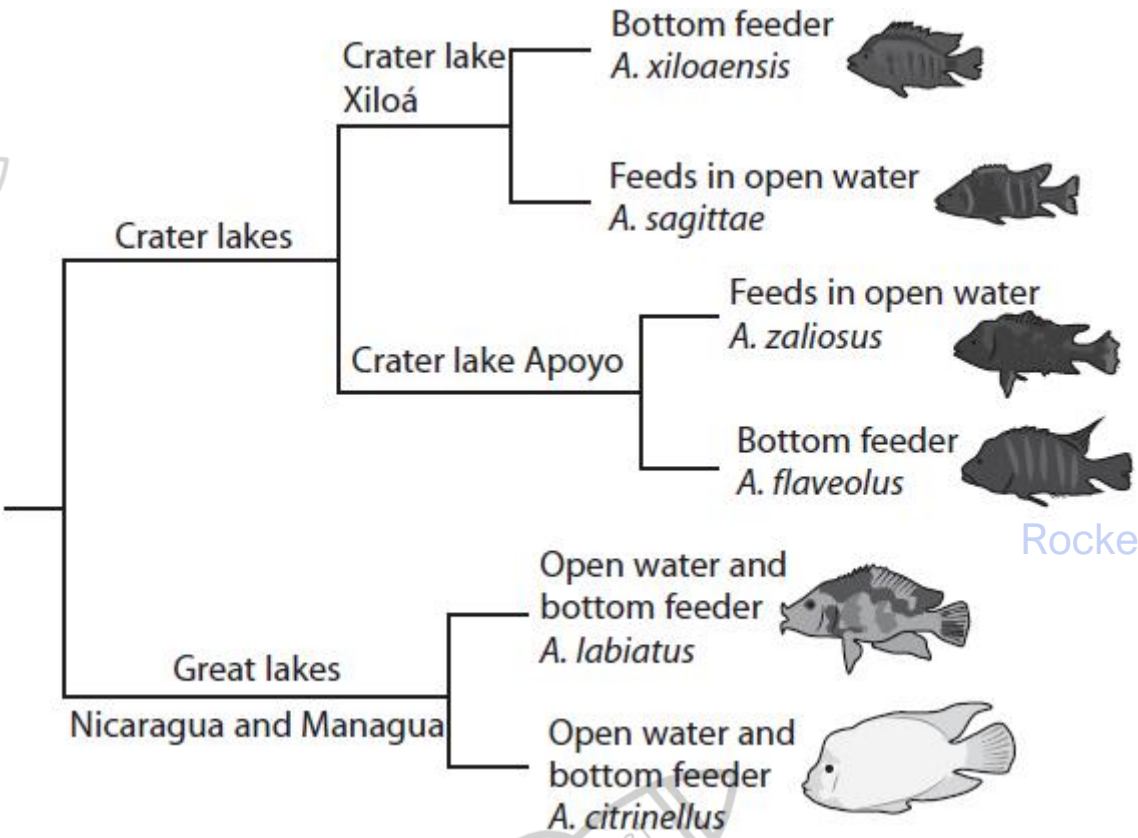
Crater lake

(Source: © John Mitchell/Alamy Stock Photo)

The diagram shows a proposed relationship for some species of cichlid fish and the location of the lakes they inhabit.

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The table shows the estimates of when cichlid fish species first inhabited the crater lakes and when speciation occurred.

Crater lake	First inhabited / number of years ago	First speciation occurred / number of years ago
Apoyo	4700	3700
Xiloá	4300	2700

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Describe how new cichlid species may have been formed in these four lakes and how the scientists would have determined that these cichlids were six different species.
Use the information in the question and your own knowledge to support your answer.

(6)

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(c) A study compared the diversity of cichlid species in two crater lakes.

The index of diversity (D) for Lake Xiloá was 2.8.
The table shows the data obtained from Lake Apoyo.

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Species	Number of individuals (n)	n(n – 1)
A. astorquii	156	24 180
A. chancho	45	1 980
A. flaveolus	78	6 006
A. globosus	8	56
A. supercilius	17	272
A. zaliosus	12	
	N = 316	$\sum n(n-1) =$

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An index of diversity (D) is calculated using the formula:

$$D = \frac{N(N-1)}{\sum n(n-1)}$$

Determine which of the lakes has the higher biodiversity of cichlids.
Use the table and the formula to help you.

(4)

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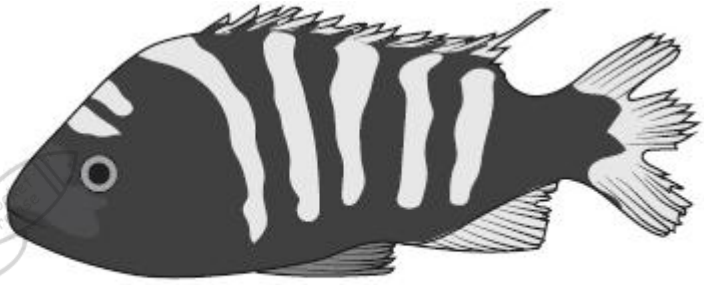
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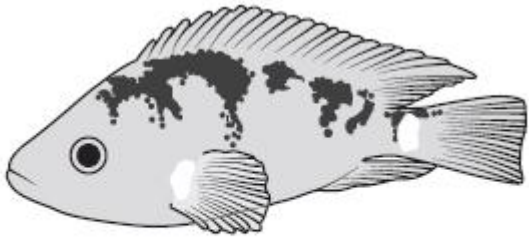
(d) The drawing shows two *A. citrinellus* cichlids.

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melanic phenotype –
black with white stripes



gold phenotype –
orange with black mottling

The gold phenotype is caused by inheriting a dominant allele and the melanic phenotype is caused by two recessive alleles at the same locus.
A scientist studied the change in allele frequencies in the *A. citrinellus* population over time.
In the first study, the allele frequencies in the population were p = 0.6 and q = 0.4.
In a later study, 672 fish of this species were gold and 128 were melanic.
Determine if the allele frequencies have changed, using the equations:

$$p^2 + 2pq + q^2 = 1$$

$$p + q = 1$$

(3)

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(Total for question = 14 marks)

Q28.

The photograph shows a titan arum plant (*Amorphophallus titanum*).

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(Source: © Khairil Azhar Junos / Alamy Stock Photo)

This plant is native to Sumatra, Indonesia, and is critically endangered.

There are fewer than 1 000 individuals in the wild and 500 in botanical gardens around the world.

(a) Sumatra is described as having a high species richness.

State what is meant by the term **species richness**.

(1)

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(b) Seed banks often conserve endangered plants by storing samples of their seeds.

The seeds are dried and then frozen.
Explain the advantages of drying seeds before storage.

(2)

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(c) Scientists have stated that more than a third of critically endangered plant species cannot be saved from extinction by storing seeds in seed banks.

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The titan arum is one of these plants.
This plant flowers once every 10 years and the flowers last for two to three days.
The plant can reproduce asexually or sexually. In order to reproduce sexually it relies on insects to transfer pollen to other titan arum plants.
Scientists are developing ways to conserve this plant without losing genetic diversity.
Some of the suggestions include:

- grow more plants produced by asexual reproduction in botanical gardens
- collect pollen when an individual plant flowers and store it in a seed bank
- create a studbook for the species
- artificially pollinate plants in the wild and in botanical gardens.

(i) The heterozygosity index can be calculated using an equation.
Write this equation.

(1)

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(ii) The heterozygosity index of the 500 titan arum plants in botanical gardens was found to be 0.166.
Calculate the number of heterozygotes in this population.

(1)

Answer .....

*(iii) Discuss the suggestions, proposed by these scientists, for conserving the titan arum.

(6)

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(Total for question = 11 marks)

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Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A mimic octopus can change shape to look like other organisms.

The photographs show a venomous sea snake, a mimic octopus looking like a venomous sea snake and a mimic octopus looking like a flat fish.

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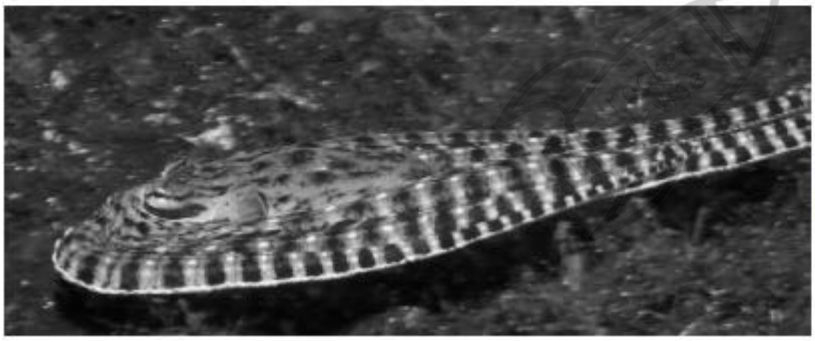
venomous sea snake

(Source: Reinhard Dirscherl / Alamy Stock Photo)



mimic octopus looking like a venomous sea snake

(Source: David Fleetham / Alamy Stock Photo)



mimic octopus looking like a flat fish

(Source: Hal Beral / VWPics / Alamy Stock Photo)

(a) This octopus is adapted to its niche.

(i) State what is meant by the term **niche**.

(1)

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(ii) The mimic octopus has the same banding pattern as the venomous sea snake. Name this type of adaptation.

(1)

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(iii) Suggest why the mimic octopus would change its appearance to look like a flat fish and a venomous sea snake.

(2)

Flat fish .....

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Venomous sea snake .....

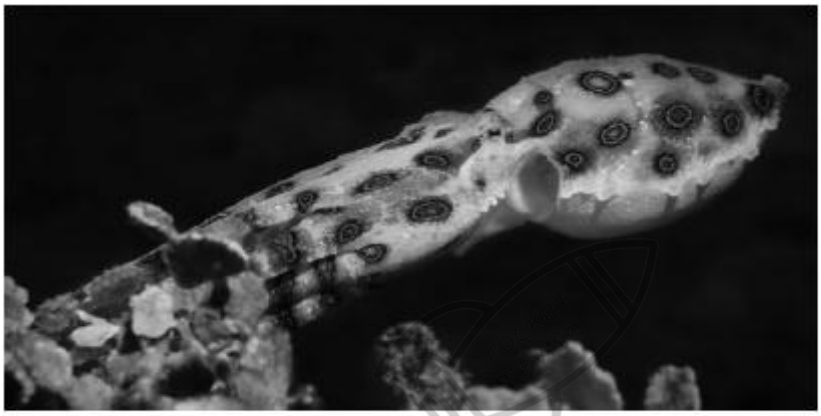
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(b) The photograph shows a blue-ringed octopus.

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(Source: Oksana Golubeva / Alamy Stock Photo)

This octopus produces a toxin called tetrodoxin. This toxin affects neurones by blocking sodium channels.

The pain–relieving properties of this toxin were investigated in a drug trial.

Stage II of this drug trial compared the tetrodoxin with the current pain–relieving drug in a group of cancer patients.

(i) Which of the following describes the purpose of the current pain–relieving drug in stage II of the drug trial?

(1)

- ☒ **A** control
- ☒ **B** digitalis
- ☒ **C** placebo
- ☒ **D** solvent

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(ii) Describe the processes that would have occurred before and after stage II in this tetrodoxin drug trial.

(4)

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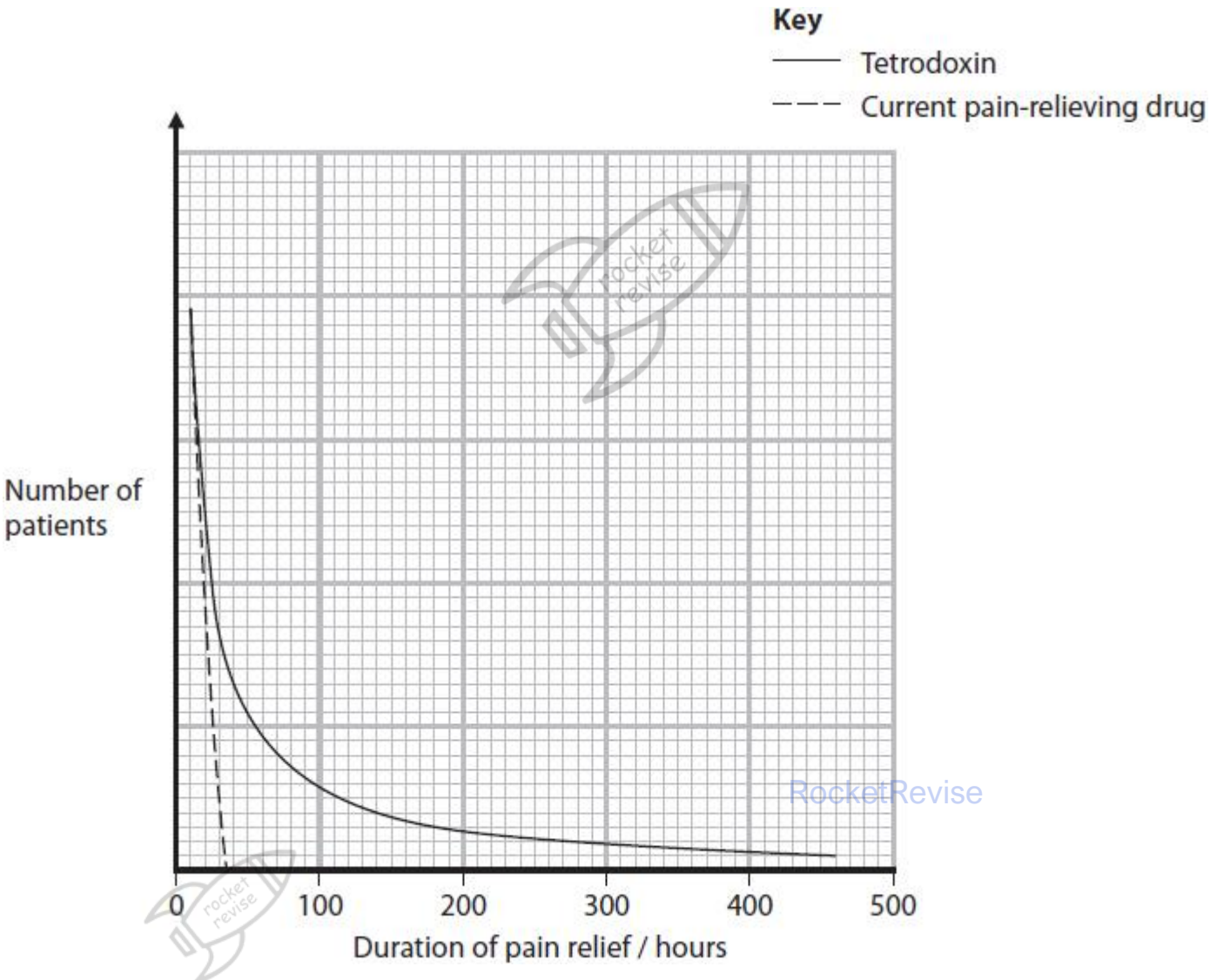
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(c) The graph shows the duration of pain relief using both tetrodoxin and the current pain–relieving drug in stage II of this drug trial.

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Comment on the results of this investigation.

(4)

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(Total for question = 13 marks)

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows different varieties of eggplant fruits.



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(Source: © Leon Werdinger/Alamy Stock Photo)

The table shows the mass of eggplants imported by four countries in 2020.

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Country	Mass imported in 2020/ tonnes
Austria	1911
France	52892
Italy	25150
Slovenia	48

(a) It is expected that the import of eggplants by France would increase by 1.6% per year.

Calculate the expected mass of eggplants imported by France in 2023.
Give your answer to the nearest whole number.

(2)

Answer ..... tonnes

(b) *R. solanacearum* bacteria infect some varieties of eggplant.

These bacteria reproduce in the xylem vessels and block them. This causes the disease bacterial wilt, which results in the death of the plant.
The extent of bacterial disease can be recorded as the disease index. The greater the extent of disease, the larger the value of the disease index.
The diagram shows a healthy plant and a wilted plant.



healthy plant



wilted plant

Scientists at a seed bank investigated the wilt percentage and disease index of some eggplant varieties.

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Varieties	Wilt percentage (%)	Disease Index (DI)	Crop yield
A	95.8	45.8	high
B	66.7	61.3	medium
C	100.0	89.2	medium
D	33.3	9.2	low

The resistance of the plant variety was categorised based on its mean DI value as shown in the table.

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Resistance category	DI value
Resistant	0–30
Moderately resistant	>30–40
Moderately susceptible	>40–50
Susceptible	>50

(i) How many of the varieties tested were categorised as susceptible to the *R. solanacearum* bacteria?

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(ii) Seed banks contain the seeds of closely related wild plants.
Explain how seed banks would prepare, store and assess viability of these seeds.

(4)

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(iii) Explain how scientists could breed new high-yielding varieties of eggplants which are resistant to *R. solanacearum* bacteria.
Use the information in the tables to support your answer.

(4)

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(Total for question = 11 marks)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

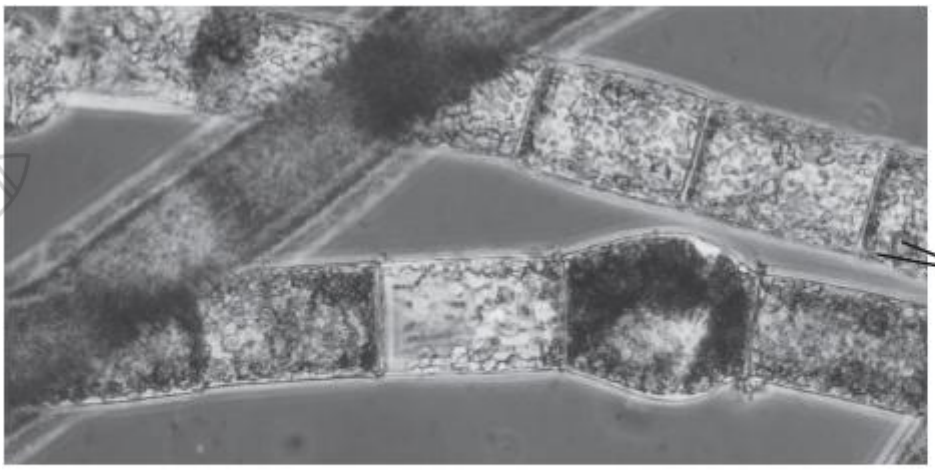
The three-domain system is used to classify organisms.

Photograph A shows some cells in one type of algae, as seen using one type of microscope.

The cells of this organism contain chloroplasts.

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(Source: © Steve Taylor ARPS/Alamy Stock Photo)

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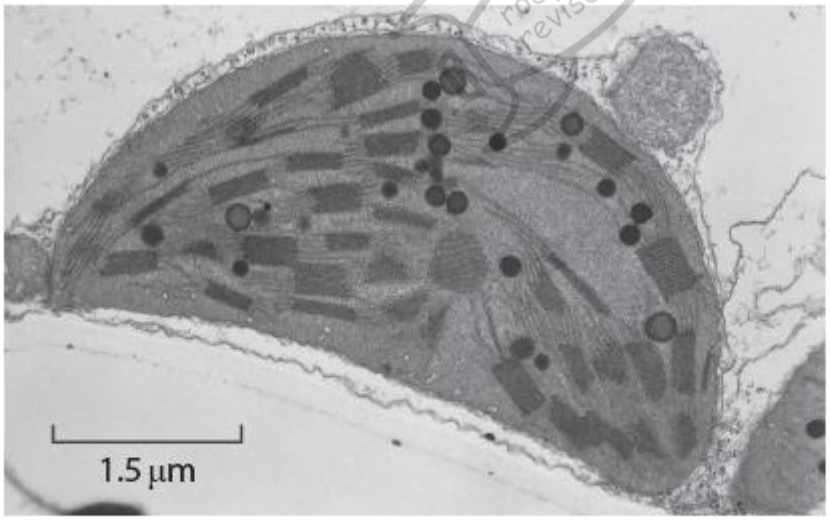
Photograph A

(a) Name the domain which would contain this organism.

(1)

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(b) Photograph B shows a chloroplast as seen using a different type of microscope.



(Source: © Science History Images/Alamy Stock Photo)

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Photograph B

(i) Calculate the magnification of the chloroplast shown in photograph B.
Give your answer to **two** significant figures.

(2)

Answer .....

(ii) Explain why the microscope used for photograph B shows more detail than the microscope used for photograph A.

(2)

(iii) Chloroplasts contain starch grains.
Which of the following is another structure that contains starch?

(1)

- ☒ **A** amyloplast
- ☒ **B** plasmodesmata

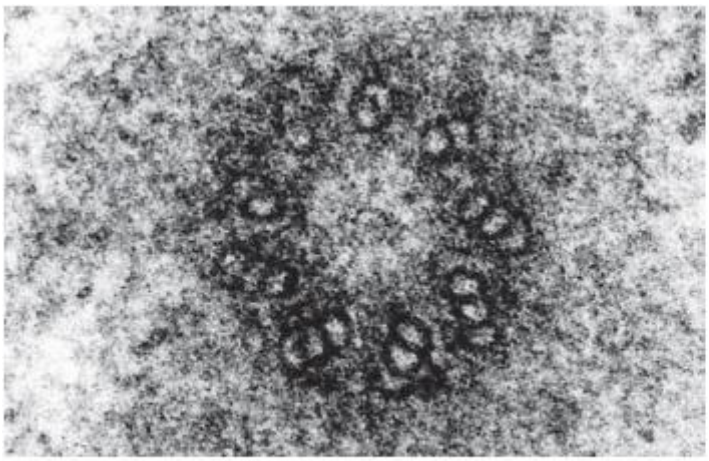
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☒ **C** tonoplast
☒ **D** vacuole

(c) The cells from algae contain some structures that are characteristic of animal cells.

The photograph shows one of these structures.



(Source: © DON W. FAWCETT/SCIENCE PHOTO LIBRARY)

The structure has the same function in the algal cell as in an animal cell.
Give the name and function of this structure.

(2)

Name

.....

Function

.....

.....

.....

(Total for question = 8 marks)

Q32.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a humphead cichlid fish found only in Lake Tanganyika in Africa.

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(Source: © Martin Lindsay/Alamy Stock Photo)

A behavioural adaptation shown by the female humphead cichlid is that fertilised egg cells are kept in her mouth until they hatch.

(a) (i) Which of the following is a term used to describe the fact that this fish is found only in Lake Tanganyika?

(1)

- ☒ **A** diversity
- ☒ **B** endemic
- ☒ **C** homozygous
- ☒ **D** specific

(ii) Suggest why the female humphead cichlids have this behavioural adaptation.
Give a reason for your answer.

(2)

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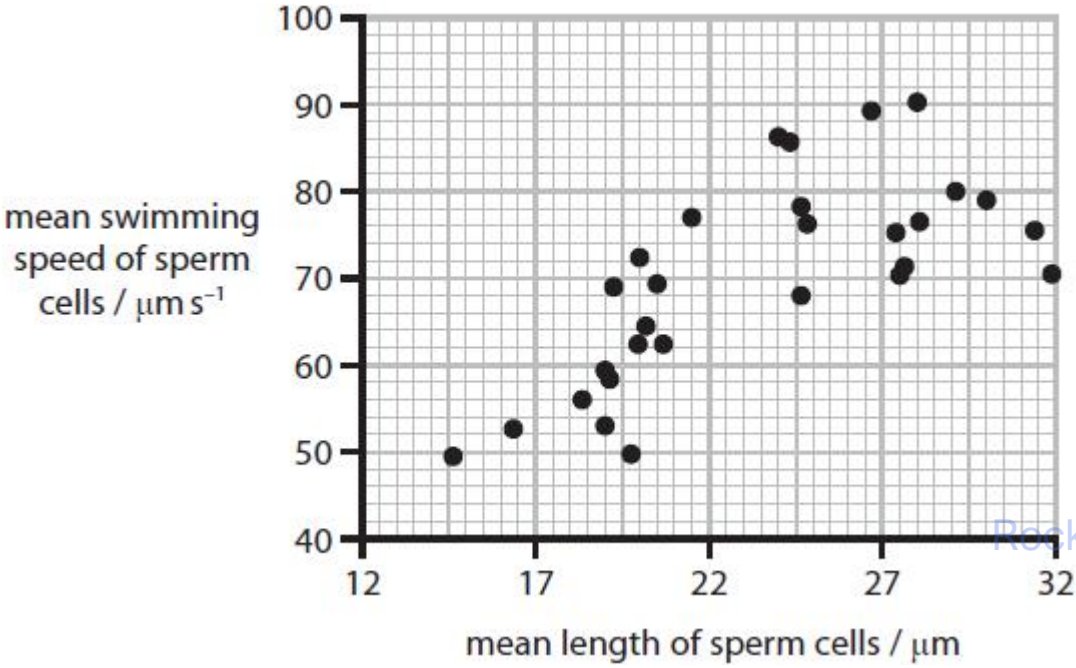
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(b) The relationship between the length and swimming speed of the sperm cells of 30 cichlid species was investigated.

The graph shows the results of this investigation.

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(i) State the relationship between the mean length and swimming speed of the sperm cells of these cichlid species.

(1)

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(ii) A scientist wanted to assess the strength of the relationship between these two variables. Which statistical test should be used to assess the strength of the relationship between these two variables?

(1)

- ☒ **A** chi-squared test
- ☒ **B** correlation coefficient
- ☒ **C** index of diversity
- ☒ **D** Student's t-test

(iii) Suggest **one** advantage for a male cichlid of producing sperm cells with a faster swimming speed.

(1)

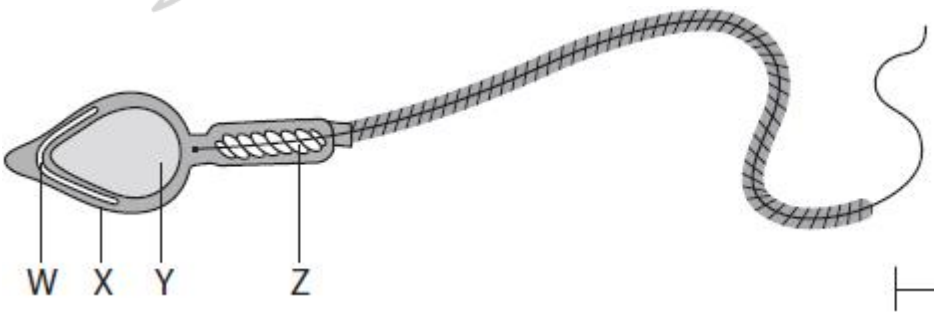
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(c) A sperm cell contains an acrosome, a specialised lysosome.

The diagram shows a sperm cell.



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(i) Which labelled structure is an acrosome?

(1)

- ☒ **A** W
- ☒ **B** X
- ☒ **C** Y
- ☒ **D** Z

(ii) Calculate the magnification of the sperm cell shown in the diagram.
Give your answer to **two** significant figures.

(2)

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Answer .....

(iii) Compare and contrast the structure and function of lysosomes and acrosomes.

(3)

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(Total for question = 12 marks)

Q33.

The photograph shows mangroves in the Caribbean.

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(Source: Stefania Barbier / Alamy Stock Photo)

A new type of bacteria was discovered living in these mangroves.

Scientists called this new type of bacteria *Thiomargarita magnifica*.

One cell of these bacteria is 20 mm in length.

The smallest known type of bacteria, *Mycoplasma genitalium*, is 300 nm in length.

(a) (i) Both of these are found in the domain Bacteria.

Give the names of the other **two** domains.

(1)

1 .....

2 .....

(ii) Calculate how many times larger *T. magnifica* is than *M. genitalium*.
Give your answer in standard form.

(2)

Answer .....

(b) The cytoplasm of *T. magnifica* contains structures called pepins.

The function of pepins is protein synthesis.
Pepins contain circular DNA and ribosomes and are surrounded by a membrane.
Explain how these **three** structures contribute to the function of pepins.

(3)

DNA .....

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Ribosomes .....

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Membrane .....

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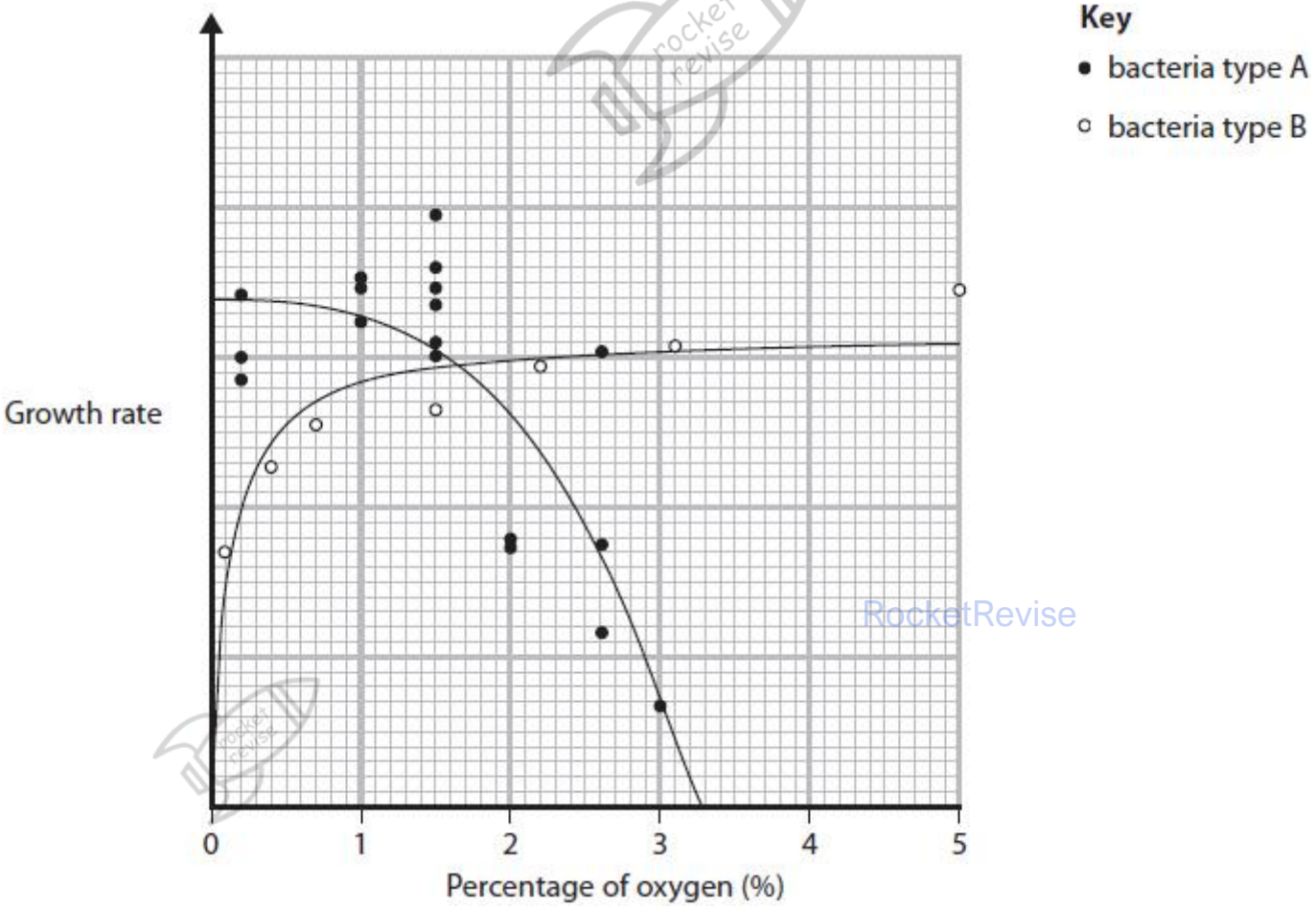
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(c) Some bacteria require oxygen in their surroundings whereas other bacteria do not.

The graph shows the growth rate of two types of bacteria, A and B, which were grown in different concentrations of oxygen.
The bacteria were supplied with water and grown at their optimum temperatures.



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(i) Explain why an optimum temperature and water are needed for bacterial growth.

(2)

Optimum temperature .....

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Water .....

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(ii) Explain the results for these two types of bacteria shown in the graph.

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(Total for question = 11 marks)

Q34.

Snow leopards (*Panthera uncia*) live in mountainous regions in Asia.

The photograph shows a snow leopard on a snow-covered mountain.

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(Source: © Natural Visions/Alamy Stock Photo)

(a) Give **one** reason for a named anatomical adaptation of this snow leopard that enables it to occupy its niche.

Use the information in the photograph to support your answer.

(1)

Adaptation

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Reason

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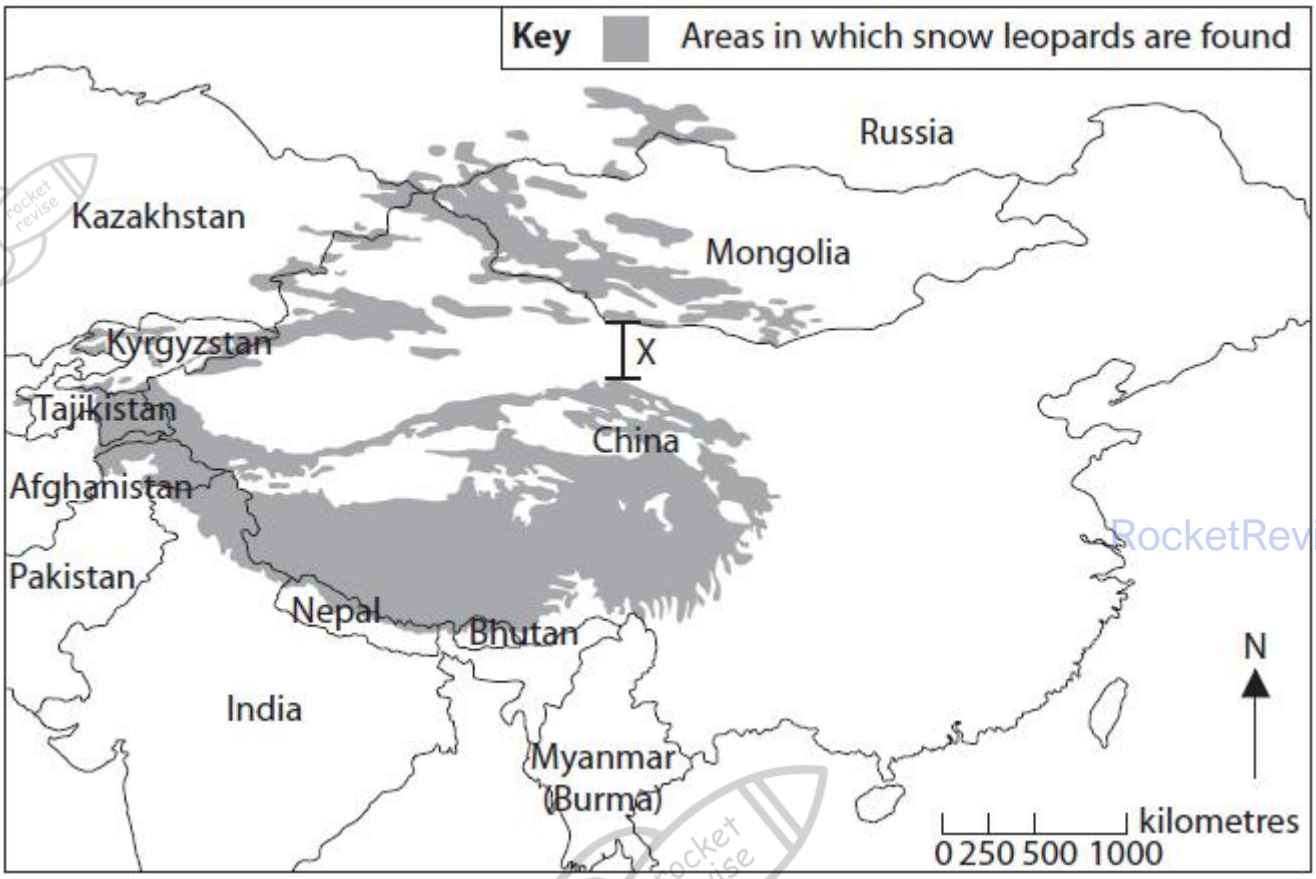
(b) Snow leopards are found in several areas in Asia.

The map shows the areas where snow leopards can be found.

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Describe how the biodiversity of habitats in these areas could be compared.

(3)

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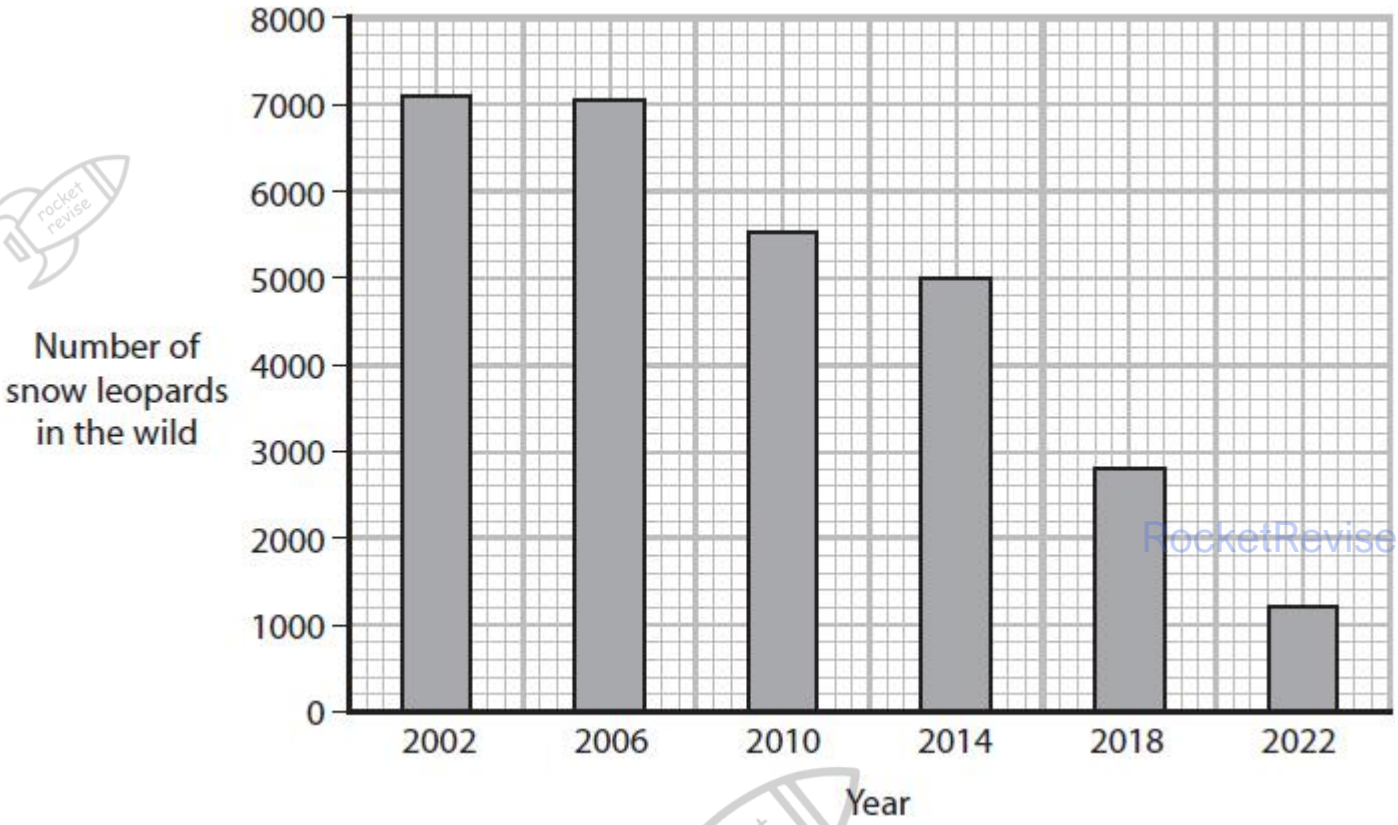
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(c) The graph shows the numbers of snow leopards in the wild since 2002.

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(i) Calculate the percentage decrease in snow leopards from 2002 to 2022.
Give your answer to one decimal place.

(2)

.....%

(ii) Suggest **one** reason for this decrease in the number of snow leopards.

(1)

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(d) Populations of snow leopards, living in different areas of Asia, sometimes develop different characteristics. These populations are called subspecies.

Individuals from different subspecies are able to produce fertile offspring.
Some scientists have proposed that there are three subspecies of snow leopard.
The table shows the names and locations of the snow leopard range for the three subspecies.

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Subspecies	Location of snow leopard range
P. uncia irbis	Northern
P. uncia uncia	Western
P. uncia uncioides	Central

(i) The northern and central locations are separated by a desert region.
The width of the region is shown by the line labelled X on the map.
Calculate the width of this desert region using the scale shown on the map.
Give your answer in standard form.

(2)

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*(ii) Explain how these subspecies of snow leopard could form.
Use the information in the question to support your answer.

(6)

(iii) Give a reason as to why these different populations are described as subspecies and not different species of snow leopard.

(1)

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(Total for question = 16 marks)

Q35.

There are 18 species of puffer fish found in the Maldives.

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The photograph shows one of these species, *Canthigaster valenti*.



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© kaschibo/Shutterstock

Magnification ×0.5

(a) The markings on the skin of *Canthigaster valenti* are warnings to predators. It also protects itself from predators by producing poisons and by inflating its body.

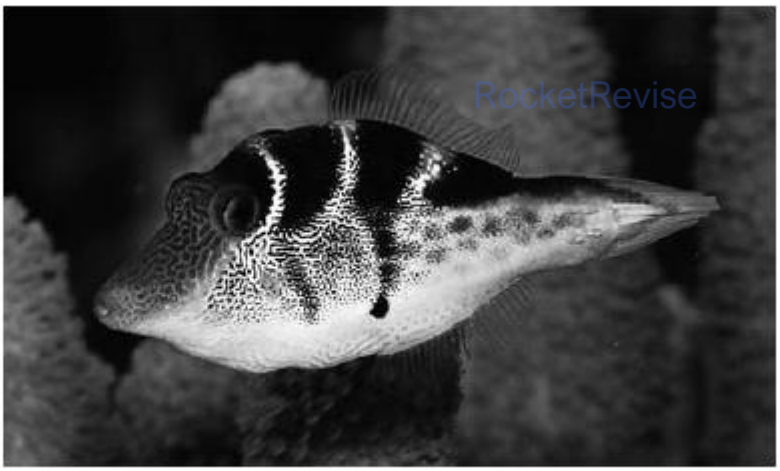
Which row of the table describes these types of adaptations?

(1)

	Markings on the skin	Production of poison	Inflating the body
☐ **A**	anatomical	behavioural	physiological
☐ **B**	anatomical	physiological	behavioural
☐ **C**	physiological	anatomical	behavioural
☐ **D**	physiological	behavioural	anatomical

(b) Another fish found in the Maldives is *Paraluteres prionurus*.

This fish is not poisonous. It grows to about 10 cm in length.
The photograph shows *Paraluteres prionurus*.



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Source: http://www.underwaterkwaj.com/uw-misc/file/Paraluteres-prionurus.htm

Explain how the appearance of *Paraluteres prionurus* shows it is adapted to its habitat.

(3)

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(c) Explain why *Canthigaster valenti* and *Paraluteres prionurus* are unable to reproduce with each other.

(2)

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(Total for question = 6 marks)

Q36.

Arginase is an enzyme produced in liver cells.

It breaks down arginine into urea and ornithine.

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A mutation in a gene called ARD1 can cause arginase deficiency (AD), which can be fatal. The mutant allele is recessive.

(a) It is estimated that 1 in 300 000 to 1 000 000 individuals have AD.

(i) It was suggested that this mutant allele arose in meiosis during prophase I.
Suggest how this mutant allele could be produced in prophase I.

(1)

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(ii) If the incidence of AD was 1 in 300 000, and the population was 7 800 000 000, then 26 000

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people would have AD.
Calculate how many people do not have a mutant allele for ARD1.
Use the equation:

$$p^2 + 2pq + q^2 = 1$$

(3)

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Answer .....

(b) One treatment for AD is mRNA therapy. In this therapy, people are given the mRNA molecules needed for arginase production.

The mRNA used in this therapy is active mRNA.
Active mRNA is produced by post-transcriptional changes to mRNA (pre-mRNA).
Give **two** differences between active mRNA and pre-mRNA.

(2)

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(c) Another treatment method could be stem cell therapy using either embryonic stem cells or human induced pluripotent stem cells (hiPSC).

The hiPSCs are sourced from skin cells from a person without AD.
(i) Explain how a pluripotent stem cell could become a cell that synthesises arginase.

(5)

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(ii) Explain why society may support the use of hiPSC stem cells to treat patients with AD.

(3)

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(Total for question = 14 marks)

Q37.

The biodiversity of four groups of organisms – birds, fish, mussels and aquatic plants – was studied along four sections of the Rideau River in Canada.

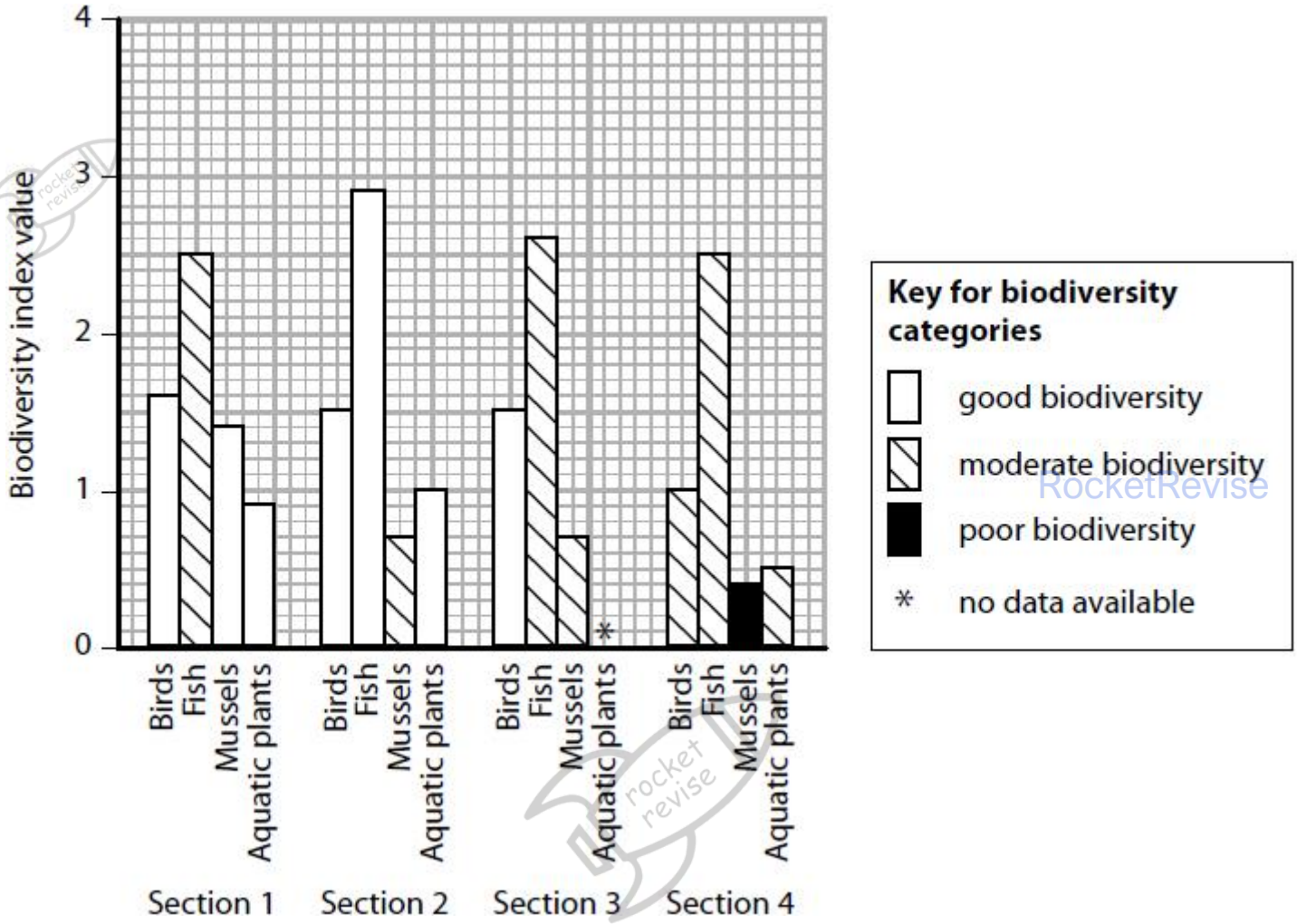
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A biodiversity index value was calculated for each group of organisms.

The graph shows the results of this study.

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The biodiversity index value can be used to compare biodiversity within one group of organisms.

The biodiversity categories (good, moderate and poor) can be used to compare biodiversity between different groups of organisms.

(a) Which statement describes biodiversity?

(1)

- ☒ **A** species richness of only the endemic species within a habitat
- ☒ **B** species richness of all the species within a habitat
- ☒ **C** the role of only the endemic species within a habitat
- ☒ **D** the role of all the species within a habitat

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*(b) Describe the changes in biodiversity along the Rideau River. Use the information in the graph to support your answer.

(6)

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(c) In Section 1, birds have a biodiversity index value of 1.6 and fish have a biodiversity index value of 2.5.

Suggest why the fish are considered to have a moderate biodiversity and the birds have a good biodiversity, but the biodiversity index value of the fish is greater.

(2)

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(d) No data were available for aquatic plants in Section 3.

A student collected some data in Section 3 to calculate a biodiversity index value.
The equation that the student used is:

$$D = \frac{N(N - 1)}{\Sigma n(n - 1)}$$

The data are shown in a table prepared by the student.

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Species of aquatic plant	Number of aquatic plants counted	(n – 1)	n(n – 1)
Coontail	8		
Tape grass	6		
Common waterweed	3		
Northern water milfoil	2		
Star duckweed	9		
White water lily	2		
Water stargrass	2		
Eurasian water milfoil	6		
Curly pondweed	5		
European frogbit	2		
Flowering rush	3		

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(i) Complete the table.

(1)

(ii) Calculate the biodiversity index value for the aquatic plants in Section 3 of this river.

(3)

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Answer .....

(Total for question = 13 marks)

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Q38.

The photograph shows a Baird's tapir.



Source: https://www.biolib.cz/IMG/GAL/171566.jpg

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(a) Baird's tapir is endemic to countries in Central America.

State what is meant by the term **endemic**.

(1)

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(b) Baird's tapir is classified as endangered.

In 2006, it was estimated that there were 5500 Baird's tapirs. This number had fallen to 3000 in 2016.

(i) Calculate the percentage decrease in the number of Baird's tapirs from 2006 to 2016.

(2)

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Answer ..... %

(ii) Explain how human activity, other than hunting, could have caused this decrease in the number of Baird's tapirs.

(3)

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(c) Preservation of sperm collected from Baird's tapir may help captive breeding programmes.

Scientists investigated the effect of freezing on sperm from Baird's tapir.
The sperm were frozen and then thawed.
The results of this investigation are shown in the table.

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Sperm	Percentage of sperm capable of moving (%)	Ability of sperm to swim in a straight line / a.u.	Percentage of sperm with an undamaged acrosome (%)
Freshly collected	63	3.5	80
Frozen and then thawed	38	2.5	48

(i) Describe how each of these effects of freezing could be determined.

(3)

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(ii) Explain how freezing sperm could affect the success of captive breeding programmes.

(4)

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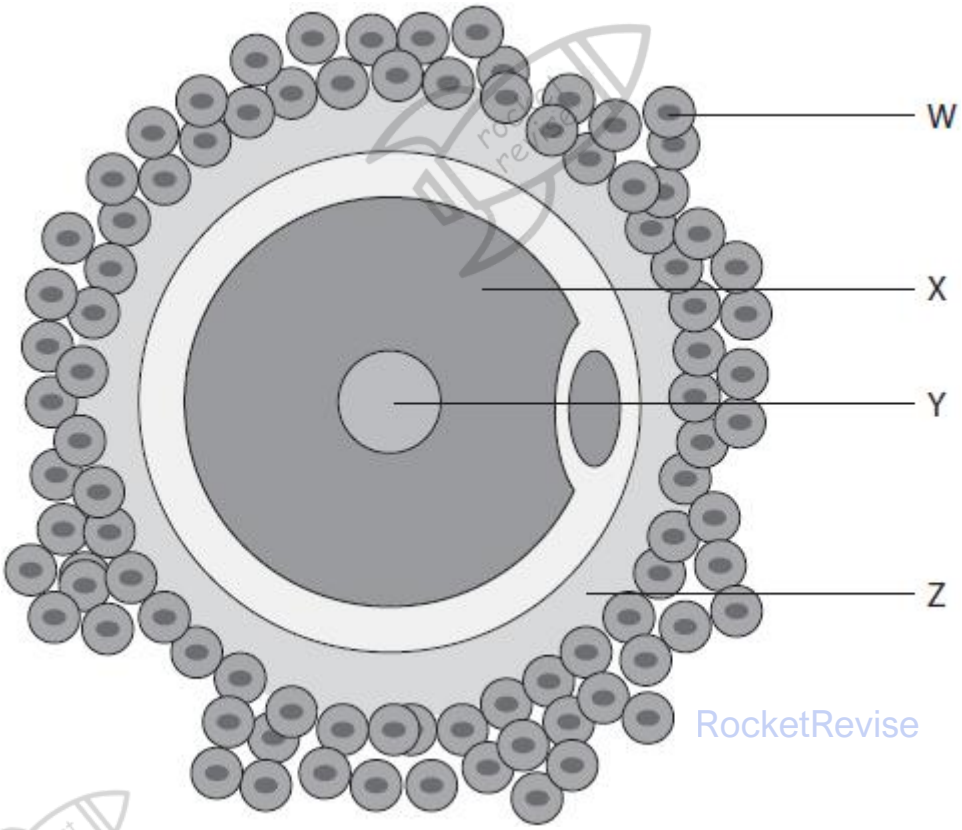
(Total for question = 13 marks)

Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some organisms can produce gametes for sexual reproduction.

(a) The diagram shows an incomplete drawing of a human egg cell, drawn by a student.



(i) State the letter which shows the zona pellucida.

(1)

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(ii) The student did not include the structures that contain enzymes involved in hardening the zona pellucida.

Name these structures.

(1)

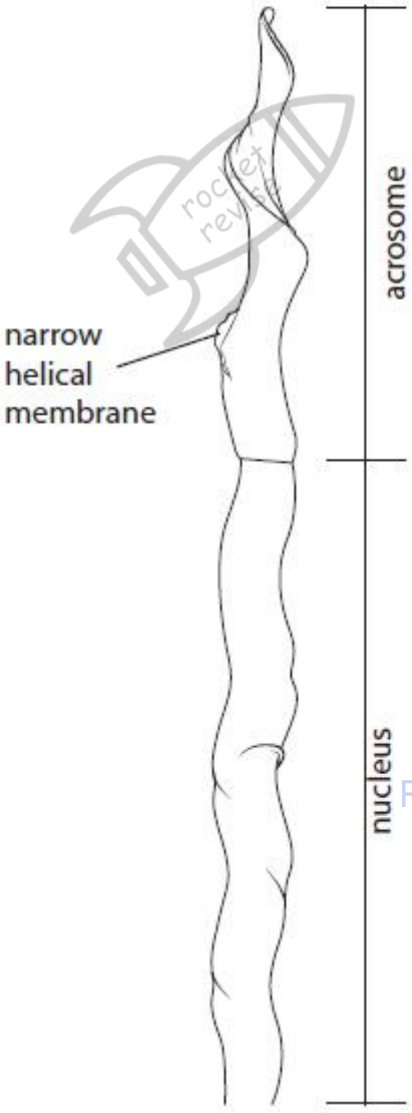
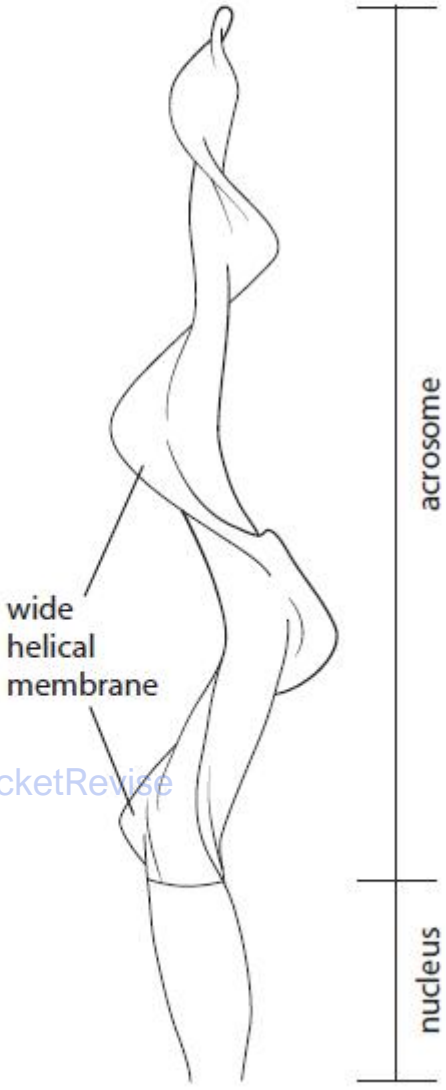
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(b) The table shows some information about two birds.

Both birds have a helical membrane around part of their sperm cells.

Information	Nuthatch	Reed bunting
Photograph	 (Source: Drew Buckley / Alamy Stock Photo)	 (Source: Angela Yates / Alamy Stock Photo)
Reproductive behaviour	Female has one male partner	Female mates with several males in a short period of time.
Drawing of head of sperm cell		

(i) How many of the following statements about sperm cells are correct?

- The mid-piece of the sperm cell contains many mitochondria for anaerobic respiration.
- The acrosome is a membrane bound sac containing digestive enzymes.

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- The nucleus contains circular DNA.

(1)

- ☒ **A** none
- ☒ **B** one
- ☒ **C** two
- ☒ **D** three

(ii) Sperm cells with a larger acrosome to nucleus ratio have an increased swimming speed.

Calculate the acrosome to nucleus ratio for the reed bunting sperm cell shown in the table.

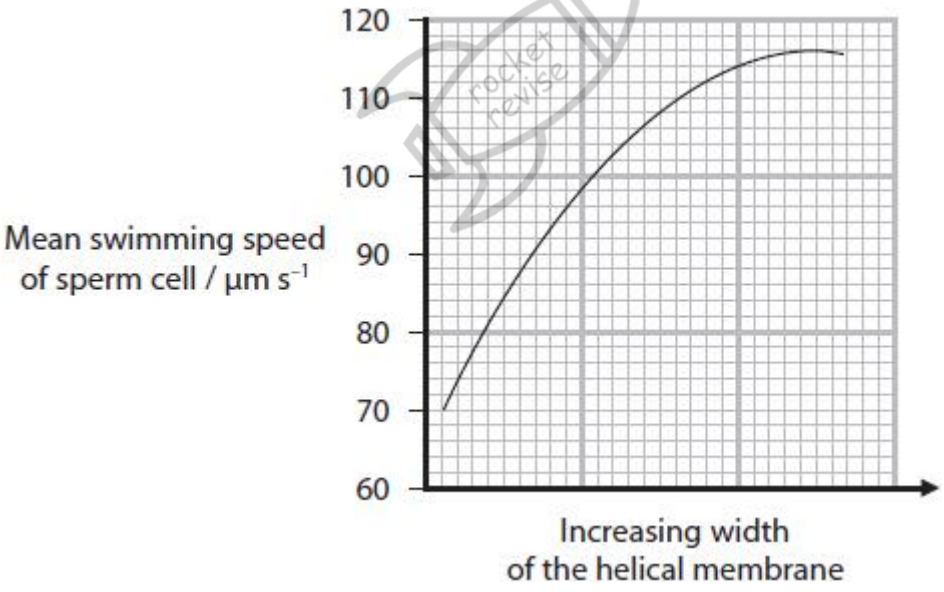
(1)

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Answer .....

(iii) The effect of the width of the helical membrane on the mean swimming speed of sperm cells from different bird species was investigated.

The graph shows the results of this investigation.



The acrosome to nucleus ratio and the width of the helical membrane affect the swimming speed of sperm cells.

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Explain how these adaptations are related to the reproductive behaviour of these birds.

Use the information in the table **and** the graph to support your answer.

(4)

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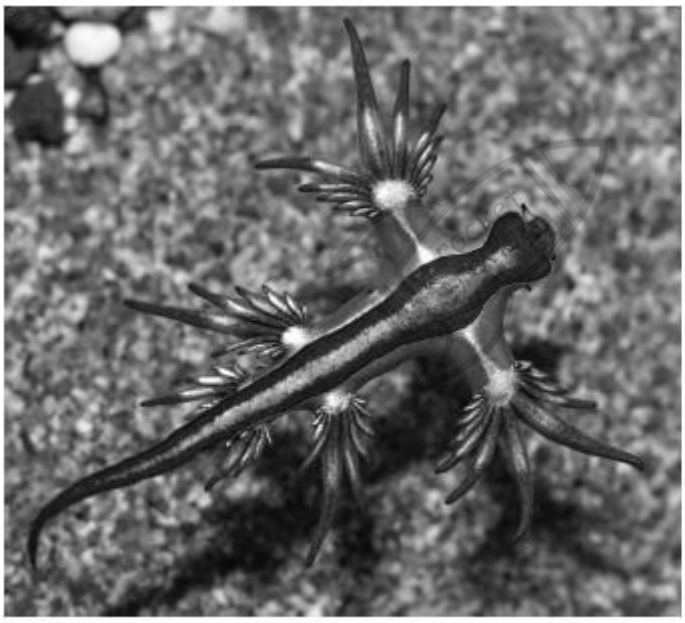
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(c) An individual organism that can produce both egg cells and sperm cells is said to be a hermaphrodite.

Blue dragons (*Glaucus atlanticus*) are sea slugs that are hermaphrodites and can be found in the Indian Ocean.
The photograph shows a blue dragon.



(Source: BIOSPHOTO / Alamy Stock Photo)

Each individual can produce both egg cells and sperm cells for sexual reproduction with another blue dragon.

Their sperm cells cannot fertilise their own egg cells.

Suggest reasons why it may be advantageous for blue dragons to be hermaphrodites.

(3)

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(d) The table shows some information about a population of 100 cats.

Colour of cat	 Black	 Black	 White
Genotype	BB	Bb	bb
Part of Hardy–Weinberg equation	p^2	2pq	q^2
Number of cats			16

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Complete the table to show how many cats would have the homozygous dominant genotype and how many cats would have a heterozygous genotype.

Use the equation

$$p^2 + 2pq + q^2 = 1$$

(3)

(Total for question = 14 marks)

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Q40.

Organisms can be classified into one of three domains.

(a) Organisms belonging to two of these domains have prokaryotic cells.

(i) Bacteria are one of these domains.
Name the other domain that has prokaryotic cells.

(1)

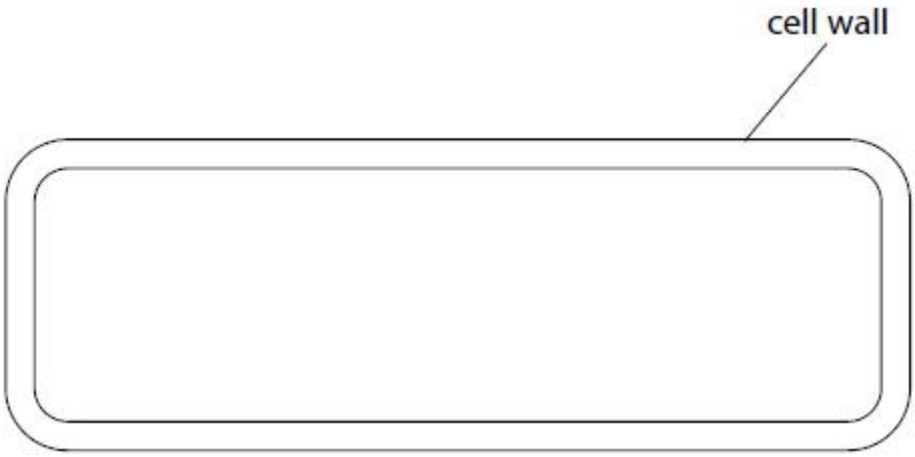
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(ii) The diagram shows the outline of a bacterial cell.
Draw **three** labelled features on this diagram that may be found in a prokaryotic cell.

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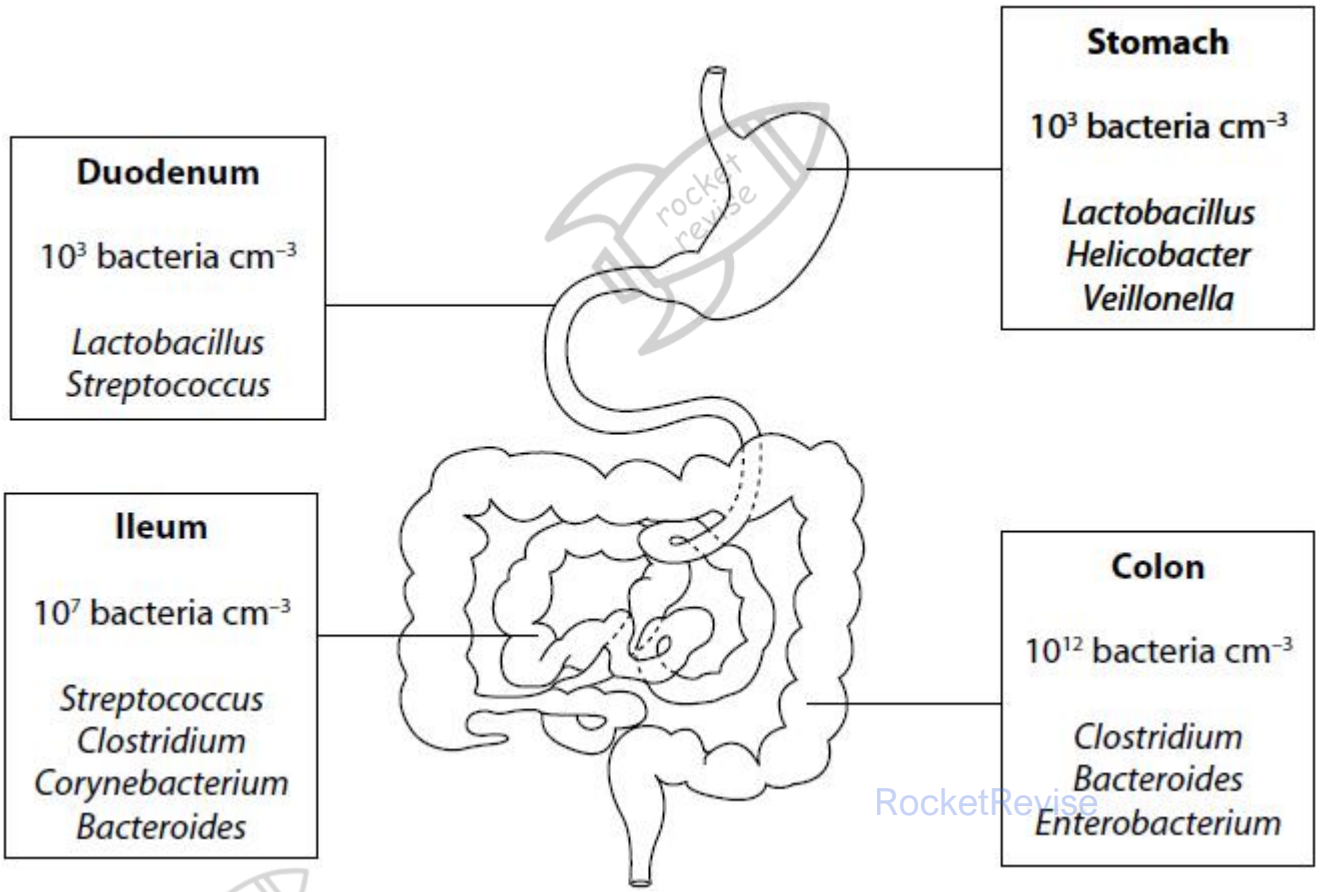
(3)



*(b) A variety of different types of bacteria is found in the human digestive system.

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The diagram shows part of the human digestive system and the number and types of bacteria that can be found in each organ.



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The table gives some information about conditions in the digestive system.

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Organ	pH	Oxygen content
Stomach	1 to 3	High ↓ Low
Duodenum	6 to 7	
Ileum	6 to 8	
Colon	5 to 7	

Explain the distribution of bacteria in the digestive system. Use the information in the diagram and table to support your answer.

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(6)

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(Total for question = 10 marks)

Q41.

The photograph shows a gentoo penguin from Antarctica.

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© robertharding / Alamy Stock Photo

These penguins swim in the very cold Antarctic waters in search of food.

They feed on fish, squid and shrimp.

(a) Explain **two** anatomical adaptations of these penguins that enable them to survive in this environment.

Use the information in the question to support your answer.

(2)

Adaptation 1 .....

Reason .....

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Adaptation 2 .....

Reason .....

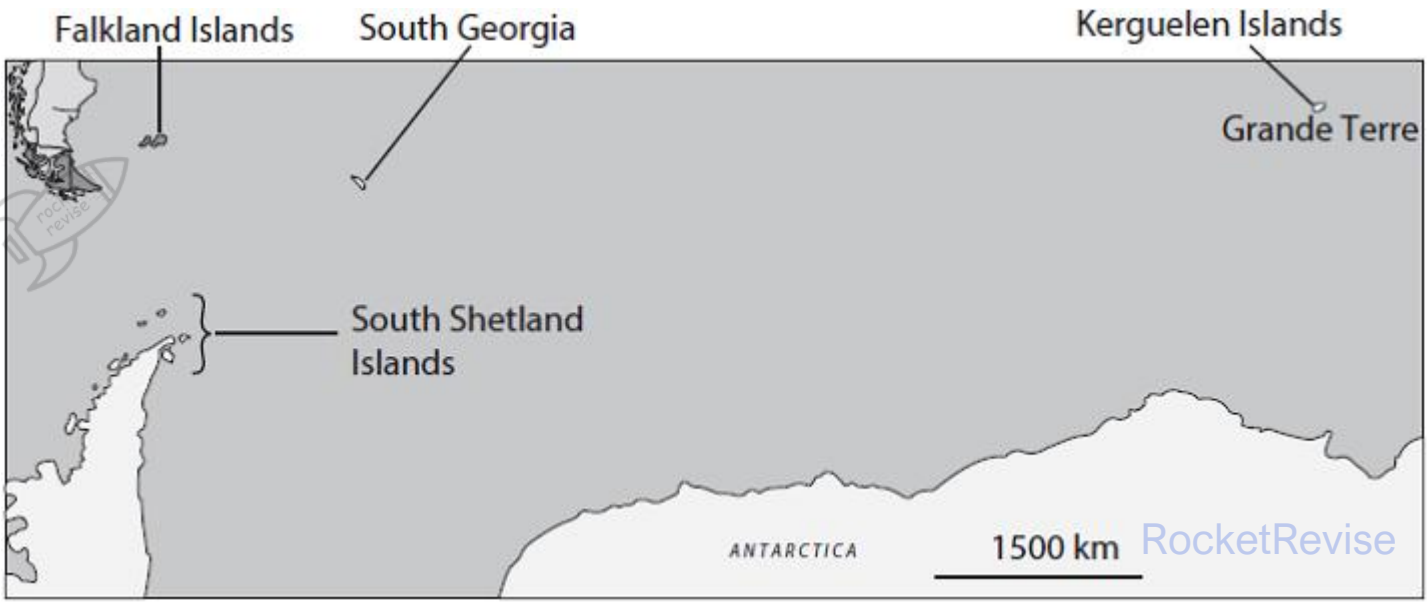
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(b) The map shows four groups of islands that have gentoo penguin populations.

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It has been proposed that the gentoo penguins on these four groups of islands are **different** species.

(i) Suggest **one** way that these penguins can be shown to belong to different species.

(1)

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(ii) Explain how different species of gentoo penguin could have formed.
Use the map to support your answer.

(5)

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(Total for question = 8 marks)

Q42.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A plant contains several types of cells, tissues and organs.

(a) State what is meant by the term **organ**.

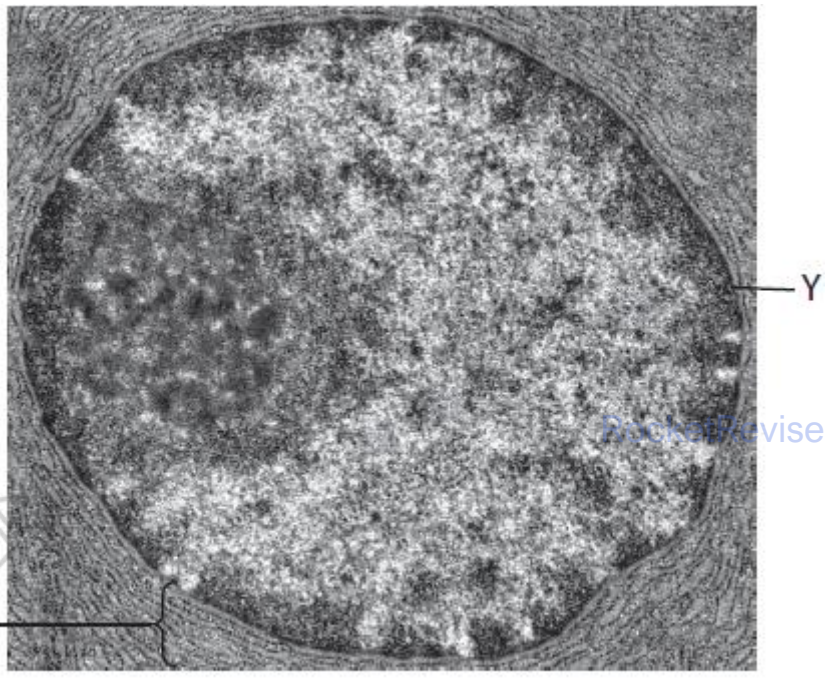
(1)

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(b) The photograph shows part of a cell.



© Jose Calvo / Alamy Stock Photo

(i) Which organelle is labelled X?

(1)

- ☐ **A** chloroplast
- ☐ **B** endoplasmic reticulum
- ☐ **C** Golgi apparatus

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☒ **D** plasmodesmata

(ii) The organelle surrounded by the membrane labelled Y contains the nucleolus.
Give the name and function of the organelle that contains the nucleolus.

(1)

Name .....

Function .....

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.....

(iii) Which of the following are functions of the nucleolus?
1. production of ribosomes
2. synthesis of DNA
3. synthesis of RNA

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(1)

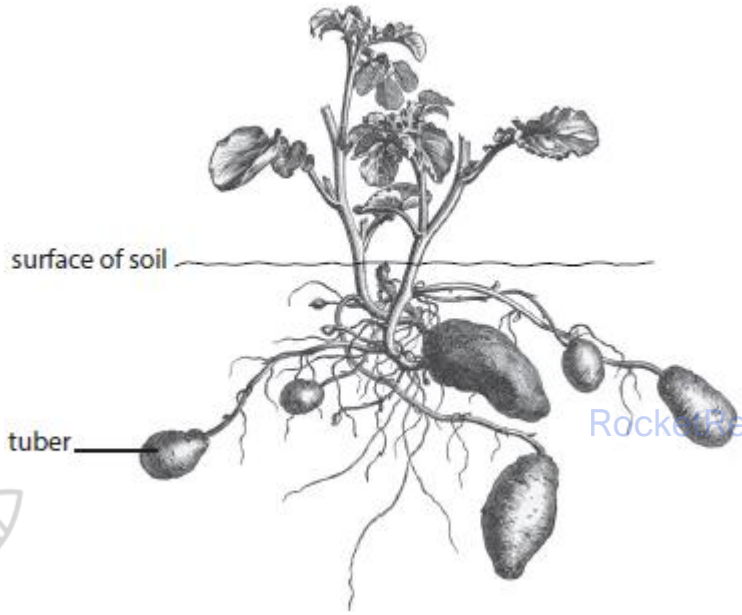
☒ **A** 1 and 2 only
☒ **B** 1 and 3 only
☒ **C** 2 and 3 only
☒ **D** 1, 2 and 3

(iv) Organisms can be classified into three domains.
State which of the three domains would contain an organism with the organelles X and Y.

(1)

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(c) The drawing shows tubers of a potato plant developing below the surface of the soil.



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Organic molecules are transported through the potato plant to the developing tubers.
These molecules can be used to form starch molecules.
(i) Which structure in the cells of these tubers would store starch molecules?

(1)

☒ **A** amyloplast
☒ **B** chloroplast
☒ **C** middle lamella
☒ **D** vacuole

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(ii) Explain how the **structure** of starch enables the potato tubers to have a high energy content.

(2)

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(Total for question = 8 marks)

Q43.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A group of horses with the same specific characteristics is called a breed.

The photograph shows a horse of the rare breed Akhal-Teke that originated in Turkmenistan.

The Akhal-Teke population is in danger from inbreeding.



(Source: © Heat Design / Alamy Stock Photo)

(a) A scientist suggested that the Akhal-Teke breed is the same as an older breed known as the Turkoman horse.

Describe **one** piece of information that would need to be collected in order to determine if the scientist

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was correct.
Give a reason for your answer.

(2)

Information

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Reason

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(b) A scientist calculated the heterozygosity index of a population of 5 457 Akhal-Teke horses to be 0.661.

How many of these horses are heterozygotes?

(1)

- ☒ **A** 1 850
- ☒ **B** 3 607
- ☒ **C** 5 457
- ☒ **D** 8 255

(c) The images show two other rare breeds of horse, the Caspian and the Arabian.



Caspian

(Source: © Irina Mavritsina / Alamy Stock Photo)



Arabian

(Source: © Juniors Bildarchiv GmbH / Alamy Stock Photo)

The Caspian breed was thought to be extinct until a small population was discovered in mountainous regions in 1965.
The Arabian breed is found in sandy deserts.
(i) The Caspian breed has small hooves in proportion to its mass, whereas the closely-related Arabian breed has large hooves in proportion to its mass.
Suggest **one** reason for this difference.

(1)

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(ii) The table shows some information about one gene for a population of 400 Caspian horses.

Genotype	two copies of the dominant allele	heterozygous	two copies of the recessive allele
Part of Hardy–Weinberg equation	p^2	2pq	q^2
Number of individuals	46		

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Complete the table to show how many horses would have the heterozygous genotype and how many would have the homozygous recessive genotype.
Use the equation:

$$p^2 + 2pq + q^2 = 1$$

(3)

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*(d) Breeding programmes have been set up for both the Akhal-Teke and Caspian breeds.

Sperm samples have been taken from male horses from both the Akhal-Teke and Caspian populations.
Some of these male horses have since died.
All of the sperm samples have been frozen.
The effect of freezing and thawing sperm cells was investigated.
The table shows the characteristics of two types of sperm cell samples.

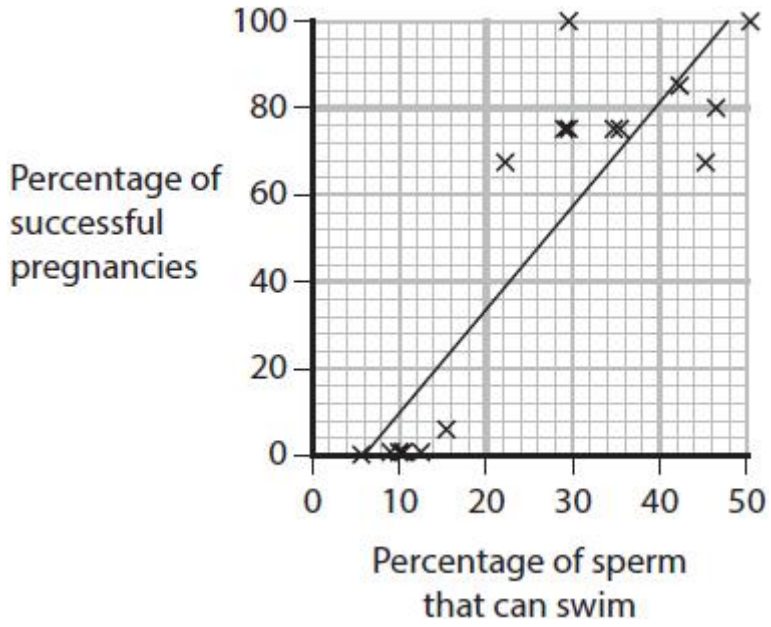
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Type of sample	Percentage of sperm cells that can swim	Sperm cell survival / hours	Percentage of sperm cells with damaged DNA
Fresh sperm cells	50.7	117.7	28.7
Frozen and thawed sperm cells	26.9	48.5	39.3

The graph shows the effect of the percentage of sperm cells that can swim on the number of successful pregnancies.

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The correlation coefficient of the data shown in the graph is +0.90.
Discuss the advantages and disadvantages of using frozen sperm in breeding programmes for these endangered breeds.
Use all of the information in this question to support your answer.

(6)

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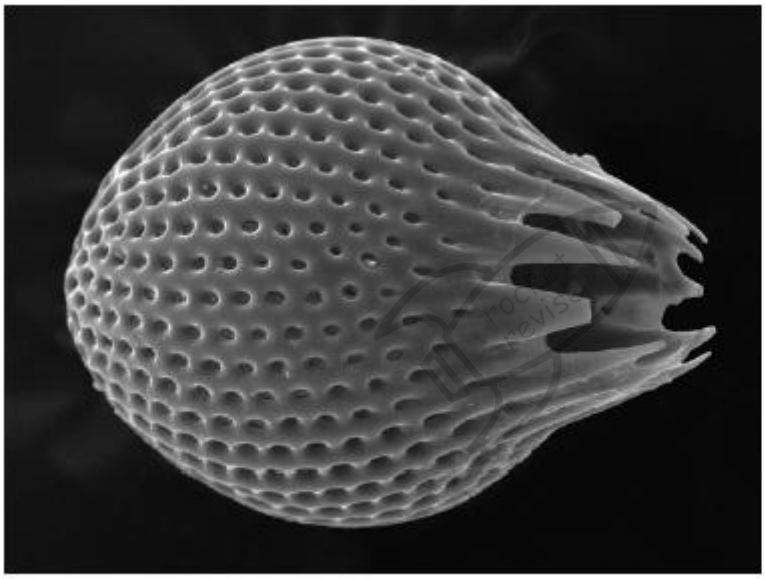
(Total for question = 13 marks)

Q44.

Diatoms are single-celled organisms important for many ocean food chains.

The photograph shows a diatom, as seen using a scanning electron microscope.

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(Source: © AMI IMAGES / SCIENCE PHOTO LIBRARY)

(a) Diatoms contain a nucleus, mitochondria and chloroplasts.

(i) State the domain which contains diatoms.

(1)

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(ii) The nucleus in a diatom contains DNA.
Name **two processes** that take place in the nucleus of a diatom.

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(2)

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(b) It is predicted that diatom populations will decrease in number due to global warming.

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Scientists have suggested that this will cause a decrease in ocean biodiversity.
Describe how a **decrease** in biodiversity within an **ocean** habitat could be determined.

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(Total for question = 6 marks)

Q45.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a Javanese edelweiss plant.



(Source: © Ciptor / Shutterstock)

(a) This plant is endemic to Indonesia.

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State what is meant by the term **endemic**.

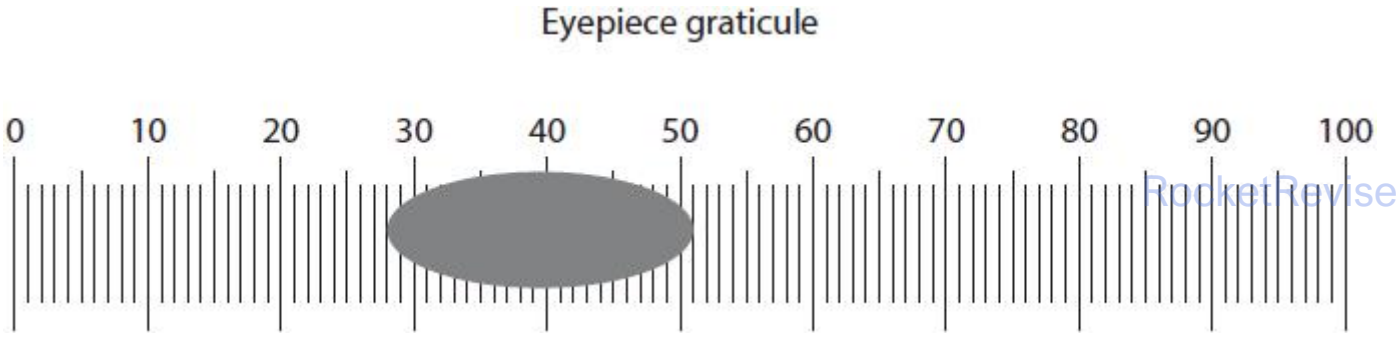
(1)

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(b) A student used a microscope and a graticule to view a pollen grain from this plant.



One hundred eyepiece graticule divisions is 0.1 mm.

(i) Which is the length of this pollen grain?

(1)

- ☒ **A** 2.3 mm
- ☒ **B** 23 μm
- ☒ **C** 230 μm
- ☒ **D** 2300 nm

(ii) State what is meant by the term **magnification**.

(1)

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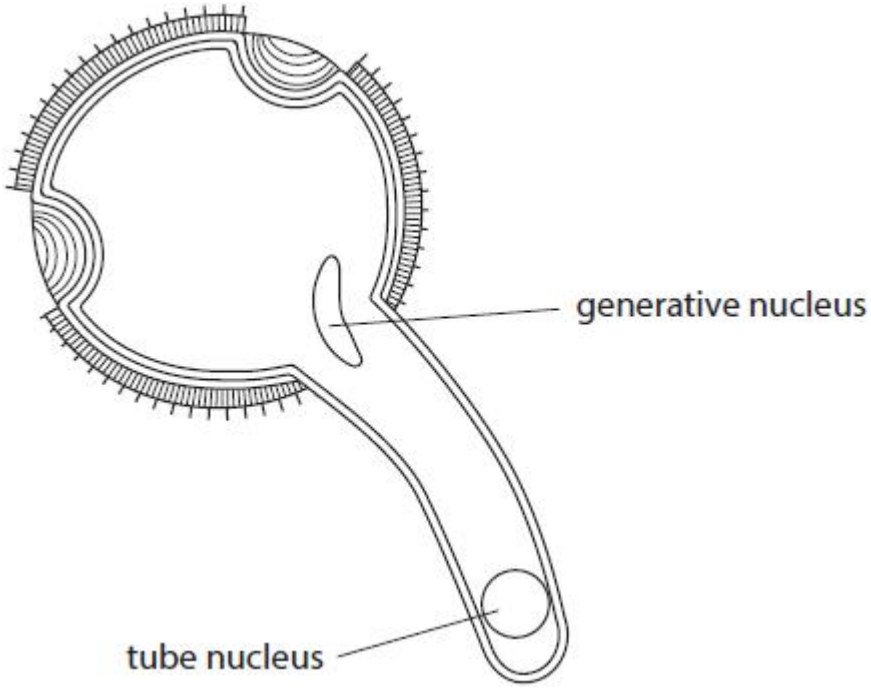
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(c) The diagram shows a germinating pollen grain.

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Describe the roles of the tube nucleus and the generative nucleus.

(4)

Tube nucleus .....

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Generative nucleus .....

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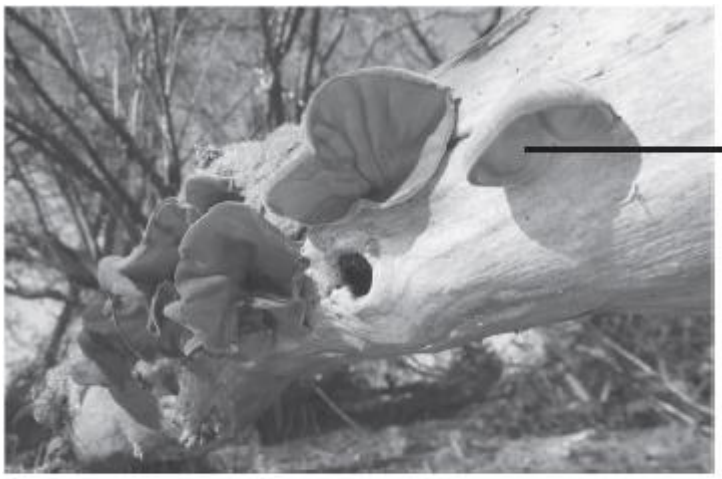
(Total for question = 7 marks)

Q46.

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The photograph shows a species of fungus growing on a dead tree in a woodland habitat.



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© www.pqpictures.co.uk / Alamy Stock Photo

Wood contains a high proportion of lignin and cellulose.

Fungi can break down molecules in wood by releasing extracellular enzymes.

(a) Describe the role of the rough endoplasmic reticulum in the formation of extracellular enzymes.

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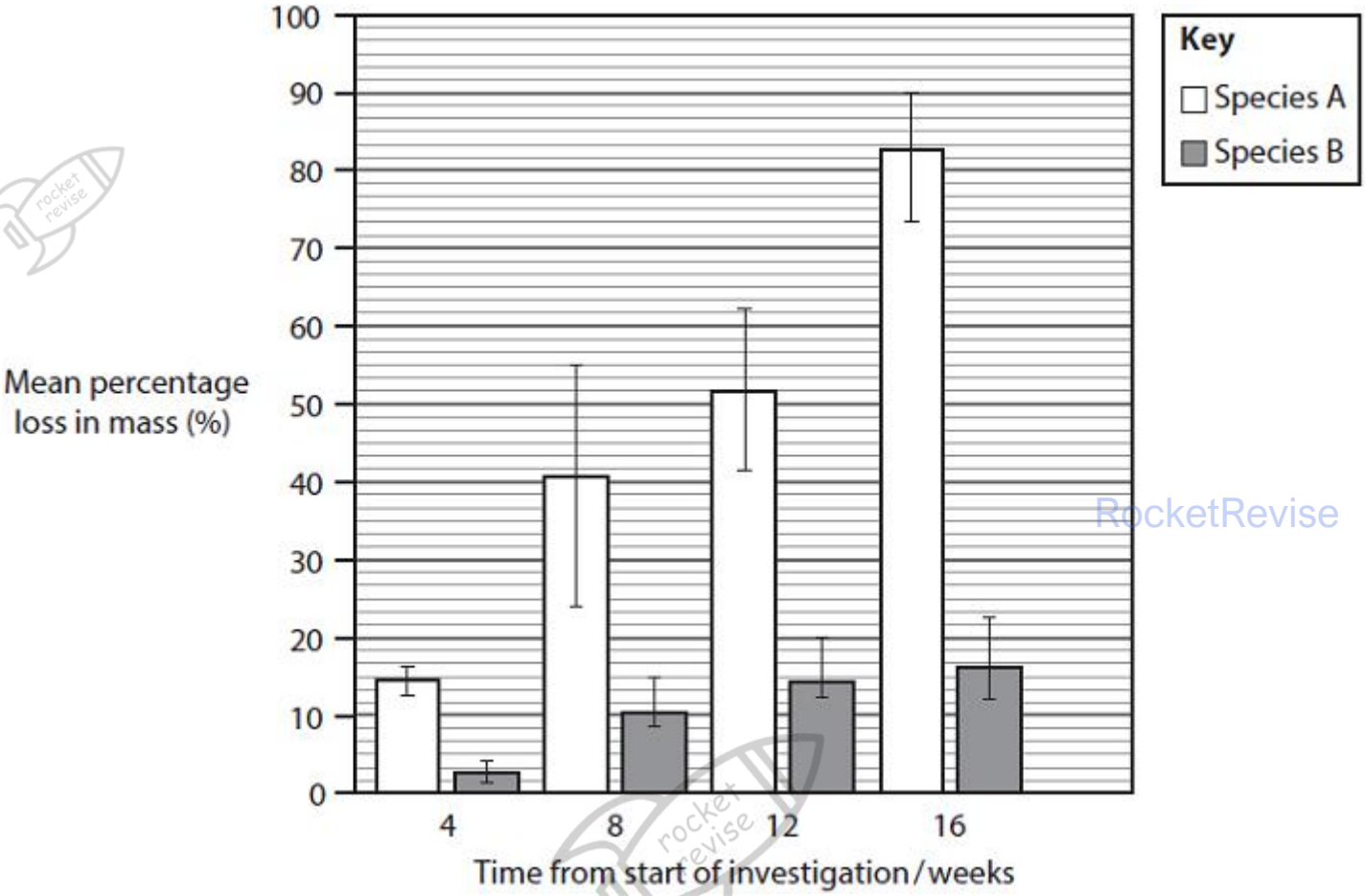
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(b) Different species of fungus can break down different molecules in wood.

The effect of two species of fungus on the percentage loss in mass of wooden blocks was investigated.
Species A was grown on one set of wood blocks and species B was grown on another set of wood blocks.
The percentage loss in mass of the blocks was recorded every four weeks.
The graph shows the results of the investigation.

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(i) Comment on the results of this investigation.

(3)

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(ii) Suggest **one** reason for the difference in breakdown of the wood by species A and species B.

(1)

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(c) A study compared the biodiversity of two woodland habitats.

The index of diversity for habitat one was 4.2.
The table shows the data obtained from habitat two.

Species	Number of individuals (n)	n(n–1)
A	34	1122
B	6	30
C	27	
D	4	12
E	9	72
F	31	930
G	120	14280

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(i) An index of diversity (D) is calculated using the formula:

$$D = \frac{N(N-1)}{\Sigma n(n-1)}$$

Calculate the index of diversity for habitat two.
Use the table and the formula to help you.

(3)

Answer .....

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(ii) State which habitat has the higher biodiversity.
Give a reason for your answer.

(1)

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(iii) Species D is an endangered plant species.
Explain the processes that could be used by a seed bank to conserve this plant species.

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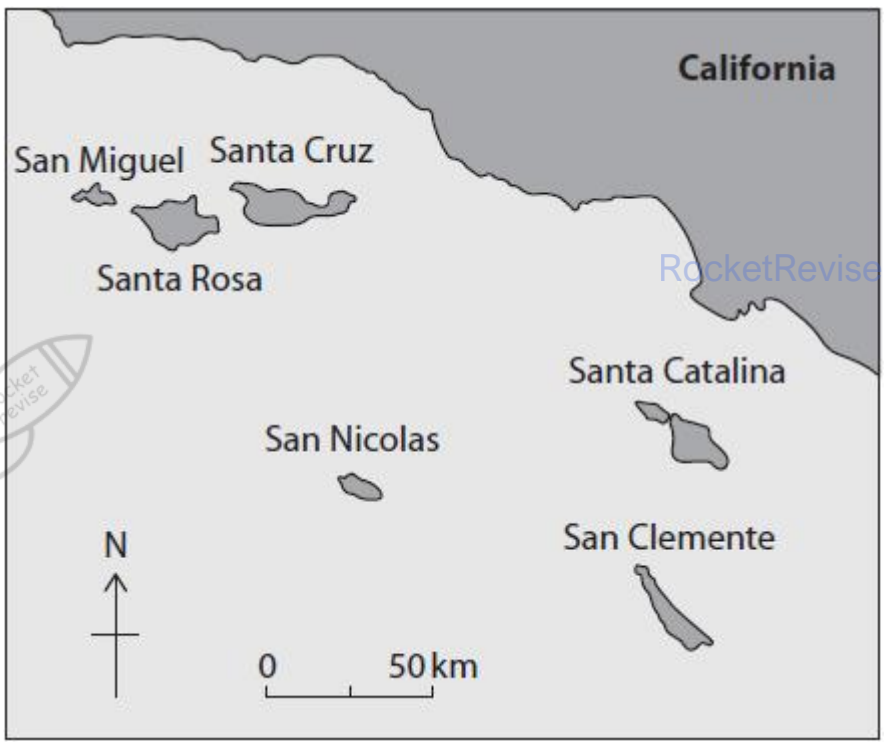
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(Total for question = 14 marks)

Q47.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The map shows six islands near the coast of California.



Each of these islands has a population of island foxes descended from the California grey fox.

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Each island fox population has developed different characteristics from the fox populations on the other islands.

The photograph shows a fox from the population found only on the island of Santa Cruz.



(Source: © Christian Loader/Alamy Stock Photo)

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(a) (i) Which of the following is a term used to describe the fact that this subspecies of island fox is found only on Santa Cruz?

(1)

- ☒ **A** diversity
- ☒ **B** endemic
- ☒ **C** mutation
- ☒ **D** specific

(ii) Explain how isolation caused the formation of six fox populations with different characteristics. Use the information in the question to support your answer.

(4)

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(iii) Explain how molecular evidence could be used to determine which of the other five fox populations is the most closely related to the fox population on Santa Cruz.

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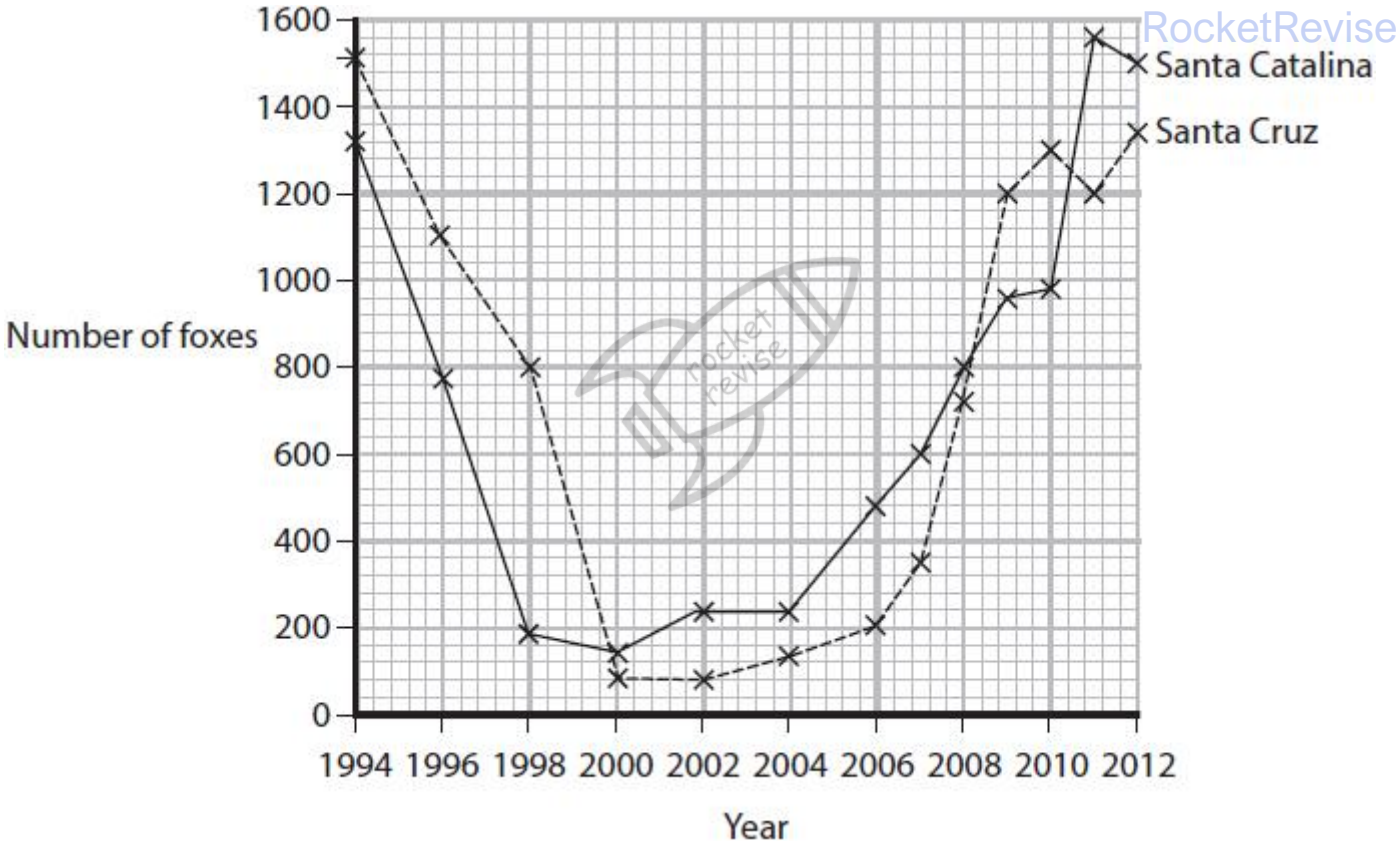
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(b) The graph shows the sizes of two fox populations from 1994 to 2012.



Two main causes for the decrease in island fox populations from 1994 to 2000 were predation by golden eagles and diseases transmitted from dogs and raccoons.

(i) Calculate the percentage decrease in the Santa Cruz fox population from 1994 to 2002.

(1)

Answer ..... %

(ii) Explain how the Hardy-Weinberg equation can be used to show the change in the frequency of a recessive allele in the Santa Cruz population from 1994 to 2000.

(2)

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(iii) In 2004, a conservation plan was produced to:

- increase the sizes of the Santa Cruz and Santa Catalina fox populations
- maintain the genetic diversity of the Santa Cruz and Santa Catalina fox populations.

Explain how conservation could achieve these goals.
Use the information in the question and your own knowledge to support your answer.

(4)

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(Total for question = 14 marks)

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Q48.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a mother and baby Malayan tapir (*Tapirus indicus*). They live in the rainforests of southeast Asia.

Each gamete of a Malayan tapir contains 26 chromosomes.

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© Sipa US / Alamy Stock Photo

(a) The body cells of the mother Malayan tapir have some genetic similarities to and differences from the cells of her baby.

Explain why the body cells of the mother and baby have genetic similarities **and** differences.

(3)

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(b) Biodiversity within a species can be measured using genetic diversity.

Three melanistic Malayan tapirs have been discovered that have black fur all over their bodies.
Melanistic tapirs have two copies of the recessive allele for fur colour and are homozygous.

(i) Scientists wanted to determine the heterozygosity index of the Malayan tapir species.
Which equation can be used to calculate the heterozygosity index?

(1)

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	Equation
☐ A	heterozygosity index = $\frac{\text{number of heterozygotes}}{\text{number of individuals in the population}}$
☐ B	heterozygosity index = $\frac{\text{number of heterozygotes}}{\text{number of species in the habitat}}$
☐ C	heterozygosity index = $\frac{\text{number of individuals in the population}}{\text{number of heterozygotes}}$
☐ D	heterozygosity index = $\frac{\text{number of species in the habitat}}{\text{number of heterozygotes}}$

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(ii) Explain how scientists could determine if a **change** in the frequency of the recessive allele was occurring over time.

(2)

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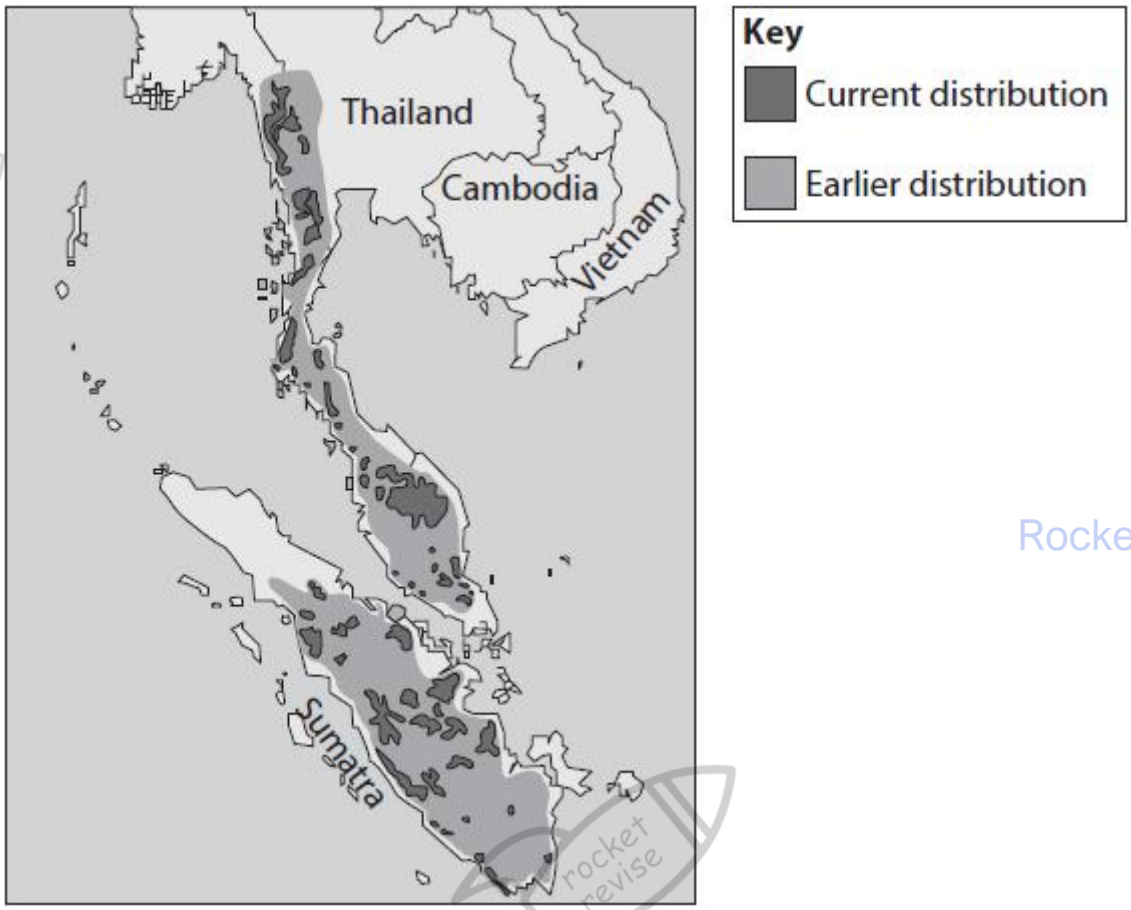
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(c) The map shows the change in distribution of the Malayan tapir over the past 20 years.

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(i) Give **one similarity** between the current and earlier distribution of the Malayan tapir.

(1)

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(ii) One difference is that the distribution has reduced due to human activity.
Suggest **one** reason why human activity has caused this change in distribution.

(1)

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(iii) A study suggested that the genetic diversity of the Malayan tapirs has decreased due to populations being separated from each other.
Explain how a captive breeding programme could maintain the genetic diversity of Malayan tapir populations.

(3)

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(Total for question = 11 marks)

Q49.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Migration is when many individuals of a species move from one geographical region to another.

Monarch butterflies migrate from Canada and North America to Mexico each autumn.

The photograph shows a cluster of migrating Monarch butterflies on a branch of a tree.



(Source: © Patrick Stoll/Alamy Stock Photo)

(a) (i) Migration is an example of which type of adaptation?

(1)

☐ **A** anatomical
☐ **B** behavioural

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☒ **C** isolational
☒ **D** physiological

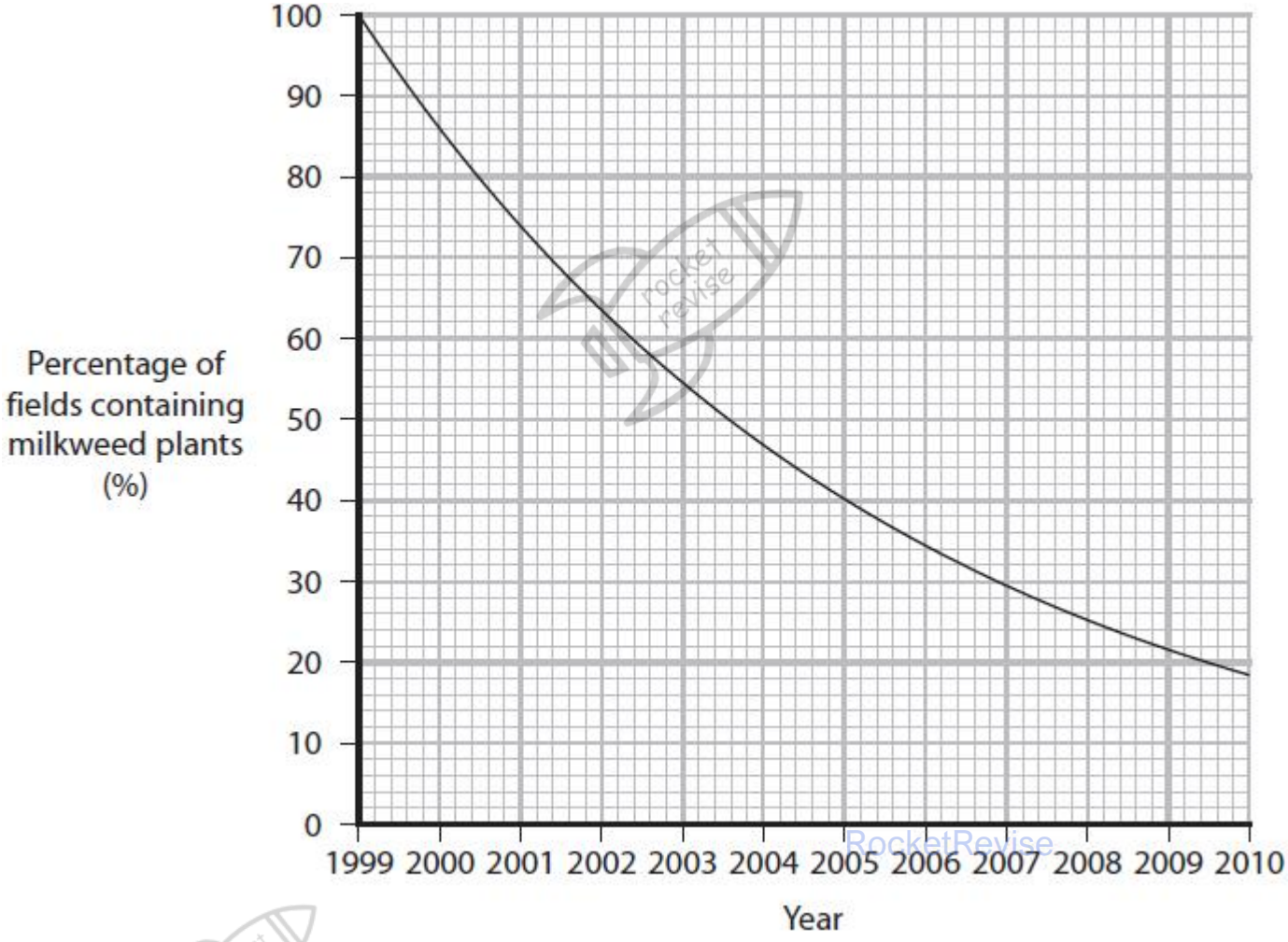
(ii) The heterozygosity index for a cluster of 8 000 Monarch butterflies was calculated as 0.166.
Calculate the number of heterozygotes in this cluster.

(1)

Answer .....

(b) Female Monarch butterflies lay fertilised eggs on the leaves of milkweed plants.

Herbicides can be used by farmers to kill plants such as milkweed. Farmers in the USA have increased the use of glyphosate herbicide since 1999 to kill plants, such as milkweed.
The graph shows the percentage of fields in the USA containing milkweed plants from 1999 to 2010.



Explain what effect this human activity would have had on populations of Monarch butterflies, including their genetic diversity, after 2010.

Use the information in the graph to support your answer.

(3)

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(c) The photograph shows a hen (female chicken) that has just laid an egg.



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(Source: © Angela Hampton Picture Library / Alamy Stock Photo)

The rate of egg laying by hens is an example of polygenic inheritance.
This phenotype shows continuous variation.

(i) State what is meant by **polygenic inheritance** with reference to this phenotype.

(1)

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(ii) Sketch a **labelled** graph to show the expected egg laying rate for a population of hens.

(2)

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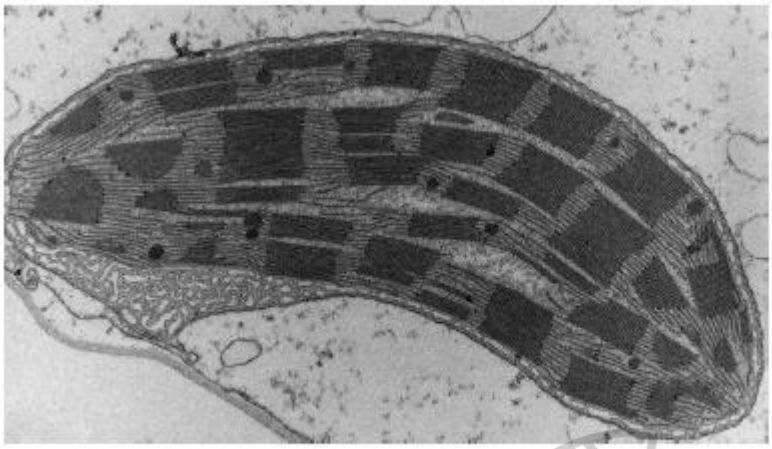
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(Total for question = 8 marks)

Q50.

(a) The photograph shows an organelle found in many plant cells, as seen using a microscope.



(Source: © Ashley Cooper pics / Alamy Stock Photo)

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(i) Give the name and function of this organelle.

(1)

Name .....

Function .....

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(ii) Name the type of microscope used for this photograph.
Give **one** reason for your answer.

(2)

Name .....

Reason .....

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(b) The photograph shows a lake partly covered with New Zealand pygmy weed plants (*C. helmsii*).

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(Source: © Science History Images / Alamy Stock Photo) RocketRevise

The effect of New Zealand pygmy weed on the biodiversity of a lake was investigated. Species richness and an index of diversity were determined.

(i) State what is meant by **species richness**.

(1)

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(ii) The number of individuals of different species in this area of the lake was measured **before** the appearance of the New Zealand pygmy weed.

The table shows these results.

Species	Number of individuals (n)	(n – 1)	n(n – 1)
A	54	53	2862
B	12		
C	3	2	6
D	31	30	930
E	7	6	42
F	46	45	
	Total (N) =		$\sum n(n-1)$ =

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Calculate the index of diversity (D) for this area of the lake using the formula:

$$D = \frac{N(N-1)}{\sum n(n-1)}$$

Use the table to help you.

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Give your answer to **two** decimal places.

(3)

Answer .....

(iii) A year **after** the appearance of the New Zealand pygmy weed in this lake the index of diversity (D) was found to be 1.8.

Suggest the effect of the New Zealand pygmy weed on the diversity of this part of the lake.

(2)

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(Total for question = 9 marks)

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Q51.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The Grandidier's baobab is endemic to Madagascar.

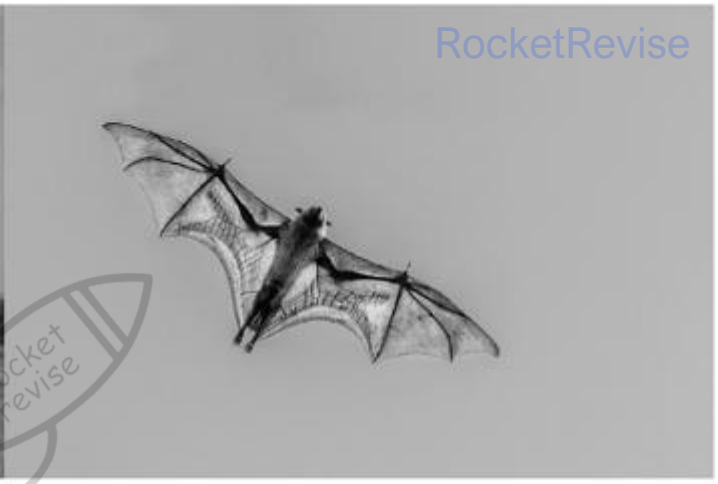
Fruit bats transfer pollen from one baobab tree to another.

The population of fruit bats in Madagascar is decreasing.

The photographs show one Grandidier's baobab tree in a field of rice and a Madagascan fruit bat.



(Source: © Graham, David / Alamy Stock Photo)



(Source: © John Dambik / Alamy Stock Photo)

(a) (i) This species of tree is endangered.

Identify **two** reasons why this species of tree is endangered.
Use the information above and the photographs to support your answer.

(2)

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(ii) Give **two** reasons how education could help conserve the Grandidier's baobab.

(2)

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(b) Calcium ions are used to make molecules in plants.

(i) Explain the role of **one** named molecule that uses calcium ions.

(2)

Name of molecule .....

Role of molecule .....

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(ii) Water is also important to plants.
Which of the following would transport water **down** the stem of a plant?

(1)

- ☒ **A** phloem
- ☒ **B** sclerenchyma
- ☒ **C** vacuole
- ☒ **D** xylem

(c) The photograph shows pods on a soybean plant.



(Source: © B Christopher / Alamy Stock Photo)

The effect of magnesium ions on the number of pods per soybean plant was investigated.

In 2019, soybean plants were grown in two fields.

One field had no magnesium ions added to the soil and the other field had magnesium ions added to the soil.

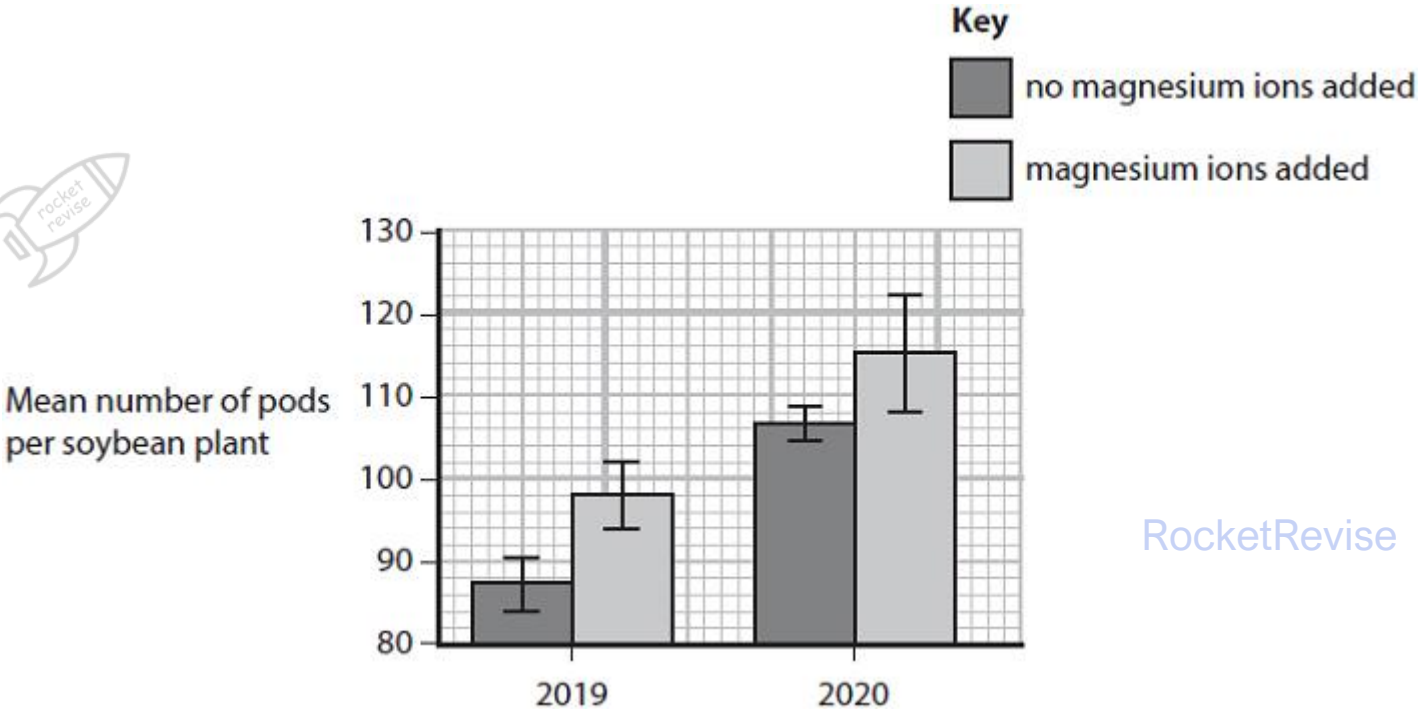
The mean number of pods per soybean plant was determined for both fields.

In 2020, the investigation was repeated using the same fields.

The graph shows the results of this investigation.

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Give **two** conclusions that can be made using this data.

(2)

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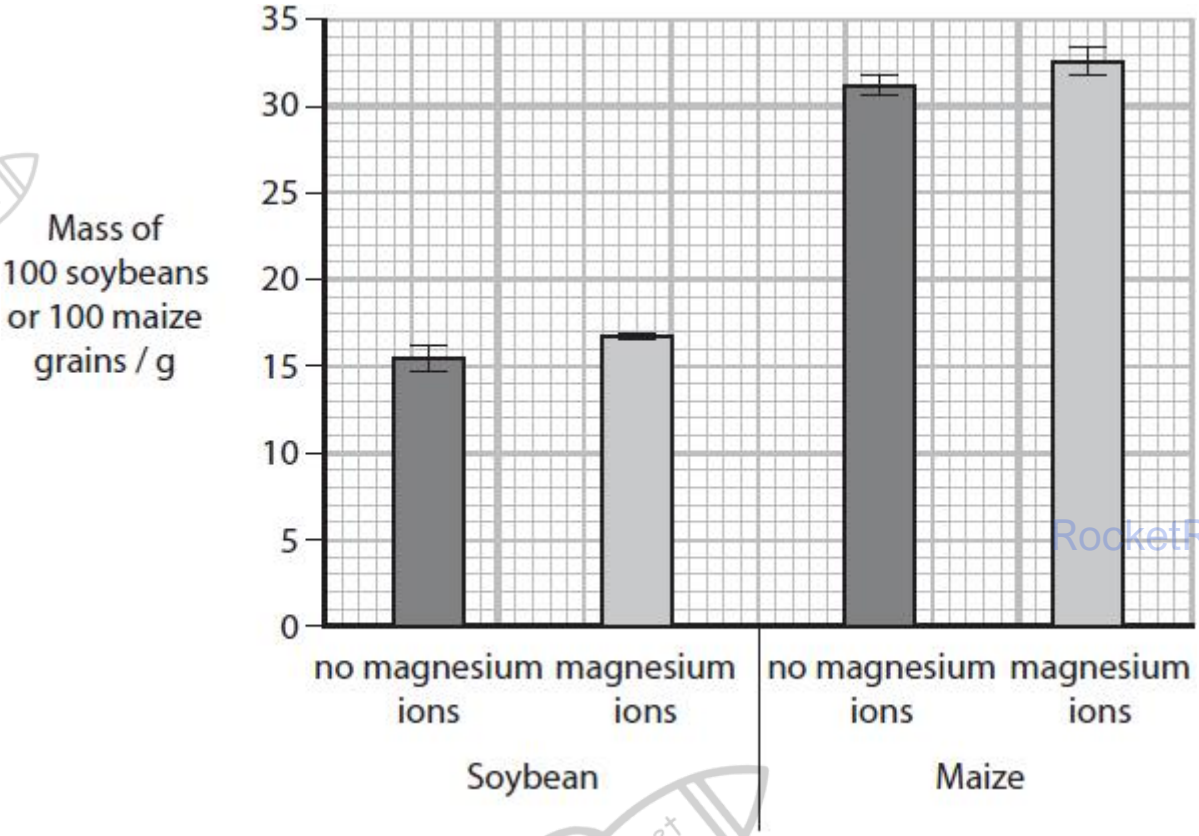
(d) In another investigation, the effect of magnesium ions on the mass of soybeans and maize grains was studied.

The graph shows the results of this investigation.

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Comment on the results from this investigation.

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(Total for question = 13 marks)

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Q52.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a rabbit with brown fur from Lokrum Island near Croatia.



(Source: © Peter Shaw / Alamy Stock Photo)

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(a) The Hardy-Weinberg equation can be used to determine changes in allele frequencies over time.

The equation can be written as $p^2 + 2pq + q^2 = 1$
Rabbits that inherit a dominant B allele have brown fur.
The proportion of homozygous dominant (BB) rabbits in the island population is represented by p^2.

(i) Which is the value for 2pq if $p^2 = 0.36$?

(1)

- ☐ **A** 0.06
- ☐ **B** 0.16
- ☐ **C** 0.48
- ☐ **D** 0.64

(ii) Albino rabbits in this population have the genotype bb.
State the part of the equation that represents this genotype.

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(1)

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(b) A viral disease has killed many of the rabbits on Lokrum Island.

Several genes are involved in protecting rabbits from this virus.
Explain how this viral disease could result in a change in allele frequencies of **these** genes in the rabbit population on Lokrum Island.

(4)

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(Total for question = 6 marks)

Q53.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Different species of millipede have different adaptations.

One shared adaptation is a hard outer layer called an exoskeleton.

The photograph shows a millipede.



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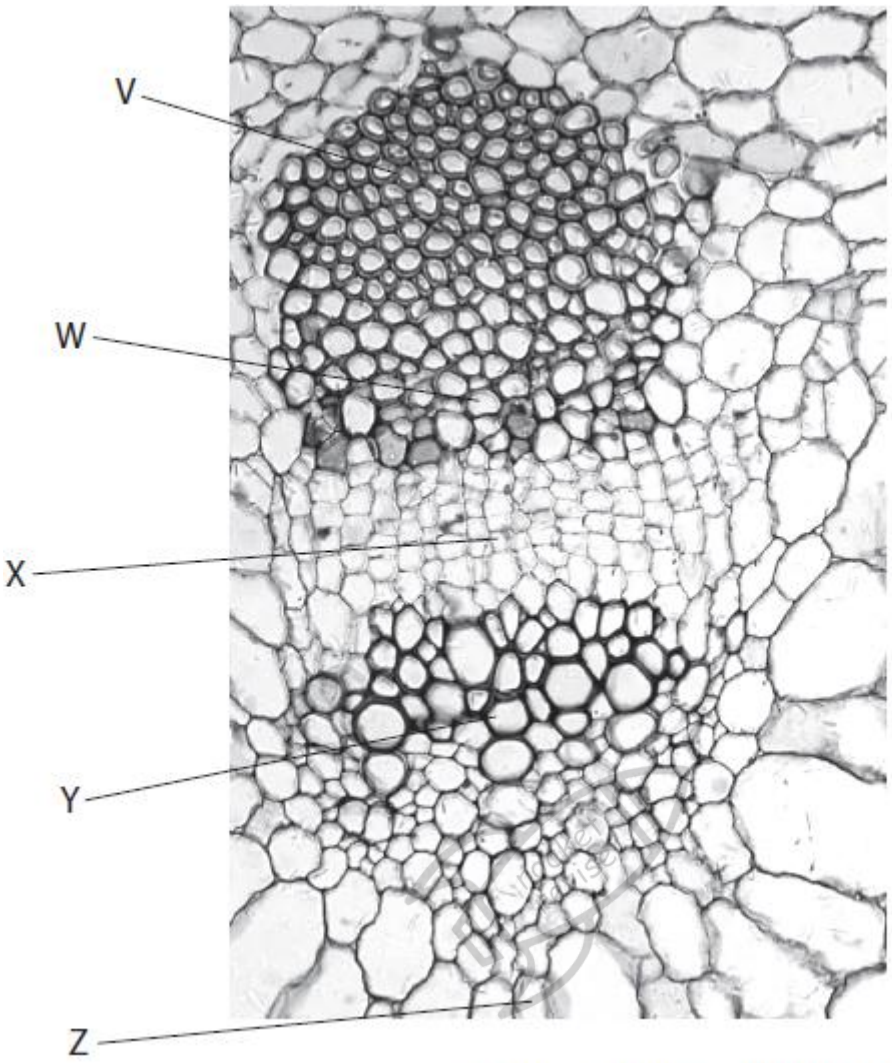
(Source: © Marina Kovacevic / Alamy Stock Photo)

(a) *Illacme plenipes* is a species of millipede that is endemic to California. It was thought to be extinct but was rediscovered in 2005.

The mouth of this millipede is adapted to make holes in xylem and phloem tissues of living plants.
The photograph shows some plant tissues, as seen using a microscope.

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(Source: © Choksawatdikorn / Shutterstock)

(i) Which of the labelled structures is a xylem vessel?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

(ii) Which of the labelled structures contains sieve plates?

(1)

- ☐ **A** V
- ☐ **B** W
- ☐ **C** X
- ☐ **D** Y

(iii) Explain the advantages for this millipede of being able to make holes in xylem and phloem.

(3)

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(b) The *Illacme tobini* species of millipede is also endemic to California.

(i) It has an exoskeleton and 200 poison glands.
Suggest why this millipede has these adaptations.

(2)

Hard exoskeleton

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Poison glands

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(ii) Explain how scientists could determine how closely related *I. plenipes* is to *I. tobini*.

(2)

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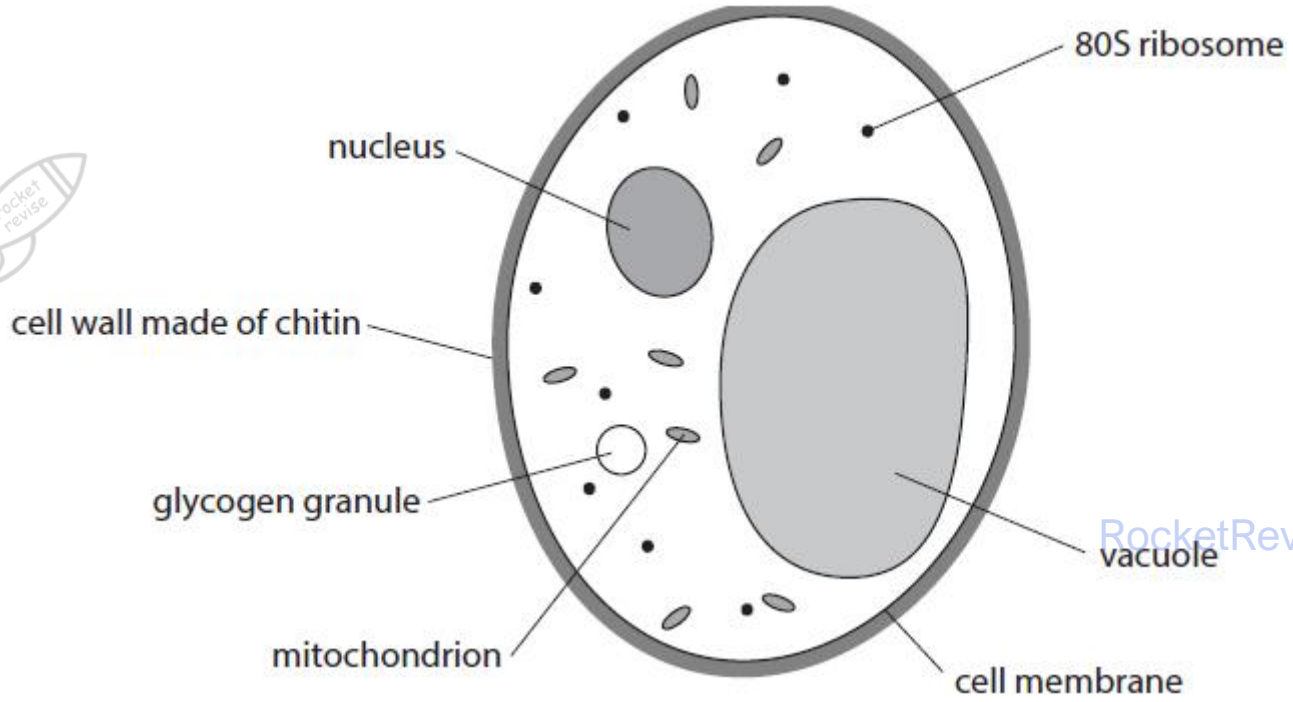
(c) Some millipedes have bacteria in their gut that can break down fungal cells.

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The diagram shows a fungal cell.

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Compare and contrast the structure of this fungal cell and a bacterial cell.

(3)

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(Total for question = 12 marks)

Q54.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

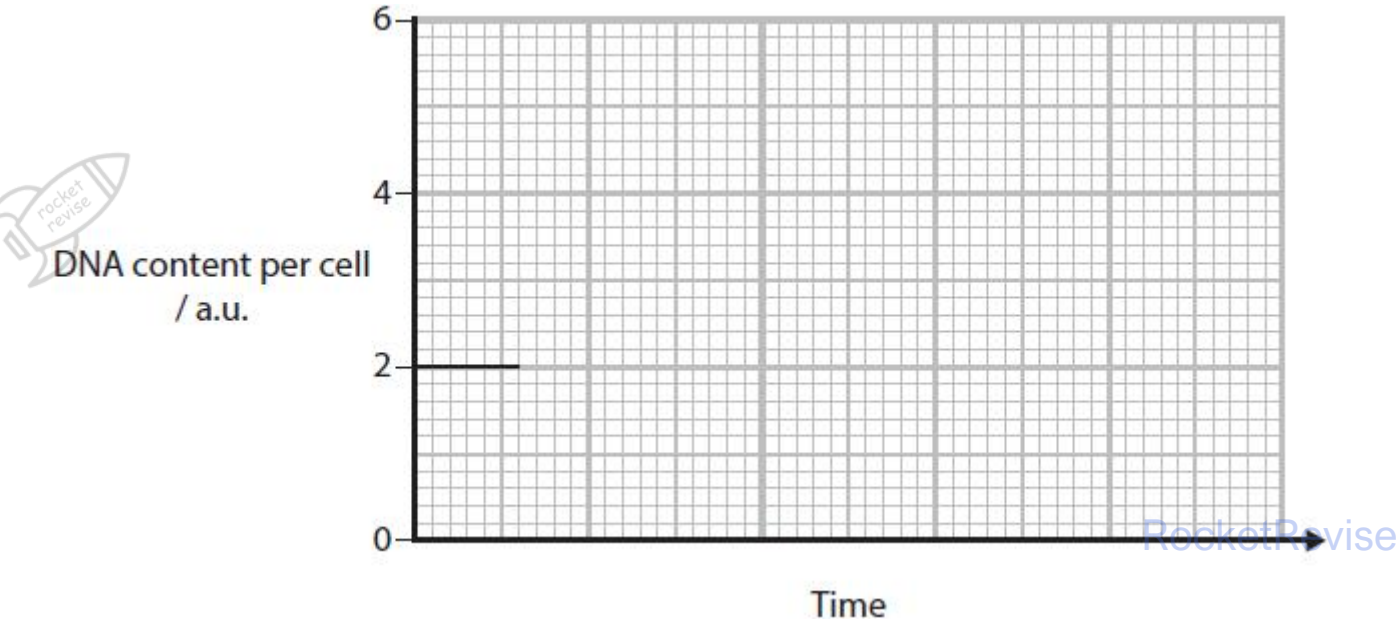
Some cells divide by meiosis to form gametes.

(a) Complete this graph to show how the DNA content per cell would change when a cell, with 2 a.u. of DNA, undergoes meiosis to produce gametes.

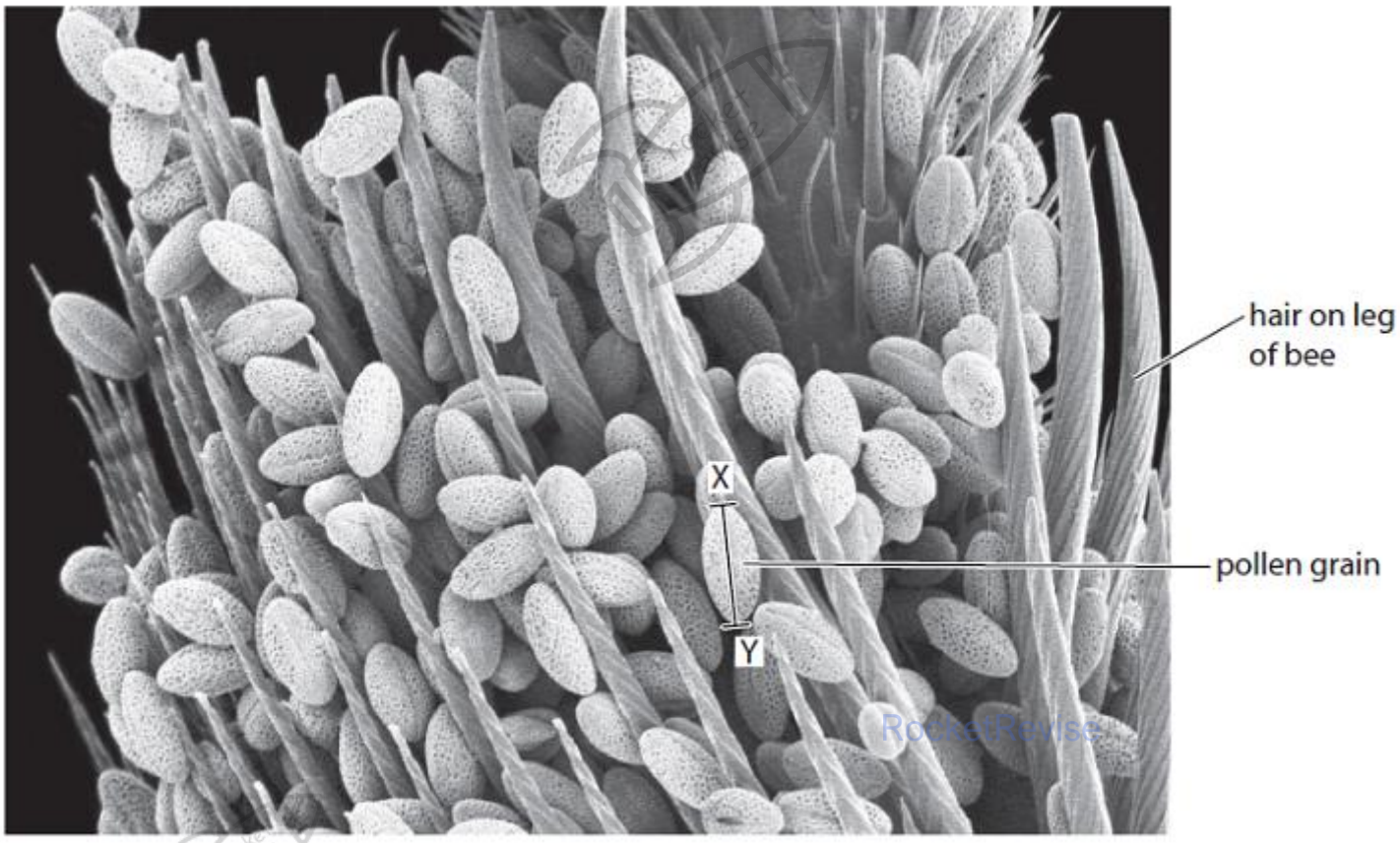
(3)

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(b) The photograph shows pollen grains on part of a leg of a bee.



(Source: © Susumu Nishinaga / Science Photo Library)

The actual length of the pollen grain between X and Y is 24 μm.
Calculate the magnification of this photograph.
Give your answer to **two** significant figures.

(2)

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Answer .....

(c) The photograph shows a bee on a peony flower.



(Source: © Boris Zhitkov / Alamy Stock Photo)

Pollen grains from the leg of the bee are transferred onto this flower.
The pollen grains will germinate and the pollen tubes will start to grow.
Describe the growth of a pollen tube and the process of fertilisation.

(5)

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(d) The genetic diversity in a population of one species of peony was investigated.

There were 418 peony plants in this population.

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Some of the genes of peony have several alleles.
The table shows the number of different alleles and the heterozygosity index for each of four genes in this population.

Gene	Number of different alleles of the gene	Heterozygosity index
PS43	10	0.422
PS103	10	0.810
PS105	4	0.036
PS147	16	0.834

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(i) How many of the peony plants in this population are heterozygous for gene PS105?

(1)

- ☒ **A** 15
- ☒ **B** 60
- ☒ **C** 340
- ☒ **D** 350

(ii) The width of peony leaves is an example of polygenic inheritance.
Which row is correct for the width of peony leaves?

(1)

	One gene determines leaf width	Leaf width is determined by multiple alleles for several genes	Leaf width shows continuous variation
☒ **A**	✓	✓	x
☒ **B**	✓	x	✓
☒ **C**	x	✓	✓
☒ **D**	x	x	x

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(Total for question = 12 marks)

Q55.

Humans have introduced new species to the island of St Helena.

Some of these species have become invasive and have reduced the biodiversity on this island.

These invasive species include cats, rabbits and rats.

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(a) The St Helena plover is a bird endemic to this island. It lays its eggs in a nest on the ground.

The photographs show a nest and offspring of a St Helena plover.



(Source: © Snapper Nick / Alamy Stock Photo)

Nest of a St Helena plover



(Source: © Buiten-Beeld / Alamy Stock Photo)

St Helena plover chick

(i) Suggest why the population of the St Helena plover decreased after the introduction of cats and rats to the island.

(2)

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(ii) Conservation programmes resulted in the population of the St Helena plover increasing from 200 adult birds in 2006 to 560 adults in 2016.

Calculate the mean rate of increase in this population.

(1)

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..... number of birds per year

(b) The St Helena gumwood tree is native to the island of Saint Helena.

The photograph shows part of a St Helena gumwood tree.

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(Source: © Prisma by Dukas Presseagentur GmbH / Alamy Stock Photo)

The St Helena gumwood tree is critically endangered.
Rabbits have contributed to this situation by eating the shoots of young trees.
A breeding programme increased the population of the gumwood trees from two trees in 2009 to 6000 in 2019.

(i) Describe how this St Helena breeding programme could have been carried out.

(2)

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(ii) Explain **one** way that young trees could be protected from rabbits.

(2)

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(iii) Scientists have proposed storing seeds from St Helena gumwood trees in seed banks to aid the conservation of this endangered species.
Describe how a seed bank would successfully select, prepare **and** store seeds from these trees to aid the conservation of this endangered species.

(4)

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(c) A study compared the biodiversity of two habitats on the island of St Helena.

The table shows the data obtained from habitat 1.

Species	Number of individuals (n)	n(n-1)
A	747	557 262
B	8	56
C	75	5 550
D	12	132
E	63	
	N = 905	$\sum n(n-1) =$

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(i) An index of diversity (D) is calculated using the formula:

$$D = \frac{N(N-1)}{\sum n(n-1)}$$

Calculate the index of diversity for habitat 1.
Use the table and formula to help you.
Give your answer to **two** decimal places.

(3)

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Answer .....

(ii) Habitat 2 has an index of diversity (D) of 2.80.
A student concluded that the second habitat had a higher biodiversity.
State whether you agree or disagree with this conclusion. Give a reason for your answer.

(1)

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(Total for question = 15 marks)

Q56.

An egg cell is a specialised cell.

(a) The acrosome reaction is stimulated when a sperm cell binds to the ZP3 protein on the zona pellucida of an egg cell.

Describe the acrosome reaction.

(3)

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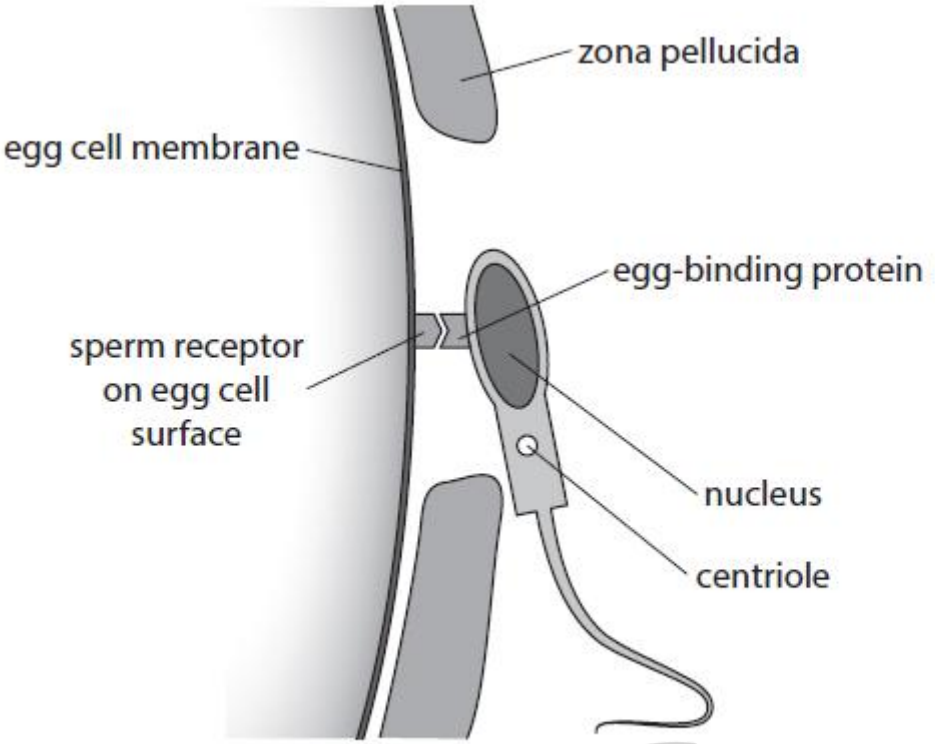
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(b) When a sperm cell arrives at an egg cell surface it binds to a receptor on the egg cell surface membrane.

The diagram shows a sperm cell binding to a complementary shaped receptor on the egg cell surface membrane.

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(Source: adapted from https://www.sciencedirect.com/topics/nursing-and-health-professions/zona-pellucida-glycoprotein)

(i) Suggest why some sperm cells may not be able to bind to the sperm receptor on the egg cell surface membrane.

(3)

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(ii) Explain how more sperm cells are prevented from entering the egg cell.

(2)

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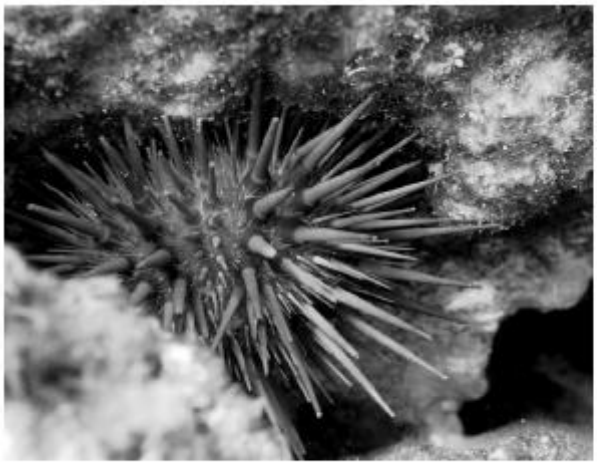
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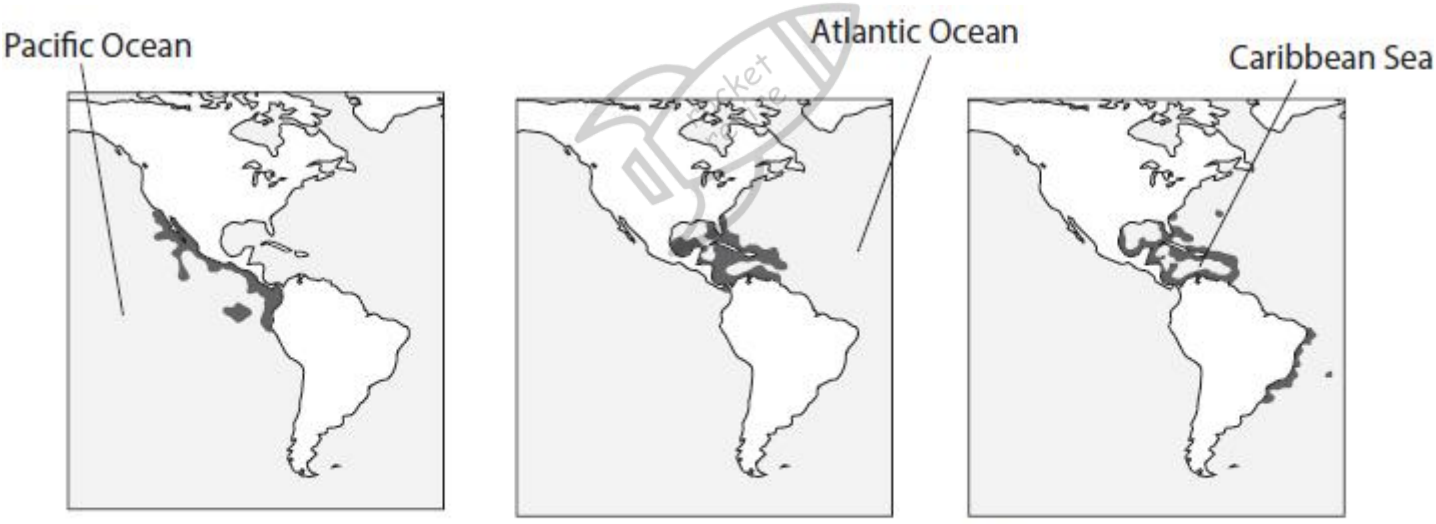
(c) The photograph shows a species of reef urchin (*E. lucunter*) found in the Caribbean Sea.



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(Source: © NomisH / Alamy Stock Photo)

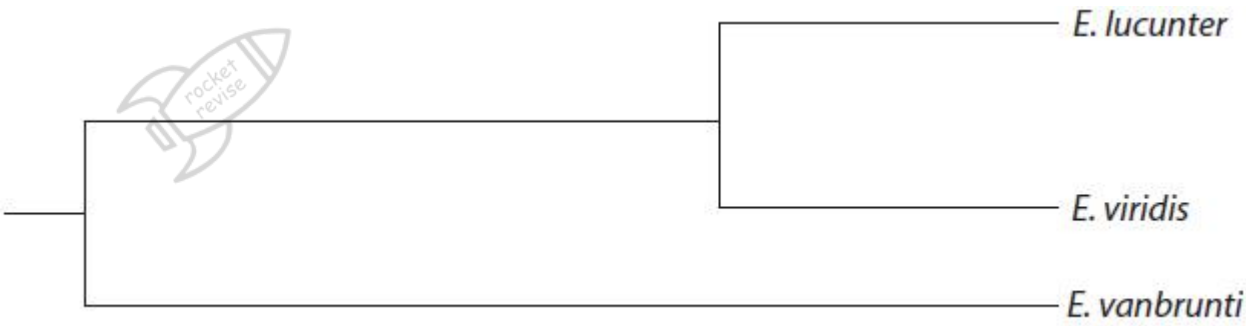
Sperm cells from *E. viridis* and *E. vanbrunti* cannot fertilise egg cells of *E. lucunter*. The maps show the distribution of these three species.



(Source: adapted from https://www.gbif.org/species/2278861)

The diagram shows the relationship between these three species.

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(i) Explain how scientists could have determined that *E. viridis* was more closely related to *E. lucunter* than to *E. vanbrunti*.

(2)

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*(ii) Explain how reproductive isolation could have led to the formation of these three species. Use your own knowledge and **all** the information from this question to support your answer.

(6)

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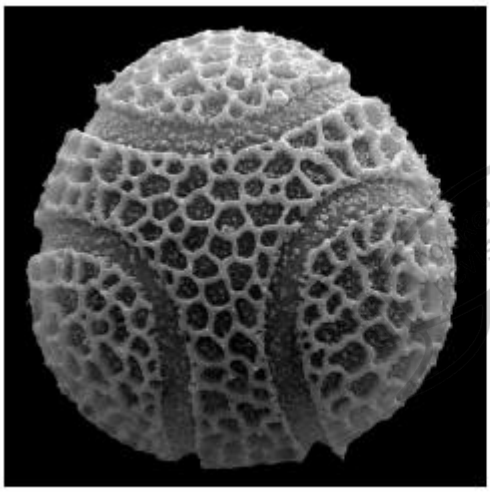
(Total for question = 16 marks)

Q57.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The photograph shows a pollen grain from a passion flower.



(Source: © Power and Syred / Science Photo Library)

(a) Pollen grains contain a generative nucleus.

The generative nucleus divides to form two male nuclei.
Which statement is correct?

(1)

- ☒ **A** one male nucleus is involved in the formation of the pollen tube
- ☒ **B** one male nucleus is needed to fertilise an egg cell nucleus to form the embryo
- ☒ **C** the male nuclei are diploid
- ☒ **D** two male nuclei are needed to fertilise a polar nucleus to form the endosperm

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(b) There is genetic diversity in the passion flower population.

Two causes of genetic diversity in this population are mutations and crossing over of non-sister chromatids on homologous chromosomes.

(i) Name the stage of meiosis when crossing over of chromatids occurs.

(1)

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(ii) Describe **two** other processes that can give rise to genetic variation in the offspring of the passion flower.

(2)

1 .....

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(iii) The heterozygosity index is a way of measuring the genetic diversity in a population.
Write an equation for the heterozygosity index.

(1)

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(c) Seeds from plant species are stored in seed banks.

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Explain how seeds would be selected for storage in a seed bank.

(2)

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(Total for question = 7 marks)

Q58.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Woese classified organisms into a three-domain system.

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(a) Which domains contain prokaryotic organisms?

1. Archaea
2. Bacteria
3. Eukarya

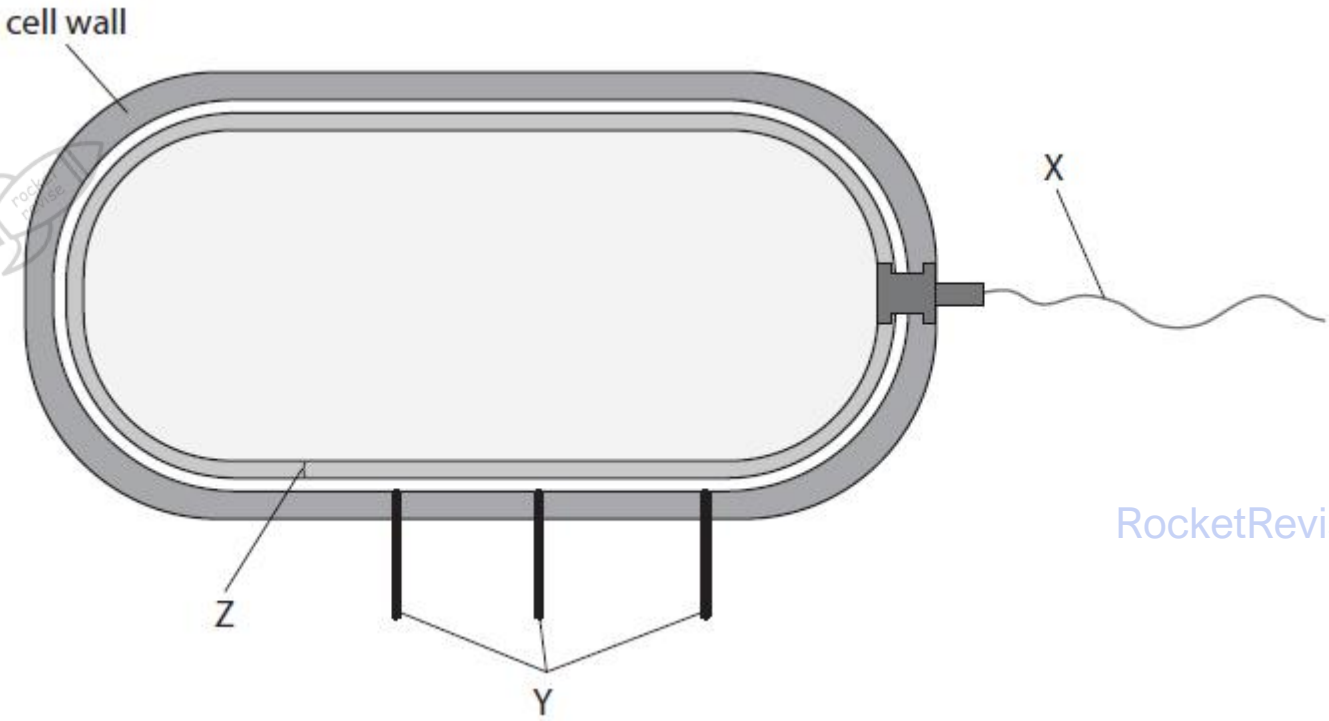
(1)

☐ **A** 1 and 2 only
☐ **B** 1 and 3 only
☐ **C** 2 and 3 only
☐ **D** 1, 2 and 3

(b) The diagram shows part of a prokaryotic cell.

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(i) Which row gives the name and function of structure Z?

(1)

		Name	Function
☐	**A**	capsule	adhesion to surfaces
☐	**B**	capsule	prevent water loss
☐	**C**	cell membrane	protein synthesis
☐	**D**	cell membrane	controls what enters and leaves the cell

(ii) Give the name and function of structure X and structure Y.

(2)

Name of structure X

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Function

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Name of structure Y

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Function

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(iii) Which structure could be found inside a **prokaryotic** cell?

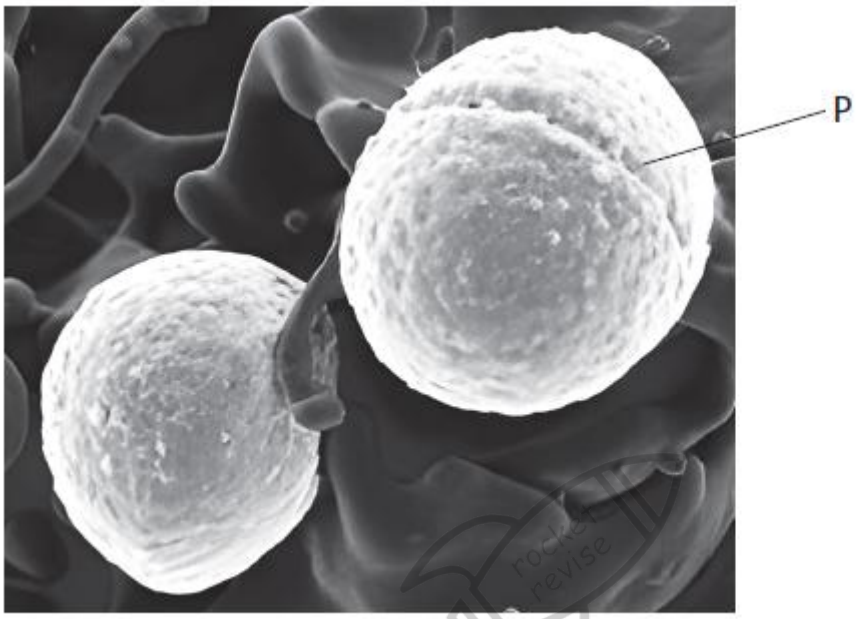
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(1)

- ☒ **A** chloroplast
- ☒ **B** linear DNA
- ☒ **C** nucleolus
- ☒ **D** ribosome

(c) The photograph shows two prokaryotic cells, as seen using an electron microscope.



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(Source: © BSIP SA / Alamy Stock Photo)

(i) The electron microscope had a resolution of 2 nm.
State what is meant by the term **resolution.**

(1)

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(ii) The cell labelled P has a diameter of 0.60 μm.
Calculate the volume of cell P.
Use the formula:

$$V = \frac{4}{3}\pi r^3$$

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Give your answer to **two** decimal places.

(2)

Answer ..... μm^3

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(Total for question = 8 marks)

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